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LENS: Layers of Evaluation of Hallucination in GenAI Systems

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***Abstract*—Large Language Models (LLMs) and Vision–Language Models (VLMs) demonstrate remarkable capabilities but remain vulnerable to hallucinations—producing plausible yet factually incorrect content—with error rates ranging from as low as 1.47% in clinical applications [1] to as high as 75% in domain-specific queries [2].**

Despite growing attention, existing hallucination evaluation frameworks remain insufficient to meet critical needs. Through a comprehensive survey of over 100 evaluation methods spanning six methodological paradigms (probe-based, adversarial testing,

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causal intervention, uncertainty-guided, internal state analysis, and online evaluation), we identify fundamental limitations: current approaches fall short of enabling objective model comparison, providing diagnostic insights into failure modes, supporting domain-evolving benchmark construction, and guiding targeted mitigation strategies.

Our analysis reveals that evaluations remain fragmented, operating either horizontally—comparing models across tasks and domains—or vertically—probing reasoning chains within single outputs. This fragmentation limits holistic assessment: horizontal evaluations provide breadth but risk superficiality, while vertical assessments deliver depth but lack generalizability. Moreover, we identify five critical gaps: (1) dimensional poverty reducing hallucinations to binary metrics, (2) failure to integrate horizontal breadth with vertical depth, (3) metacognitive blind spots overlooking when models should seek external verification, (4) adaptability crisis from static benchmarks, and (5) transparency deficits providing scores without actionable insights.

Beyond surveying the landscape, this paper articulates eight fundamental challenges confronting comprehensive evaluation—from epistemological difficulties in defining ground truth and computational complexity of scaling assessment, to attribution opacity obscuring causal mechanisms and dynamic knowledge evolution rendering benchmarks obsolete. These challenges span multimodal complexity, scale and diversity requirements, adversarial robustness, and human alignment considerations.

We present LENS (Layers of Evaluation of Hallucination in GenAI Systems), a unified framework addressing these gaps through hierarchical, tree-based query decomposition. LENS transforms complex evaluation tasks into multi-layered assessment structures via a six-stage pipeline (task formulation, decomposition, tool-augmented execution, structured generation, multi-dimensional scoring, and trace analysis), enabling MRI-like scanning of inference processes to reveal where and why hallucinations originate.

The framework introduces four key innovations: (1) *Tool Necessity Detection and Selection (TND/TSA)* – explicitly evaluating when models should consult external sources versus relying on parametric memory, addressing a fundamental hallucination source. (2) *Multi-Dimensional Metrics* – assessing degree (accuracy, faithfulness, tool appropriateness), quantity (coverage, completeness), stability (consistency, robustness), and risk (uncertainty quantification) beyond binary detection. (3) *User-Centric Benchmark Construction* – empowering organizations to design custom evaluations from their evolving knowledge bases while maintaining methodological rigor. (4) *Actionable Error Attribution* – providing hierarchical decomposition traces with causal attribution, evidence chains, and OpenTelemetry-based reproducibility for transparent auditing.

Our systematic taxonomy unifies previously fragmented approaches across evaluation targets (task-specific, modality-based, hallucination-type, domain-specific), dimensions (factuality, faithfulness, consistency, robustness, causal reasoning, interpretability), and methodologies. We introduce unified metrics transcending individual dimensions and present mitigation-aware evaluation strategies integrating RAG, parameter-efficient fine-tuning, knowledge distillation, preference optimization, and temporal intervention approaches.

By combining horizontal breadth (across domains and architectures) with vertical depth (into reasoning processes), LENS advances hallucination evaluation from post-hoc error detection to proactive risk assessment. Case studies in medical diagnosis, legal analysis, and financial reasoning demonstrate the framework's transformative impact, enabling objective model comparison, informed selection, diagnostic insights, domain-evolving benchmarks, and targeted mitigation development—fostering calibrated trust in AI systems deployed in safety-critical applications where accuracy, interpretability, and accountability are indispensable.

Keywords—Large language models, vision-language models, hallucination evaluation, hallucination mitigation, retrieval-augmented generation, comprehensive hallucination evaluation, horizontal vs vertical evaluation, generative AI evaluation, LENS framework, multi-dimensional metrics, multi-layered assessment structures, cross-domain evaluation, visual-textual misalignment, cross-modal inconsistencies, dynamic knowledge, evolving benchmarks, tool necessity detection, tool selection accuracy, hierarchical query decomposition, faithfulness evaluation, domain-specific benchmarks, user-centric benchmark construction, transparency and interpretability, surface-level accuracy, compositional complexity, ground truth ambiguity, reasoning failures, adversarial robustness, factuality assessment, uncertainty quantification, evidence grounding, tree-of-thoughts, reliability and robustness, causal reasoning capability, proactive risk assessment, computational efficiency, actionable error attribution, traceable evidence chains, post-hoc error detection, trustworthy AI, domain-specific queries, temporal drift, Universal Village, system theory, metacognitive evaluation, mechanistic interpretability, multimodal hallucination, causal attribution.

I. Introduction

Large Language Models (LLMs) and Vision-Language Models (VLMs) have emerged as transformative technologies, demonstrating remarkable capabilities in understanding and generating human-like text across diverse applications. However, these models suffer from a critical limitation that threatens their reliability: hallucination, which refers to the generation of content that appears plausible but is factually incorrect, internally inconsistent, or unfaithful to the provided context [3], [4]. As these systems are increasingly deployed in high-stakes domains, including healthcare diagnostics, legal analysis, and educational content generation, the ability to systematically evaluate and quantify hallucination becomes paramount for ensuring safe and trustworthy AI deployment.

The evaluation of hallucination in generative AI systems serves multiple critical purposes. It enables objective comparison across models and architectures, provides diagnostic insights into failure modes, establishes benchmarks for progress tracking, and informs the development of targeted mitigation strategies [5], [6]. Yet despite its importance, current evaluation approaches remain fragmented and incomplete, often focusing on either breadth, testing across multiple domains, or depth, examining complex reasoning chains, but rarely achieving both simultaneously [5]. This limitation becomes particularly acute as models grow more sophisticated and their deployment contexts more diverse.

In the following sections, we first examine the phenomenon of hallucinations and explain why their evaluation plays a critical role in Section I-A. We then outline the current landscape of hallucination evaluation in Section I-B, and highlight the significance of horizontal and vertical assessment in Section I-C. Building on this foundation, Section I-D specifies the stringent requirements and design principles that effective hallucination evaluation must satisfy to ensure both scientific rigor and practical applicability. These principles encompass

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Safety and Risk Awareness, Factuality, Faithfulness, Causal Fidelity, Interpretability, Actionability, Consistency, Reliability, Robustness, and Computational Cost and Efficiency, together forming a reliable foundation for the development of evaluation metrics across multiple dimensions.

Section I-E then elaborates the fundamental challenges that hinder the realization of a truly comprehensive evaluation framework, including compositional complexity, ground truth ambiguity, computational efficiency, cross-modal consistency, and dynamic knowledge. These challenges underscore the limitations of existing approaches and the necessity of methodological innovation.

Collectively, this analysis motivates our proposed evaluation strategy—LENS (Layers of Evaluation of Hallucination in GenAI Systems)—a comprehensive framework designed to address the identified gaps and challenges by enabling full-spectrum, diagnostic evaluation of hallucinations in generative AI systems. LENS establishes a robust foundation and standardized protocols for the development of evaluation metrics across multiple dimensions. Crucially, its full-spectrum benchmark construction not only supports objective comparison across models and architectures, but also facilitates tracking domain-knowledge evolution, provides diagnostic insights into underlying failure modes, and guides the design of targeted mitigation strategies.

A. The Hallucination Challenge: Definition and Impact

We define hallucination as the generation of content that deviates from factual accuracy, logical consistency, or faithfulness to input context. This encompasses three primary categories: **factual hallucinations** where models generate objectively false information such as incorrect dates or nonexistent entities; **logical inconsistencies** where outputs contain self-contradictory statements within the same response or reasoning chain; and **contextual infidelity** where generated content deviates from or contradicts the provided source material [3].

Recent empirical studies reveal the severity of this challenge. Healthcare applications exhibit hallucination rates of 1.47% in clinical note generation [1], while legal information retrieval shows 6.4% error rates even in state-of-the-art models [7]. More alarmingly, when queried about specific court rulings, LLMs can hallucinate up to 75% of the time [2].These statistics underscore that hallucination is not merely a technical curiosity but a fundamental barrier to reliable AI deployment.

The consequences of undetected hallucinations extend far beyond individual errors. In medical settings, hallucinated symptoms or treatments could lead to misdiagnosis or inappropriate interventions, potentially endangering patient lives [8], [9]. Legal professionals relying on AI-generated case citations risk professional sanctions for citing nonexistent precedents, as has already occurred in several high-profile cases [10], [11]. Educational systems using LLMs for content generation may inadvertently propagate misinformation to students, undermining educational objectives [12]. These risks necessitate robust, comprehensive evaluation frameworks that can not only detect hallucinations but also provide insights into their origins and patterns [13].

B. Current Evaluation Landscape: Categories and Limitations

The evaluation of hallucinations in generative AI systems has gained increasing attention in recent years, underscoring the urgency of this challenge. Effective assessment requires comprehensive coverage across diverse tasks, domains, and modalities. However, the research community has thus far produced a fragmented landscape of evaluation approaches. The proliferation of benchmarks and methodologies highlights both the significance of the problem and the absence of consensus on how best to conduct evaluation. Although this diversity has generated valuable insights into specific aspects of hallucination, it has also resulted in a scattered ecosystem in which findings are difficult to compare, synthesize, or translate into actionable improvements.

Existing hallucination evaluation approaches can be categorized into several primary paradigms, each with distinct focuses and fundamental limitations that prevent comprehensive assessment.

Task-format evaluations concentrate on specific operational modes, including question-answering benchmarks like TruthfulQA [4], summarization evaluators such as FactCC and QAGS [14], [15], and dialogue assessment tools like FaithDial [16], [17]. These approaches focus primarily on surface-level task performance rather than examining the underlying reasoning processes. They fail to capture multistep inference patterns and provide limited ability to identify where in the reasoning chain hallucinations originate, making it difficult to develop targeted mitigation strategies [18].

Modality-Based Evaluation introduces an additional axis of hallucination assessment, distinguishing between text-only and multimodal evaluation settings. Text-only evaluation primarily targets factuality and faithfulness in purely linguistic contexts, while multimodal evaluation extends the scope to include visual grounding, temporal reasoning, and cross-modal alignment, thereby addressing hallucinations that arise from incomplete or inconsistent integration of heterogeneous modalities.

Hallucination-Type–Based Evaluation further categorizes evaluation efforts according to representative manifestations of hallucinations, each illuminating distinct aspects of LLM behavior. These categories include: factual hallucinations (fabricated or incorrect world knowledge), reasoning hallucinations (errors in logical inference), memory-based hallucinations (inaccurate recall of conversational or contextual history), contextual hallucinations (misinterpretation of provided inputs), multimodal hallucinations (fabricated or mismatched cross-modal content), temporal or knowledge staleness (outdated or temporally inconsistent claims), and intrinsic versus extrinsic hallucinations (stemming from model-internal reasoning versus external contextual mismatch). Systematically evaluating these categories not only exposes the breadth of failure modes but also clarifies their differential impacts on reliability, interpretability, and safety across applications.

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Domain-specific evaluations test factual accuracy within specialized knowledge domains through benchmarks such as MedQA for medical knowledge [19], LegalBench for legal reasoning, and ScienceQA for scientific facts. While these provide valuable domain-specific assessment, they suffer from limited cross-domain generalization, inability to assess reasoning transfer between domains, and vulnerability to data contamination as static test sets become incorporated into training data over time [20], [21].

Furthermore, evaluation methods vary widely across operational dimensions, including granularity (e.g., token-level, sentence-level, or document-level assessment), reference availability (reference-based, reference-free, or hybrid approaches), the balance between automation and human involvement, as well as the core evaluation dimensions—such as factuality, faithfulness, consistency, robustness, causal reasoning capabilities, and interpretability.

In particular, capability-oriented evaluations represent a more recent development which introduces Benchmarks such as HotpotQA for multi-hop reasoning [22] and chain-of-thought–based evaluations [23] to probe models' ability to handle complex reasoning tasks. However, despite these advances, such approaches remain limited in several ways: they do not systematically decompose reasoning chains, they offer only partial interpretability in attributing errors to specific reasoning steps, and they provide minimal flexibility for user-defined evaluation criteria tailored to deployment-specific needs [24].

As a result, existing methods provide uneven coverage across these dimensions, leading to fragmented and incomplete assessment landscapes [5]. This underscores the need for evaluation frameworks that integrate both horizontal coverage across domains and vertical decomposition of reasoning chains, thereby enabling interpretable error attribution and customizable criteria that reflect real-world deployment contexts.

Methodologically, researchers have developed diverse techniques, such as probe-based evaluation (e.g., POPE [25]), adversarial and stress testing (e.g., ReEval [26]), causal intervention and counterfactual analysis (e.g., TruthPrint [27]), uncertainty-guided and self-consistency checks (e.g., SelfCheckGPT [28]), and internal state analysis (e.g., attention patterns) [29]. While these methods offer unique strengths, the lack of a unified framework makes results difficult to compare [3].

The design of evaluation tools and frameworks also faces challenges, including benchmark construction requirements (e.g., generalization, mitigating human bias), dataset construction phases (e.g., raw data acquisition, QA pairing, annotation), and the development of automated pipelines. Current tools often focus on specific aspects, lacking end-to-end solutions that limit their practicality [30].

C. The Evaluation Gap: Horizontal vs. Vertical Assessment

Current evaluation methods operate primarily at what we term the "horizontal" level, assessing performance breadth across various domains or tasks without examining the underlying reasoning process. This horizontal evaluation, while useful for benchmarking and model comparison, misses the critical "vertical" dimension: the hierarchical, step-by-step inference chain that leads to final outputs and determines where hallucinations originate. The detailed comparison between Horizontal Assessment and Vertical Assessment is summarized in Table I.

TABLE I. COMPARISON OF HORIZONTAL AND VERTICAL EVALUATION

Dimension	Horizontal Evaluation	Vertical Evaluation
Definition	Measures model performance breadth across domains, tasks, or datasets.	Examines inference chains step-by-step, focusing on intermediate reasoning.
Focus	Final output correctness across different contexts.	Process-level correctness at each reasoning step.
Strengths	Useful for benchmarking, model comparison, and cross-domain assessment.	Reveals where hallucinations originate and how errors propagate.
Limitations	Ignores reasoning process; cannot localize errors.	Narrower scope; requires detailed process tracing and interpretability.

Consider a complex query that requires multiple reasoning steps. Horizontal evaluation simply measures whether the final answer is correct across different domains or task types. In contrast, vertical evaluation examines each intermediate reasoning step, identifying precisely where errors emerge and how they propagate through the inference chain. This distinction is crucial because a model might produce a correct final answer despite flawed intermediate reasoning, or conversely, make a single early error that cascades into complete failure.

Current frameworks lack three critical capabilities that prevent comprehensive evaluation. First, they cannot accommodate user-oriented customization, preventing organizations from creating benchmarks from their own information sources tailored to specific deployment contexts [31]. Second, they provide minimal process transparency, offering scores without interpretable insights into the reasoning process, making it impossible to identify where and why hallucinations occur [32], [33]. Third, no existing framework effectively combines horizontal domain coverage with vertical reasoning analysis, forcing evaluators to choose between breadth and depth rather than achieving both.

This gap is particularly problematic for organizations deploying LLMs in specialized domains. A healthcare provider cannot simply rely on general medical benchmarks but needs evaluation tailored to their specific patient populations, treatment protocols, and documentation requirements [34]. Similarly, a legal firm requires assessment focused on their practice areas and jurisdictional requirements, not generic legal reasoning tests.

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D. Evaluation Requirements and Design Principles

After outlining the current landscape of hallucination evaluation in Section I-B and emphasizing the significance of horizontal and vertical assessment in Section I-C, this section turns to the stringent requirements and design principles that effective hallucination evaluation must satisfy to ensure both scientific rigor and practical applicability. These principles establish a reliable foundation for the development of evaluation metrics and protocols, which will be elaborated in detail in Section III-B4 and Section III-B.

Safety and Risk Awareness requires that evaluation incorporate explicit *safety likelihood and risk assessment*, quantifying not only the frequency of hallucinations but also their potential severity and impact. This probabilistic orientation is particularly crucial for deployment decisions in high-stakes domains, where even infrequent but high-severity failures can cause tangible harm in critical real-world applications. By prioritizing risk-sensitive evaluation, the framework ensures that the most consequential errors receive heightened scrutiny, thereby aligning evaluation outcomes with real-world safety imperatives.

Factuality demands reliable differentiation between factual and hallucinated content while minimizing false positives and false negatives [35]. Multisource verification and confidence-weighted scoring are employed to manage uncertainty in ground truth determination, including challenges related to *static knowledge* and *temporal factuality*, where outdated or evolving knowledge increases error risk in a dynamic world.

Faithfulness is a quantifiable and interpretable requirement that ensures models remain grounded in the evidence they are provided, rather than generating unsupported but plausible-sounding content. Faithfulness refers to the degree to which a model's output is consistent with the input source material, in contrast to factuality, which measures alignment with broader world knowledge. Faithfulness evaluation aims to verify whether each claim is explicitly supported by, or logically inferred from, the provided context, computes a faithfulness score as the proportion of supported claims, and directly captures a model's ability to avoid introducing extraneous, distorted, or contradictory content, thereby providing a critical safeguard against hallucinations.

Causal Fidelity underscores the necessity of assessing a model's causal reasoning capabilities. Beyond detecting surface-level correlations, evaluation must establish whether models correctly infer and articulate cause–effect relationships, while distinguishing between errors that arise from spurious correlations, temporal knowledge cutoffs, or structural flaws in reasoning. This capability is especially critical in high-stakes domains such as medicine, law, and public policy, where causal misattributions can lead to severe downstream consequences [36]. By incorporating causal fidelity into evaluation, frameworks ensure that models are not only factually accurate but also conceptually sound in their reasoning processes.

Interpretability elevates evaluation from a mere measurement exercise to a diagnostic process that provides transparency and actionable insights. In ideal framework, each evaluation node is accompanied by explicit evidence chains and confidence calibration, ensuring that every flagged instance of hallucination is supported by interpretable reasoning. This design enables evaluators to identify not only what content is incorrect, but also why it was classified as hallucinatory and which evidence supports that determination. By offering this level of explanatory depth, interpretability serves a dual purpose: it increases transparency and trust for end users while also equipping developers with targeted insights for model refinement.

Actionability necessitates that evaluation yields more than scalar performance scores; it must provide traceable insights that guide concrete improvements in model design and deployment. This requires a *vertical reasoning traceability* that dissects the model's inference chain layer-by-layer, localizing the origin of hallucinations to specific faulty premises, flawed synthesis steps, or error propagation pathways. This granular diagnosis is paramount for developing targeted mitigation strategies. Such actionable feedback transforms evaluation from a purely diagnostic tool into a driver of iterative system refinement.

Consistency ensures that evaluation performance is stable and reproducible across runs, configurations, and environments, making it indispensable for both scientific validity and real-world deployment. Without consistency, evaluation outcomes risk reflecting random noise rather than genuine model performance. Standardized protocols, seed-controlled sampling, and rigorous replication practices safeguard against such variability, ensuring that results can be independently verified and meaningfully compared across systems [29].

Reliability extends beyond consistency to evaluate the stability of model performance across time, contexts, and user populations. It explicitly accounts for factors such as *temporal drift*, *contextual variability*, and demographic differences that may introduce instability in evaluation outcomes. A reliable evaluation framework must therefore not only measure correctness at a single point in time but also capture how model behavior fluctuates under varying conditions. By systematically tracking these sources of variability, reliability assessment ensures that evaluations provide a trustworthy characterization of real-world performance rather than an overly static or idealized snapshot.

Robustness requires resilience against adversarial inputs, distributional shifts, and edge cases that can expose weaknesses in simplistic evaluation metrics [26]. A robust evaluation framework must diagnose how models respond under stress—whether to perturbations in input phrasing, adversarial prompts, or domain-specific challenges—without a disproportionate increase in hallucination rates. Our use of hierarchical decomposition strengthens robustness by breaking complex or ambiguous queries into smaller, verifiable components, thereby isolating vulnerabilities and improving the framework's capacity to withstand challenging evaluation conditions.

Computational Cost and Efficiency requirement reflects

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the inherent tradeoff between diagnostic depth and practical feasibility. Comprehensive hallucination evaluation is often computationally intensive, requiring multiple sampling rounds, retrieval operations, or complex verification pipelines. The principle of cost–efficiency ensures that hallucination evaluation remains scalable, accessible, and practicable across diverse scenarios, preventing rigorous evaluation from becoming the privilege of only well-resourced organizations. To significantly reduce computational burden while preserving diagnostic quality, evaluation frameworks must therefore incorporate optimizations—such as adaptive sampling (which adjusts evaluation depth based on uncertainty) and cascaded pipelines (which apply lightweight filters before costly deep checks).

These design principles will translate into the core dimensions we will use to evaluate specific model outputs, which will be discussed in detail in Section III-B4.

E. Major Challenges in Comprehensive Evaluation

Developing a truly comprehensive hallucination evaluation framework faces several fundamental challenges that calls for novel architectural and methodological innovations.

Developing a truly comprehensive hallucination evaluation framework entails addressing several fundamental challenges, which in turn necessitate novel architectural designs and methodological innovations.

Compositional Complexity emerges as queries become more sophisticated, with the number of possible reasoning paths growing exponentially [37]. Our tree-based approach manages this complexity through intelligent pruning strategies and verification gates that focus computational resources on the most promising reasoning paths while maintaining comprehensive coverage.

Ground Truth Ambiguity presents a philosophical and practical challenge, as many real-world queries lack definitive correct answers [38]. We address this through multi-source verification that aggregates evidence from multiple authoritative sources and confidence-weighted consensus mechanisms that acknowledge uncertainty rather than forcing binary judgments.

Computational Efficiency becomes critical for practical deployment, as comprehensive evaluation can be prohibitively expensive without optimization. We implement sophisticated caching strategies that avoid redundant computations and parallel evaluation paths that leverage modern computing infrastructure while maintaining evaluation integrity.

Cross-Modal Consistency adds another layer of complexity for vision–language models, where visual and textual modalities must align coherently [39], [40]. Our framework extends naturally to multimodal evaluation through modal-specific verification nodes that assess both within-modality accuracy and cross-modal alignment [41].

Dynamic knowledge presents an ongoing challenge as information changes over time, making static benchmarks obsolete [42], [43]. Our user-centric approach allows continuous benchmark updates from current sources, ensuring evaluations reflect contemporary knowledge rather than outdated training data.

Additionally, evaluating mitigation strategies (e.g., Retrieval-Augmented Generation [RAG], Low-Rank Adaptation [LoRA]) poses a challenge, as it requires measuring whether these strategies reduce hallucinations while avoiding new errors or performance degradation [44]. Hallucinations caused by language priors and biases are difficult to detect, as models may generate plausible but incorrect content based on statistical regularities in training data [45]. Prompting strategies also significantly impact hallucination rates, necessitating systematic assessment of prompt variations [46].

F. Proposed Solutions: The LENS Framework

To address these fundamental gaps, we present **LENS** (**L**ayers of **E**valuation of hallucinatio**N** in GenAI **S**ystems), a comprehensive evaluation framework that introduces a hierarchical, tree-based query decomposition system. Our framework transforms complex evaluation tasks into multilayered assessment structures, providing what we term "MRI-like scanning" of the model's inference process. Just as medical MRI technology enables layer-by-layer visualization of internal structures, our framework exposes each reasoning layer independently, identifies patterns across different reasoning paths, monitors how information propagates through the inference chain, and pinpoints exact locations where hallucinations emerge.

The key innovation lies in our tree-based decomposition approach, where complex queries are hierarchically broken down into atomic verification units. Each node in this tree represents a discrete reasoning step, while edges indicate logical dependencies between steps. This structure enables us to trace the exact path of reasoning and identify precisely where hallucinations originate, whether from faulty initial assumptions, incorrect intermediate inferences, or flawed final synthesis.

Our framework introduces four fundamental contributions that collectively advance the state of hallucination evaluation. First, we implement **Tool Necessity Detection and Selection (TND/TSA)**, explicitly evaluating whether models correctly identify when external verification is needed and select appropriate tools. This transforms evaluation from post hoc error detection to proactive risk assessment, addressing a critical source of hallucinations where models confidently generate information from parametric memory when they should instead consult external sources.

Second, we develop **Multidimensional Evaluation Metrics** that go beyond binary hallucination detection. Our framework assesses degree through accuracy, faithfulness, and tool appropriateness; quantity through coverage and completeness of information; stability through consistency across multiple evaluations; and risk through uncertainty quantification and confidence calibration. This comprehensive approach captures the multifaceted nature of hallucination that simple accuracy metrics miss.

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Third, we enable **User-Centric Benchmark Construction**, allowing organizations to create custom evaluation datasets from their own information sources. This ensures evaluation relevance to specific deployment contexts while maintaining standardized protocols that enable meaningful comparison across different implementations.

Fourth, our framework generates **Actionable Insights** beyond simple hallucination detection. We provide causal attribution of errors to specific reasoning steps, enabling developers to understand not just that a hallucination occurred but why it happened. This leads directly to suggested mitigation strategies based on observed error patterns and informed model selection guidance for specific use cases.

G. Paper Organization and Impact

The remainder of this paper systematically presents our framework, its theoretical foundations, empirical validation, and practical applications. Section II first analyzes the fundamental challenges for hallucination evaluation. Section III then surveys how existing evaluation methods and benchmarks address the discussed challenges, while Section IV examines the key limitations of current approaches and highlights the gap between the present state and the desired direction, thereby positioning our contributions within the broader research landscape. Section V introduces the innovative LENS architecture, which integrates a six-stage evaluation pipeline with a tree-based decomposition methodology. Section V-B further elaborates on the multidimensional evaluation metrics and their theoretical underpinnings. Finally, Section V-A and IV-E emphasize actionability, illustrating how evaluation results can be translated into practical improvements for both model developers and end users. Section V-H presents the Universal Village case study, showcasing real-world deployment and measurable impact, while Section V-H1 extends the framework to multimodal models, addressing the unique challenges of cross-modal hallucination detection. Section VI and Section VII conclude with a candid discussion of limitations, promising future directions, and the broader implications for trustworthy AI development.

Through systematic evaluation that combines breadth and depth, interpretability and automation, standardization and customization, LENS advances the field toward more reliable and trustworthy AI systems. Our framework enables informed model selection where organizations can choose models based on comprehensive performance profiles tailored to their specific needs rather than generic benchmarks. It facilitates targeted mitigation development by identifying precise failure points, allowing developers to create focused solutions rather than broad approximations. Most importantly, it enables trust calibration, giving users nuanced understanding of when and how to trust model outputs based on transparent evaluation results.

The ultimate goal transcends mere hallucination detection. We aim to understand hallucination origins, predict their occurrence, and systematically eliminate them, transforming LLMs from impressive but unreliable systems into trustworthy partners suitable for deployment in critical applications where accuracy, reliability, and interpretability are non-negotiable requirements.

II. Fundamental Challenges in Hallucination Evaluation

The evaluation of hallucinations in LLMs and VLMs confronts multiple interconnected challenges that span technical, epistemological, and practical dimensions [3]. These challenges emerge from the fundamental tension between the probabilistic nature of generative models and the deterministic requirements of factual accuracy, creating a complex landscape where traditional evaluation paradigms prove insufficient [47]. Understanding these challenges in their full depth is essential for developing effective evaluation frameworks and, ultimately, for building trustworthy AI systems capable of deployment in critical applications.

A. The Epistemological Challenge: Defining Ground Truth in Complex Domains

At the core of hallucination evaluation lies a profound epistemological challenge: establishing what constitutes ground truth in domains characterized by uncertainty, ambiguity, and evolving knowledge [48]. This challenge transcends simple fact verification and touches on fundamental questions about knowledge representation and validation in artificial intelligence systems [5].

The nature of ground truth varies dramatically across different knowledge domains and query types. In mathematical reasoning, ground truth appears deceptively clear through formal proofs and computational verification. Yet even here, complexity emerges when models must choose between multiple valid solution paths or when problems admit approximate solutions within acceptable tolerance bounds. The evaluation framework must distinguish between mathematically equivalent expressions that differ in notation, between optimal and merely correct solutions, and between errors in computation versus errors in reasoning approach [5].

Medical and scientific domains introduce temporal dynamics where ground truth evolves with advancing research. What constitutes accepted medical practice today may be considered outdated or even harmful tomorrow [8]. Clinical guidelines undergo continuous revision based on emerging evidence, creating a moving target for evaluation frameworks [1]. A model's response about treatment protocols might be simultaneously correct according to one professional organization's guidelines and incorrect according to another's, or accurate for one patient population but inappropriate for another [1], [34]. This variability demands evaluation frameworks that can accommodate multiple authoritative sources while acknowledging legitimate disagreements within expert communities.

Legal domains present unique challenges where ground truth depends heavily on jurisdictional context and temporal specificity. Legal precedents evolve through case law, statutes undergo amendments, and interpretations shift with changing judicial philosophies. An evaluation framework must navigate

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the distinction between black letter law and its practical application, between majority opinions and dissenting views, and between different legal traditions across jurisdictions. The same legal question might have fundamentally different answers depending on whether it pertains to common law or civil law systems, federal or state jurisdiction, or historical versus contemporary interpretation [49].

Cultural and linguistic contexts further complicate ground truth determination. Historical events may be interpreted differently across cultures, with varying emphasis on causes, significance, and outcomes. Social norms and acceptable practices differ across societies, making universal ground truth problematic for questions involving human behavior, ethics, or social conventions. Language itself introduces ambiguity through polysemy, metaphor, and contextual meaning that resist reduction to simple true or false classifications [49].

The challenge extends to handling contested knowledge and minority viewpoints. Scientific controversies, historical debates, and philosophical disagreements represent areas where multiple perspectives coexist without clear resolution. An evaluation framework must decide whether to privilege consensus views, acknowledge dissenting opinions, or treat controversial topics as inherently unsuitable for binary truth assessment. This decision has profound implications for how models learn to handle uncertainty and express appropriate epistemic humility [48].

B. The Computational Complexity Challenge: Scaling Comprehensive Evaluation

Comprehensive hallucination evaluation faces severe computational constraints that limit both the depth and breadth of assessment possible within practical resource limits [50]. The computational burden grows exponentially with the complexity of queries and the sophistication of verification required, creating fundamental tradeoffs between evaluation thoroughness and feasibility [51].

The combinatorial explosion in reasoning paths presents the first major computational hurdle. Complex queries that require multi-hop reasoning generate exponentially many possible inference chains, each potentially leading to different conclusions [52]. Consider a seemingly simple question about causal relationships in historical events: evaluating all plausible reasoning paths requires exploring numerous intermediate connections, alternative interpretations, and counterfactual scenarios [53]. Each path must be verified independently, with computational cost scaling exponentially with reasoning depth.

Verification itself imposes substantial computational overhead, particularly when external knowledge sources must be consulted. Real-time verification against authoritative databases, scientific literature, or legal documents requires not just retrieval but also interpretation and relevance assessment. The cost multiplies when multiple sources must be cross referenced to establish consensus or identify contradictions. Modern evaluation frameworks must balance the thoroughness of verification against practical constraints on API calls, database queries, and processing time [50].

The challenge intensifies for multimodal evaluation where visual and textual modalities must be assessed simultaneously. Vision language models require evaluation of object detection accuracy, attribute correctness, spatial relationship validity, and semantic consistency between visual and textual elements [25], [30]. Each modality demands specialized processing pipelines and verification methods, with computational cost scaling superlinearly when cross modal consistency must be verified [54], [55]. A single image caption evaluation might require object detection, scene graph construction, attribute extraction, and semantic parsing, each computationally intensive in its own right [56].

Temporal dynamics add another dimension of computational complexity. Knowledge bases require continuous updates to remain current, but revalidating existing evaluations against updated knowledge becomes computationally prohibitive at scale. The challenge is not merely storing new information but propagating updates through dependent facts and reassessing previously evaluated claims. This creates a fundamental tension between maintaining current knowledge and computational efficiency [42].

Statistical significance in evaluation results requires large sample sizes, further amplifying computational demands. Establishing that observed differences in hallucination rates reflect genuine model capabilities rather than random variation necessitates evaluating thousands or millions of examples. The sample size requirements grow when evaluating across multiple domains, tasks, and model configurations, quickly exceeding available computational budgets for thorough evaluation [51].

C. The Attribution and Interpretability Challenge: Understanding Causality in Hallucinations

Understanding why hallucinations occur proves as crucial as detecting them, yet current evaluation frameworks struggle to provide interpretable insights into the causal mechanisms underlying erroneous outputs. This attribution challenge limits both scientific understanding of model behavior and practical ability to develop targeted mitigations.

The opacity of neural network decision making creates fundamental barriers to attribution. Modern language models process information through billions of parameters organized in complex, non linear transformations that resist simple interpretation [57].When a model generates a hallucinated fact, determining whether the error originated from training data artifacts, architectural limitations, or inference time processing requires sophisticated analysis techniques that remain computationally expensive and often inconclusive [32]. The distributed nature of knowledge representation in neural networks means that no single component can be identified as solely responsible for a particular hallucination.

Distinguishing between different types of errors requires understanding their distinct causal pathways. Memory-based hallucinations, where models confidently generate false information from parametric memory, have different origins than reasoning errors where correct facts are combined incorrectly.

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Contextual hallucinations arising from misinterpretation of prompts differ from capability limitations where models lack necessary knowledge or computational capacity [58]. Current evaluation frameworks typically identify symptoms without diagnosing these underlying causes, limiting their utility for model improvement [57].

The propagation of errors through reasoning chains presents particular challenges for attribution. In multi step reasoning, an early error can cascade through subsequent steps, amplifying into complete hallucination. Identifying the initial point of failure requires tracking information flow through the entire reasoning process, understanding how uncertainty compounds, and distinguishing between error propagation and error correction attempts. This becomes especially complex when models engage in implicit reasoning not reflected in their outputs.

Interaction effects between different model components further complicate attribution. Attention mechanisms, feed forward networks, and layer normalization interact in complex ways that can either amplify or suppress hallucination tendencies. The same factual error might manifest differently depending on prompt formulation, context length, or decoding parameters, suggesting that hallucinations emerge from system level interactions rather than isolated component failures. Understanding these interactions requires evaluation frameworks that can systematically vary conditions while tracking internal model states.

The challenge extends to attributing hallucinations in retrieval augmented systems where models access external knowledge sources. When hallucinations occur despite access to correct information, determining whether the failure lies in retrieval, interpretation, or integration becomes critical for improvement. The model might retrieve irrelevant information, misinterpret relevant sources, or fail to appropriately weight external evidence against parametric knowledge. Each failure mode requires different mitigation strategies, yet current evaluation frameworks rarely provide sufficient granularity to distinguish between them.

D. The Dynamic Knowledge Challenge: Evaluating Against Evolving Information

Knowledge is not static, and the temporal dynamics of information present fundamental challenges for hallucination evaluation [42]. Facts change, scientific understanding evolves, and social contexts shift, creating a moving target that static evaluation frameworks cannot adequately address [43].

The most visible aspect of this challenge appears in rapidly evolving domains like current events, scientific research, and technology. A model's statement about political leadership, pandemic statistics, or technological capabilities might be accurate when generated but become hallucination within days or hours. Evaluation frameworks must grapple with temporal scoping, determining the appropriate timeframe for knowledge currency and handling the transition from current fact to historical information. This requires not just updating evaluation datasets but also understanding how models should appropriately caveat time sensitive information [43].

Scientific knowledge evolution presents particular complexities beyond simple fact updates. Scientific understanding progresses through hypothesis, experimentation, and theory refinement, with periods of uncertainty and contested findings. A model's statement about medical treatments might reflect outdated consensus, emerging research, or controversial findings, each requiring different evaluation treatment. The framework must distinguish between superseded theories that are now considered false and historical theories that were reasonable given available evidence at the time [8].

The challenge extends to evaluating model handling of conflicting information during transition periods. When new research challenges established understanding, multiple viewpoints coexist temporarily in scientific literature. Models must navigate this uncertainty, and evaluation frameworks must assess not just factual accuracy but also appropriate expression of uncertainty and acknowledgment of ongoing debates. The binary classification of hallucination versus truth proves inadequate for capturing the nuanced reality of evolving knowledge [42].

Knowledge deprecation poses additional challenges, as information becomes not just outdated but actively misleading. Legal precedents get overturned, medical procedures become contraindicated, and technical standards become deprecated. Evaluation frameworks must track not just what is currently true but what was previously accepted and why it changed. This historical context proves essential for understanding whether a model's outdated information reflects training data artifacts or failure to incorporate recent updates [43].

Cultural and social knowledge exhibits distinct evolutionary patterns that resist simple temporal modeling. Social norms, linguistic conventions, and cultural practices change gradually and non uniformly across different communities. What constitutes acceptable terminology, appropriate behavior, or accurate cultural description varies not just over time but across contemporary social groups. Evaluation frameworks must navigate these dynamics while avoiding both cultural essentialism and relativistic paralysis [49].

E. The Multimodal Complexity Challenge: Cross Modal Consistency and Verification

Vision language models introduce unique evaluation challenges that go beyond simple extension of text based approaches. The interaction between visual and linguistic modalities creates new forms of hallucination and requires sophisticated evaluation methods that can assess both within modal accuracy and cross modal consistency.

Object hallucinations in visual contexts manifest differently than textual factual errors. Models might describe objects not present in images, misattribute properties to existing objects, or invent plausible but incorrect spatial relationships [25], [59]. Evaluating these requires not just object detection capabilities but understanding of visual common sense, physical constraints, and scene semantics [54]. The challenge intensifies when dealing with ambiguous, occluded, or artistically

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rendered visual content where even human annotators might disagree on ground truth [60].

Cross modal grounding presents fundamental challenges in establishing correspondence between visual and textual elements. When a model describes a complex scene, evaluation must determine which visual regions correspond to which textual descriptions, whether relationships are accurately captured, and if the overall semantic meaning is preserved. This mapping problem becomes especially complex for abstract concepts, metaphorical language, or culturally specific visual symbols that resist literal interpretation.

The compositional nature of visual understanding creates evaluation challenges absent in text only contexts. Understanding a visual scene requires integrating information about objects, attributes, relationships, and context into coherent interpretation [56]. Hallucinations can occur at any level of this hierarchy, from misidentifying basic objects to misunderstanding complex compositional relationships [55]. Evaluation frameworks must assess accuracy across these multiple levels while understanding how errors at one level affect interpretation at others [30].

Temporal dynamics in video understanding add another layer of complexity. Video based models must track objects across frames, understand temporal relationships, and maintain consistency in their descriptions over time [61]. Hallucinations might manifest as phantom objects appearing and disappearing, incorrect temporal ordering of events, or inconsistent object tracking. Evaluating these requires frameworks that can assess both frame level accuracy and temporal coherence.

The interaction between modalities creates unique forms of hallucination not present in unimodal systems. Models might generate textually plausible but visually impossible descriptions, or visually reasonable but semantically meaningless interpretations. These cross modal inconsistencies prove particularly challenging to evaluate as they require deep understanding of both modalities and their interaction constraints [55].

F. The Scale and Diversity Challenge: Representative Evaluation Across Domains

Creating evaluation frameworks that adequately represent the full spectrum of potential model applications presents enormous challenges in scale, diversity, and representativeness. The space of possible queries, contexts, and use cases is effectively infinite, yet practical evaluation must work with finite resources and limited samples [62].

Domain coverage represents the first dimension of this challenge. Language models are applied across hundreds of specialized fields, each with unique terminology, reasoning patterns, and accuracy requirements [62]. Medical applications demand different evaluation approaches than creative writing, legal analysis requires different verification methods than customer service, and scientific reasoning needs different assessment criteria than casual conversation [63], [64]. No single evaluation framework can adequately cover this diversity without becoming unwieldy and computationally intractable [65].

The long tail distribution of knowledge presents particular challenges for representative evaluation. While models might perform well on common topics covered extensively in training data, performance on rare or specialized topics often degrades dramatically. Yet these edge cases often represent critical use cases where accuracy is most important. Evaluation frameworks must balance coverage of common scenarios against the need to test boundary conditions and rare but important cases [65].

Task diversity adds another dimension requiring careful consideration. The same model might be used for question answering, summarization, translation, creative generation, code writing, and reasoning tasks, each with distinct hallucination patterns and evaluation requirements [66], [67]. A model that excels at factual question answering might hallucinate extensively when generating long form explanations, or vice versa [68]. Comprehensive evaluation must span this task diversity while maintaining comparability across different evaluation modes.

Geographic and cultural representation poses ongoing challenges for global deployment of language models. Evaluation datasets typically exhibit strong geographic and cultural biases, overrepresenting certain regions, languages, and cultural contexts while neglecting others [49]. This creates blind spots where models might hallucinate extensively about underrepresented cultures or geographic regions without detection [69]. Building truly representative evaluation frameworks requires intentional effort to include diverse perspectives and knowledge sources [70].

The interaction between scale and statistical validity creates practical constraints on evaluation design. Achieving statistical significance requires substantial sample sizes, but comprehensive coverage across domains, tasks, languages, and cultures would require millions of evaluation examples. This forces difficult tradeoffs between breadth of coverage and depth of evaluation within each area. Stratified sampling and adaptive evaluation strategies offer partial solutions but cannot fully resolve the fundamental tension between comprehensive coverage and practical constraints [51].

G. The Adversarial Robustness Challenge: Evaluation Under Distribution Shift

Evaluation frameworks must contend with the reality that deployed models face inputs deliberately crafted to induce hallucinations or that naturally deviate from training distributions. This adversarial robustness challenge requires evaluation approaches that go beyond benign test cases to assess model behavior under challenging conditions [26].

Adversarial prompting represents an immediate challenge where carefully crafted inputs can induce hallucinations even in otherwise reliable models [26]. These prompts might exploit known model biases, use unusual phrasings that confuse natural language understanding, or combine information in ways that trigger incorrect pattern matching [45]. Evaluation frameworks must systematically test model robustness to such adversarial inputs while distinguishing between legitimate

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edge cases and artificial constructs unlikely to occur in practice [26].

Distribution shift poses a more subtle but equally important challenge. Models deployed in production inevitably encounter inputs that differ from their training distribution in vocabulary, style, topic, or format. These shifts can trigger hallucinations as models extrapolate beyond their training experience. Evaluation must assess not just performance on in distribution test sets but also degradation patterns under various types of distribution shift. This requires frameworks that can systematically vary input characteristics while maintaining semantic validity [45].

The evolution of adversarial techniques creates an arms race between attack and defense. As evaluation frameworks identify and test for certain types of adversarial inputs, new attack vectors emerge that exploit previously unknown vulnerabilities. This dynamic nature means that evaluation frameworks must continuously evolve, incorporating new adversarial strategies while maintaining backward compatibility for tracking improvement over time [26].

Prompt injection and manipulation present particular challenges for models deployed in interactive settings. Users might intentionally or accidentally include instructions that override system prompts, leading to hallucinations or other undesired behaviors. Evaluation frameworks must test model robustness to various forms of prompt manipulation while assessing whether safety mechanisms appropriately balance security against functionality [45].

The challenge extends to evaluating model behavior under composed adversarial conditions. Real world deployments might face inputs that combine multiple challenging characteristics: ambiguous phrasing, unusual formatting, mixed languages, and implicit adversarial intent. These composed challenges often trigger failure modes not apparent when testing individual factors in isolation. Evaluation frameworks must assess model robustness to realistic combinations of challenging conditions while maintaining interpretability about which factors contribute most to observed failures [26].

H. The Human Alignment Challenge: Evaluating Subjective and Value Laden Dimensions

Beyond factual accuracy, hallucination evaluation must grapple with subjective dimensions of human preference, cultural values, and contextual appropriateness [31]. This human alignment challenge introduces evaluation complexities that transcend simple truth verification.

The subjective nature of quality assessment creates fundamental challenges for standardized evaluation. What constitutes a helpful, harmless, and honest response varies across individuals, cultures, and contexts. A response deemed appropriately detailed by one user might be considered verbose by another; information considered essential in one cultural context might be irrelevant or offensive in another. Evaluation frameworks must navigate these subjective dimensions while maintaining sufficient standardization for meaningful comparison.

Value alignment presents particular challenges when evaluating responses to ethically charged questions. Models must balance factual accuracy with social responsibility, potentially refusing to provide technically accurate information that could enable harm. This creates situations where hallucination, in the form of deflection or refusal, might be preferable to truthful responses. Evaluation frameworks must distinguish between protective hallucination and harmful misinformation while respecting diverse ethical frameworks and cultural values.

Context appropriateness adds layers of complexity beyond factual accuracy. The same information might be appropriate for adult audiences but not children, relevant in academic contexts but not casual conversation, or acceptable in some cultures but not others. Models must modulate their responses based on implicit and explicit contextual cues, and evaluation frameworks must assess this contextual sensitivity. This requires understanding not just what models say but how they adapt their communication style and content selection to different contexts.

The challenge of evaluating implicit bias and fairness intersects with hallucination detection in complex ways. Models might generate statistically accurate but socially harmful stereotypes, or conversely, might hallucinate in attempting to avoid bias. Evaluation frameworks must navigate the tension between statistical accuracy and social fairness, assessing both whether models generate factually correct information and whether they do so in ways that perpetuate or mitigate social biases.

User expectation alignment presents ongoing challenges as mental models of AI capabilities evolve. Users increasingly expect models to express appropriate uncertainty, acknowledge limitations, and provide metacognitive insights about their own reasoning [46]. Evaluation frameworks must assess not just whether models provide correct information but whether they do so in ways that calibrate user trust appropriately [71]. This includes evaluating whether models appropriately convey uncertainty, acknowledge when they lack information, and guide users toward reliable verification when needed.

I. Synthesis: The Imperative for Revolutionary Change

The challenges explored throughout this section converge on an inescapable conclusion: hallucination evaluation represents not a technical problem amenable to incremental refinement but a fundamental crisis requiring revolutionary reconceptualization. The epistemological impossibility of universal ground truth, the exponential computational complexity of comprehensive assessment, the opacity of causal attribution, the relentless evolution of knowledge, the irreducible complexity of multimodal interaction, the infinite diversity of deployment contexts, the persistent threat of adversarial exploitation, and the incommensurable variety of human values together create a problem space that transcends the capabilities of any existing evaluation paradigm.

Current frameworks fail not through poor implementation but through conceptual inadequacy. They assume hallucination can be reduced to binary classification when it manifests

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as a multidimensional phenomenon. They treat evaluation as static measurement when knowledge and requirements evolve continuously. They provide scores without explanations when understanding causation proves essential for improvement. They impose rigid benchmarks when deployment contexts demand adaptability. They operate as black boxes when transparency determines trustworthiness. These are not bugs to be patched but fundamental architectural flaws that necessitate complete reimagination of how we approach evaluation.

The path forward demands more than technical innovation; it requires paradigmatic transformation in how we conceptualize the relationship between AI systems and truth. We must abandon the fiction that evaluation can be separated from deployment context, that universal benchmarks can capture domain specific requirements, or that static assessment can address dynamic systems. Instead, we must embrace evaluation as an ongoing dialogue between human values and machine capabilities, where frameworks adapt continuously to evolving requirements while maintaining methodological rigor.

This transformation cannot emerge from isolated research efforts but requires coordinated action across the entire AI ecosystem. Machine learning researchers must develop architectures that expose reasoning processes rather than hiding them behind inscrutable parameters. Cognitive scientists must illuminate how human evaluation of truth can inform machine assessment. Philosophers must guide our understanding of knowledge, truth, and justified belief in the context of probabilistic systems. Domain experts must specify what constitutes acceptable uncertainty in their fields. Affected communities must articulate their requirements for trustworthy AI. Only through such comprehensive collaboration can we develop evaluation frameworks that match the sophistication of the systems we seek to deploy.

As we transition from understanding these fundamental challenges to examining how current evaluation approaches fail to address them, we carry forward a critical insight: the problem is not that we lack good evaluation frameworks but that we have been asking the wrong questions. The following section reveals how existing approaches, despite their technical sophistication and widespread adoption, remain trapped within conceptual paradigms that cannot accommodate the true complexity of hallucination evaluation. This examination will expose not just what current frameworks do wrong but why their fundamental assumptions doom them to insufficiency, setting the stage for understanding why revolutionary change through frameworks like LENS becomes not just desirable but essential.

III. Current Evaluation Landscape: Fragmentation and Insufficiency

Given the fundamental challenges outlined above, including epistemological complexities in defining ground truth [48], computational constraints in scaling evaluation, and attribution difficulties in understanding causality [32], the research community has responded with a diverse but fragmented landscape of evaluation approaches [5]. This proliferation of benchmarks and methodologies reflects both the urgency of the hallucination problem and the absence of consensus on how best to address it. While this diversity has yielded valuable insights into specific aspects of hallucination, it has also created a fragmented ecosystem where results across different evaluations remain difficult to compare, synthesize, or translate into actionable improvements [5].

The current evaluation landscape reveals a fundamental tension: while the hallucination problem demands comprehensive assessment across diverse contexts, practical constraints have led to a patchwork of specialized benchmarks that often operate in isolation. Recent surveys indicate that many distinct hallucination benchmarks have emerged since 2022 [3], yet their heterogeneous evaluation protocols and metrics make cross-benchmark comparison nearly impossible [3]. This fragmentation manifests in three critical ways: (1) Task divergence, with evaluation contexts spanning from constrained QA to open-ended generation, making performance correlation analysis infeasible; (2) Metric inconsistency, where benchmarks employ incompatible scoring systems ranging from binary classification accuracy to fine-grained semantic similarity scores; and (3) Ground truth ambiguity, where different benchmarks adopt conflicting definitions of what constitutes a hallucination.

Another dimension of fragmentation arises from the inherent limitations of manually constructed datasets, which often fail to capture the complexity of real-world use cases and user interactions. As discussed in Section III-G2, most widely adopted benchmarks are designed for narrow contexts and therefore lack generalizability across domains, modalities, and user populations. Moreover, these datasets frequently show weak alignment with human judgments and inadequately represent the complexities of subjective evaluation, including cultural variation, contextual nuance, and the multifaceted nature of human reasoning. Manual annotation also introduces systematic human biases and exposes datasets to adversarial noise or annotation artifacts, thereby reducing their robustness and reliability. Furthermore, such datasets provide only a partial representation of model performance and fall short of capturing the full spectrum of thought processes underlying LLMs.

This fragmentation not only impedes systematic progress but also creates an illusion of advancement where improvements on specific benchmarks may not translate to real-world reliability—a phenomenon typically referred to as "benchmark overfitting" where models achieve high scores on curated datasets while exhibiting not low hallucination rates in deployment.

To provide a structured summary of the current research landscape on hallucination evaluation, we organize existing studies under six key themes. First, evaluation targets and taxonomy clarify what aspects of hallucination researchers have focused on, as discussed in Section III-A. Second, evaluation dimensions and metrics outline the criteria and measurement standards that have been proposed, as discussed in Section III-B. Third, a mechanistic understanding of decision-

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making highlights efforts to connect evaluation outcomes to the underlying causes of hallucinations , as discussed in Section III-C. Fourth, evaluation methodologies describe the practical approaches that have been implemented to carry out assessments, as discussed in Section III-D. Fifth, evaluation across core dimensions reflects the broader scope of work addressing reasoning, safety, robustness, and reliability , as discussed in Section III-E. Sixth, mitigation-aware evaluation captures attempts to directly link evaluation to strategies for reducing hallucinations as discussed in Section III-F. Finally, tools and frameworks summarize the infrastructure developed for evaluation practices, as discussed in Section III-G. Together, these categories provide a comprehensive overview of how the field has approached hallucination evaluation to date, as discussed in Section III-H.

A. Evaluation Targets and Taxonomy (What to Evaluate)

This section investigates existing evaluation efforts along four complementary categories: Task-Specific, Modality Coverage, Hallucination Typology, and Domain Expertise. Each category represents a distinct lens through which hallucinations can be understood and measured, providing insights into how different forms of evaluation capture particular aspects of model behavior.

However, examining these dimensions in isolation also exposes a broader challenge: the current evaluation landscape is fragmented, with specialized benchmarks excelling in depth within narrow contexts and unified benchmarks aiming for breadth but often sacrificing contextual rigor. This tension highlights the importance of discussing The Integration Challenge – Unified vs. Specialized Benchmarks. Understanding how to balance standardization and comparability against domain sensitivity and diagnostic depth is essential for advancing hallucination evaluation. A careful analysis of this integration challenge not only clarifies trade-offs between coverage and precision but also lays the groundwork for hybrid or modular approaches that can reconcile these competing demands.

1) Task-Specific Evaluation - From Simple Queries to Complex Reasoning: Hallucination patterns vary dramatically across task contexts. We organize these tasks along a spectrum of complexity, from well-defined question-answering to open-ended generation, reflecting increasing degrees of freedom in both input interpretation and output space.

a) Question Answering: The QA evaluation paradigm represents the most controlled setting for hallucination assessment, yet reveals surprising complexity in failure modes. TruthfulQA [4] pioneered adversarial question design with 817 questions across 38 categories, specifically crafted to trigger human-like false beliefs. Its key innovation lies in distinguishing between truthful and informative responses—the best model achieves only 58% truthfulness, exposing how fluency masks factual errors [4].

HaluEval [7] scaled evaluation to 35,000 samples, revealing task-specific hallucination patterns: including general queries, knowledge-grounded dialogue, and summarization tasks [7]. Its automated generation pipeline using ChatGPT achieved high correlation with human judgments while reducing annotation costs [7]. HaluEval-Wild [42], the first benchmark designed to assess LLM hallucinations in the wild, first prescreens user queries from the ShareGPT dataset to form the initial pool of challenging queries via classifiers, then selects and categorizes into five types of 500 challenging user queries [42].

OnionEval [58], a multi-layer structured framework with a specific metric, context influence score (CI), is constructed to evaluate the fact-conflicting hallucination tendencies of small LLMs across different contextual levels. The experimental results showcase one important feature for SLLMs that SLLMs perform fine on factual analysis while struggling with context reasoning while simple Chain-of-thought strategy help enhance the practical applicability of SLLMs in real-world scenarios [58].

b) Conversational Tasks: Dialogue evaluation exposes temporal consistency challenges absent in single-turn interactions. DiaHalu [17] is a dedicated dialogue-level hallucination evaluation benchmark for LLMs, which includes four common multi-turn dialogue domains, and five hallucination subtypes, extending from factuality and faithfulness hallucination. To construct Dialogue, the system first combines the collected topics into system prompts, facilitates a dialogue between two LLMs, manually modifies the information not adhering to human language conventions, and then has LLMs re-generate, simulating authentic human-machine interaction scenarios. At the end, professionals annotate all samples from the dataset. Experiments on all well-known LLMs and detection methods show that DiaHalu is a challenging and valuable benchmark for research communities.

HalluDial [16] is also a comprehensive large-scale benchmark for automatic dialogue-level hallucination evaluation, comprising 4,094 dialogues with a total of 146,856 samples that contain both spontaneous and induced hallucination scenarios. The benchmark, including both factuality and faithfulness hallucinations, enables a comprehensive meta-evaluation of LLMs' hallucination evaluation capabilities in information-seeking dialogues, and introduces a specialized judge language model, HalluJudge, which achieve competitive performance in hallucination evaluation, facilitating the automatic assessment of dialogue-level hallucinations in LLMs and providing valuable insights into this phenomenon.

c) Instruction Following: Instruction-following is a critical dimension in hallucination evaluation, as it tests whether a model can faithfully interpret user intent and produce outputs that are accurate and appropriate. Evaluation methods typically measure both instruction adherence (fulfilling all aspects of the request) and content veracity (ensuring factual correctness and contextual relevance).

Current evaluation approaches typically employ carefully designed instruction sets that probe specific vulnerability points. These include: (1) Precision-critical instructions requiring exact numerical, temporal, or spatial reasoning where minor inaccuracies may cascade into significant errors [72]; (2) Multi-step procedural instructions that test the model's

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ability to maintain consistency throughout extended reasoning chains [73]; and (3) Context-bound instructions that examine whether models appropriately respect provided constraints without introducing external assumptions [74].

The relationship between instruction following capability and hallucination propensity exhibits complex patterns. While reinforcement learning from human feedback (RLHF) significantly improves instruction adherence—with InstructGPT demonstrating superior performance despite smaller parameter counts compared to base GPT-3 [75]—its impact on hallucination mitigation shows nuanced trade-offs. As highlighted by [51], improved instruction following sometimes comes at the cost of increased overfaithfulness, where models may prioritize pleasing users with confident responses over acknowledging knowledge limitations. Furthermore, [32] reveals that instruction-following errors frequently originate from attribution difficulties in complex reasoning scenarios, where models struggle to correctly allocate confidence between parametric knowledge and instructional constraints.

Specialized benchmarks for instruction following evaluation continue to evolve, incorporating increasingly sophisticated tasks such as counterfactual reasoning, constraint satisfaction, and ethical boundary adherence. These evaluations consistently demonstrate that even state-of-the-art models exhibit non-trivial failure rates when confronted with instructions requiring precise domain expertise, complex constraint integration, or careful uncertainty calibration. The ongoing challenge remains developing evaluation frameworks that can simultaneously assess instruction fidelity while maintaining factual accuracy across diverse domains and instruction types.

d) Summarization: Summarization represents a critical yet challenging domain for hallucination evaluation, where models must distill key information from source documents while preserving factual accuracy and avoiding fabrications. Traditional single-document summarization benchmarks, like RAGTruth, provide annotated samples to evaluate faithfulness to source material, yet even state-of-the-art detectors achieve a low accuracy on comprehensive benchmarks like FaithBench, highlighting fundamental limitations in current evaluation methodologies [76], [77]. These challenges intensify in multi-document summarization (MDS), where models must synthesize information across multiple sources—evaluations reveal alarmingly high hallucination rates exceeding 75% in LLM-generated summaries, with errors disproportionately occurring in later sections where synthesis demands are greatest [78].

The integration of insights from related evaluation paradigms offers promising directions for improvement. Dialogue-level evaluation frameworks like DiaHalu demonstrate how multi-turn interactions expose compounding error patterns that mirror the sequential risks in extended summaries [17]. Similarly, domain-specific benchmarks such as Med-HALT highlight the critical need for specialized evaluation in high-stakes fields like healthcare, where hallucination consequences are severe [1]. Furthermore, human-in-the-loop auditing frameworks like LLMAuditor provide methodologies for scalable validation of model outputs, suggesting pathways toward more reliable and contextually grounded summarization evaluation [31]. Together, these approaches underscore the necessity of moving beyond generic benchmarks toward compositional evaluation frameworks that address the complex, context-dependent nature of summarization hallucinations.

e) Translation: Although fewer dedicated benchmarks exist, translation evaluation reveals whether models preserve semantic meaning across language boundaries [63]. Translation tasks uniquely expose hallucinations through cross-linguistic consistency checks, where models must preserve semantic meaning across language boundaries without inventing cultural context or idiomatic expressions that don't exist in the source [63].

f) Text Generation: Text-to-knowledge-graph generation, data-to-text writing, and news writing examine whether models produce faithful and factually accurate outputs. Recent analyses reveal that hallucination rates in open-ended text generation can be high when models generate content beyond their training distribution [79].

g) Code Generation: Recent work includes CodeBench [80] for predicting hallucination likelihood and Codehalu [81] for execution-based verification of generated code. It primarily concerns hallucinations that arise in programming tasks, such as generating incorrect logic, fabricated APIs, or non-executable code [81]. As language models are increasingly used as coding assistants, evaluating their reliability in this domain is becoming critical, since even small errors can lead to significant functional failures [82].

h) Reasoning Tasks: This category includes knowledge graph reasoning, causal reasoning through benchmarks like CausalBench [52], temporal and spatial reasoning, math word problems (MWPs), and logical tests. Recent studies demonstrate that reasoning tasks are particularly vulnerable to cascading hallucinations, where early errors compound through multi-step inference chains [83]. These evaluations are designed to push models beyond simple recall, probing whether they can handle complex reasoning and problem-solving without drifting into hallucinations.

i) Multimodal Generation: Multimodal evaluation reveals emergent failure modes absent in unimodal systems. HallusionBench [40] identifies "visual illusion vulnerability"—models hallucinate 89% of the time when presented with carefully crafted visual illusions that humans correctly interpret [40]. This suggests models rely on superficial visual features rather than robust scene understanding. **Bingo** [39] exposes "modality competition"—when visual and textual signals conflict, models exhibit great bias toward textual priors regardless of image content [39]. This benchmark uniquely tests interference effects, showing that irrelevant text in images increases hallucination rates [39].

j) RAG & Tool-use: **Retrieval-Augmented Generation (RAG)** has become an important area of evaluation in tasks such as question answering, data-to-text generation, and summarization [41]. The central aim is to assess whether incorporating external information improves factual accuracy or, conversely, introduces new inconsistencies. Paradoxically, while

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RAG reduces certain types of hallucinations by grounding responses in retrieved content, it can introduce new failure modes when models misinterpret or over-rely on retrieved information [26]. Tool-use scenarios present additional challenges, as models must correctly invoke APIs and interpret responses without fabricating functionality [84].

However, it is crucial to note that the current evaluation paradigm predominantly focuses on assessing the outcome of tool usage—i.e., whether the final answer is correct after the tool has been invoked. This overlooks a critical component of robust tool integration: the model's ability to make the correct decision on whether to use a tool in the first place. This capability, which we refer to as Tool Necessity Detection (TND), is a precursor to successful tool use. A model might generate a factually correct output without using a tool when it should have (a TND failure leading to potential hallucination due to overconfidence), or it might incorrectly call a tool for a simple task it should handle internally (inefficiency and potential error propagation). Current benchmarks and metrics are not designed to systematically evaluate this decision-making step independently from the final output accuracy. This gap in evaluation leaves an important aspect of model reliability unmeasured and underscores the need for evaluation frameworks that can dissect and assess the tool-use pipeline into its constituent parts, such as TND and **Tool Selection Accuracy (TSA)**.

2) Modality-Based Evaluation - From Text to Multimodal Complexity: Beyond task complexity, the modality of information processing introduces another critical dimension for hallucination evaluation. **Text-only Evaluation** has established relatively mature methodologies to assess factuality and faithfulness in purely textual settings. **Multimodal Evaluation** is to study visual, temporal, and cross-modal reasoning, which creates exponentially more complex evaluation challenges [85].

Text-Only Evaluation represents the most mature segment of hallucination evaluation and has enabled the development of standardized evaluation metrics, including factual consistency scores, entailment-based validation, and human-in-the-loop assessments. Benchmarks in this category generally focus on purely linguistic tasks where ground truth can often be established through textual sources, though challenges persist in addressing ambiguity, context-dependence, and knowledge evolution over time [86]. Text-only benchmarks have primarily targeted core generative tasks such as question answering, summarization, retrieval-augmented generation (RAG), and instruction following. For instance, RAGTruth [76] provides nearly 18,000 naturally generated responses with annotations, enabling fine-grained assessment of hallucination in RAG scenarios. This benchmark has been used to examine hallucination prevalence across different LLMs and has shown that smaller models, when fine-tuned, can achieve competitive performance in hallucination detection.

Another representative benchmark is HaluEval [7], which consists of over 35,000 annotated samples, including real user queries and synthetic task-specific data across question answering, knowledge-grounded dialogue, and summarization. It has been used to measure hallucination detection performance and shows that integrating external knowledge or chain-of-thought reasoning can reduce hallucination rates [7]. Furthermore, HaluEval-Wild [42] extends this work by collecting naturally occurring prompts from ShareGPT and constructs fine-grained hallucination categories for more realistic evaluation.

By contrast, Multimodal Evaluation, which encompasses vision-language, temporal reasoning, and cross-modal integration, poses substantially greater challenges than text-only settings. Multimodal models must align heterogeneous inputs across domains (e.g., image, video, and text), making them vulnerable to errors such as spurious visual grounding, temporal drift in video summarization, and inconsistent reasoning across modalities. Recent benchmarks like POPE and MMHal illustrate that evaluating multimodal hallucinations requires not only correctness within individual modalities but also the consistency and coherence across them.

Multimodal benchmarks reveal emergent hallucination mechanisms unique to cross-modal processing. The fundamental challenge, which models must reconcile, is potentially conflicting signals across modalities, leading to three distinct failure modes: (1) Modal dominance, where one modality overrides others regardless of reliability; (2) Cross-modal leakage, where patterns from one modality inappropriately influence another; and (3) Binding errors, where correct elements from different modalities are incorrectly associated.

The complexity of multimodal evaluation scales exponentially with the number of modalities involved, as evaluators must account for interdependencies, contextual grounding, and alignment across different sensory channels. This creates open questions around scalability, annotation cost, and reliability of current benchmarks, underscoring the need for more comprehensive, modality-aware evaluation frameworks that can robustly capture hallucination risks in both unimodal and multimodal generative systems.

3) Hallucination-Type–Based Evaluation: Understanding Typology of Hallucinations and Failure Modes: Hallucination-Type–Based Evaluation further categorizes evaluation efforts according to representative manifestations of hallucination. Hallucinations in generative AI emerge through distinct mechanistic pathways, each reflecting fundamental limitations of current model architectures [87]. Understanding these pathways is essential for diagnosing error modes, designing mitigation strategies, and improving model reliability. Below, we briefly introduce several representative categories of hallucinations that illuminate different aspects of LLM behavior, including Factual hallucinations, Reasoning hallucinations, Memory-based hallucinations, Contextual hallucinations, Multimodal hallucinations, Temporal or knowledge staleness, and Intrinsic and Extrinsic Hallucinations. Systematically assessing these categories helps showcase the breadth of failure modes and clarify their differential impacts on reliability, interpretability, and safety across applications.

Factual hallucinations refer to errors in which models generate statements that are factually incorrect or inconsistent with verifiable knowledge sources and typically arise from three pri-

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mary mechanisms: (1) Parametric knowledge conflicts, where contradictions within the training corpus lead to stochastic selection among competing facts, resulting in inconsistencies observed in tested cases [78]; (2) Retrieval misalignment, where spurious correlations in embedding space cause incorrect entity associations, will affect factual queries [78]; and (3) Temporal degradation, where knowledge accuracy declines progressively following the training cutoff.

Reasoning hallucinations capture failures in logical or mathematical inference, such as erroneous multi-step arithmetic or flawed causal reasoning, which highlight the limitations of LLMs in performing structured reasoning beyond surface-level pattern recognition. These hallucinations manifest in several distinct failure signatures: Logical failures often stem from "shortcut reasoning" where models rely on superficial pattern matching rather than performing genuine inference, leading to errors in many complex reasoning tasks; and causal reasoning fragility refers to the disproportionate difficulty models face when handling counterfactual reasoning, achieving only low accuracy on counterfactual scenarios compared to high accuracy on factual reasoning tasks [52].

Memory-based hallucinations denote errors stemming from a model's limited or unstable memory mechanisms, often manifesting as contradictions across dialogue turns or inconsistencies in long-form text generation. These failures reflect the difficulty of maintaining coherent internal state representations and have emerged as a growing concern in recent research [1] since a model may incorrectly recall information from earlier dialogue turns, fabricate details absent from the interaction history, or conflate information across multiple user sessions. Such phenomena highlight the fragility of memory mechanisms in generative models and their inability to track and manage evolving conversational or contextual dependencies robustly [1].

Contextual hallucinations arise when models misinterpret or over-extend the given context, generating content not entailed by input instructions or source material. These errors underscore the models' sensitivity to prompt design and contextual cues, extending beyond conventional factual and logical mistakes. This category encompasses inconsistencies with prior dialogue turns as well as factual inaccuracies triggered by contextual complexity.

Recent studies highlight that **Small Large Language Models** (SLLMs) are particularly vulnerable: while they maintain relatively high accuracy in isolated fact-checking tasks, their performance deteriorates significantly in multi-layered contextual scenarios [58]. To capture this weakness, a Context Influence Score has been proposed, quantifying the degree to which models are affected by contextual dependencies, with SLLMs showing markedly greater susceptibility than larger models.

Mitigation strategies vary in effectiveness. Structured reasoning techniques, such as chain-of-thought prompting, have demonstrated success in reducing context-driven hallucinations. In contrast, approaches that rely solely on external knowledge sources—such as retrieval-augmented generation (RAG) — may be counterproductive if not coupled with reasoning guidance, sometimes exacerbating hallucinations by introducing ungrounded or misaligned content [58].

Multimodal hallucinations refer to inaccuracies that arise in models integrating text with other modalities (e.g., vision–language systems), where outputs include fabricated or misaligned cross-modal associations. A common failure mode occurs when models describe visual elements that do not exist in an image or misattribute relationships between text and visual content. Such errors substantially complicate reliability in multimodal systems because they highlight the inherent challenges of cross-modal alignment and consistency [85]. These hallucinations not only undermine factual accuracy but also expose limitations in current multimodal fusion architectures, where models must reconcile heterogeneous signals across domains [85].

Temporal or knowledge staleness indicates outdated responses caused by training on static corpora, where models confidently present obsolete information as current fact. This phenomenon reflects a fundamental mismatch between a model's parametric knowledge and the dynamic nature of real-world information, and it remains a persistent challenge for long-term reliability [88]. Empirical evidence shows that models trained on outdated data frequently generate time-sensitive claims without awareness of temporal shifts, thereby undermining their trustworthiness in domains that demand up-to-date knowledge, such as medicine, law, and public policy [88].

In addition, *Intrinsic and Extrinsic Hallucinations* constitute two critical categories that warrant focused attention. Intrinsic hallucinations stem from deficiencies internal to the model, such as faulty reasoning processes or gaps in parametric knowledge. In contrast, extrinsic hallucinations arise when the model correctly processes input but depends on erroneous external information, for example, flawed retrieval outputs or corrupted databases [78]. This distinction has important implications for mitigation strategies: intrinsic hallucinations require architectural refinements or targeted training improvements, whereas extrinsic hallucinations necessitate more reliable source verification, retrieval alignment, and robust information filtering mechanisms [13].

4) Domain-Specific Evaluation: The Challenge of Specialized Knowledge: Domain specialization represents another critical dimension, as the consequences of errors vary dramatically across application contexts. A hallucinated fact in creative writing might be inconsequential, while the same error in medical diagnosis could be life-threatening.

In the medical field, benchmarks such as Med-HALT [1] and the recently proposed MedHallBench [9] focus on evaluating hallucinations in clinical and biomedical contexts, where the accuracy of generated information is directly tied to patient safety and clinical decision-making. **MedHallBench** [9] demonstrates domain criticality: radiological hallucinations occur in reports, with some being clinically significant errors that could impact patient care [9].

In transportation, the Traffic Incident Benchmark [64] eval-

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uates spatial and temporal reasoning using nearly 99,869 real incident records. It shows that while retrieval augmentation can reduce some hallucinations, significant performance gaps remain.

In code generation, Collu-Bench [80] introduces a framework for predicting when hallucinations are likely to occur during code generation, and Codehalu [81] evaluates hallucinations through execution-based verification, detecting cases where generated code runs incorrectly or fails to implement the intended functionality.

Besides, SciHal25 [89] is a shared task on hallucination detection in scientific question answering, providing a dataset of annotated claims.

In the financial domain, researchers present an empirical study of hallucinations in financial tasks [90]. It shows that LLMs frequently produce unreliable outputs when explaining financial concepts or retrieving stock data, and that mitigation strategies such as few-shot learning, DoLa, RAG, and tool-use offer only partial improvements.

Last but not least, multilingual and cross-lingual benchmarks, including Uhgeval [43], address the challenge of hallucination in settings where linguistic coverage and cross-language consistency are critical. Beyond these established domains, specialized evaluation benchmarks are emerging across legal, educational, and industrial applications, each with unique requirements for accuracy, interpretability, and reliability.

5) The Integration Challenge - Unified vs. Specialized Benchmarks: It is essential to develop a unified framework and benchmark for hallucination evaluation. Recently, there have been emerging unified evaluations, such as OnionEval [58], Hal-Eval [55], and Large VLM (LVLM)-EHub [65], which seeks to provide wide coverage across tasks and modalities, where models can be assessed under a single framework. It facilitates standardized reporting and benchmarking.

However, this breadth often comes at the cost of domain-specific depth. By contrast, specialized benchmarks offer more rigorous and contextually grounded assessments, but their narrower focus can limit generalizability [91]. This trade-off between coverage and precision represents more than a practical choice; it reflects fundamental questions about the nature of hallucination and whether universal evaluation principles can meaningfully span diverse application contexts.

The path forward likely requires hybrid approaches that combine the standardization benefits of unified frameworks with the contextual sensitivity of domain-specific evaluation. Recent proposal for meta-evaluation frameworks [6] offers promising directions, with modular architectures allowing for comparative analysis and specialized assessment.

The current evaluation landscape, while rich in specialized contributions, lacks the integrative frameworks necessary for systematic progress. The proliferation of benchmarks without standardized evaluation protocols has led to situations where models appear reliable on one benchmark while exhibiting severe hallucinations on another, undermining user trust and hindering deployment in critical applications.

6) Summary and Future Directions: Our comprehensive analysis of evaluation targets and taxonomy reveals a field in transition, from ad-hoc benchmark construction toward a more systematic understanding of hallucination phenomena. The key insights from this survey are as follows.

First, task-specific evaluations demonstrate that hallucination patterns are highly context-dependent. Question answering tasks exhibit a high hallucination rate driven primarily from factual errors [7]; and multimodal tasks reveal hierarchical degradation, with performance dropping from high accuracy in object detection to low in relationship inference. This heterogeneity suggests that universal hallucination mitigation strategies may face inherent limitations.

Second, the modality dimension highlights architectural limitations. While text-only evaluations achieve relatively robust results [76], multimodal systems suffer from modal dominance biases and temporal drift. These findings indicate that current fusion mechanisms inadequately integrate cross-modal information.

Third, hallucination typologies reveal mechanistic insights. Factual errors often result from parametric knowledge conflicts [78], reasoning failures scale exponentially with task complexity [83].

Fourth, domain-specific evaluations show concerning error rates in high-stakes applications [9].

Looking forward, three imperatives emerge:

- (1) Developing compositional evaluation frameworks that assess integrated capabilities rather than isolated functions;
- (2) Designing adaptive benchmarks that evolve with model progress to avoid benchmark overfitting;
- (3) Establishing risk-calibrated metrics that weight hallucinations by potential harm rather than treating all errors equally.

As discussed in Section I-C and Table I, most existing evaluation strategies remain concentrated on the horizontal axis—measuring surface-level accuracy across domains—while offering limited exploration of vertical evaluation, which examines reasoning processes and causal pathways. The long-term trajectory of the field must therefore progress beyond mere detection of hallucinations toward uncovering their underlying mechanisms. By shifting emphasis to causal understanding, evaluation frameworks can enable principled and proactive mitigation strategies, rather than relying on post hoc fixes that treat only the symptoms of model unreliability.

B. Evaluation Dimensions and Metrics (Establishing Criteria)

The field of hallucination evaluation has advanced from early token-level detection systems, such as HADES [79], to increasingly sophisticated multi-granular frameworks. Foundational works defined distinct paradigms: SelfCheckGPT [92] demonstrated zero-resource sentence-level detection through self-consistency; FActScore [93] introduced atomic claim decomposition for fine-grained factuality assessment; and Semantic Entropy [94] quantified semantic-level uncertainty

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to detect confabulations—a subclass of hallucinations involving arbitrary and incorrect generations. Collectively, these approaches revealed that hallucinations manifest differently across granularities—from token-level lexical errors to document-level inconsistencies—necessitating evaluation strategies adapted to task requirements and reference availability. Building on these foundations, hybrid frameworks such as THaMES [71] integrate retrieval grounding with parameter-efficient fine-tuning and in-context verification, underscoring how methodological innovations balance coverage, scalability, and interpretability.

Against this backdrop, the evaluation landscape has progressively crystallized around four operational dimensions: *Core*, *Granularity*, *Reference Availability*, and *Automation–Human Balance*. These dimensions jointly determine what properties are measured, at what resolution hallucinations are detected, how reliable the interpretations are, and how automation is balanced with human oversight.

- *Core Dimension:* Defines the fundamental property under evaluation—such as factuality, faithfulness, consistency, or robustness—thereby specifying the conceptual scope of hallucination.
- *Granularity Dimension:* Specifies the resolution of analysis, from token-level precision (capturing subtle lexical errors), through sentence- and claim-level assessments (balancing interpretability with practicality), to document- and dialogue-level evaluations (capturing coherence and long-range dependencies). The chosen granularity directly affects both sensitivity to subtle violations and the interpretability of results.
- *Reference Availability Dimension:* Shapes methodology by introducing trade-offs between coverage and objectivity. Reference-based approaches leverage gold-standard corpora, databases, or knowledge graphs, but face domain and coverage limits. Reference-free approaches rely on intrinsic model signals—uncertainty estimation, self-consistency, or latent probing—scaling well but lacking external validation. Hybrid methods integrate external references with internal signals to balance robustness and scalability.
- *Automation–Human Balance Dimension:* Highlights the trade-off between scalable automated metrics and nuanced human judgment. Fully automated systems enable large-scale evaluation but often overlook context-dependent subtleties; human-in-the-loop methods capture fine-grained failures but at high cost and limited scalability. This balance ultimately governs the feasibility and interpretability of evaluation outcomes.

The dimensional framework for hallucination evaluation, supported by empirical evidence across domains, underscores that hallucination assessment is inherently multi-faceted and provides a structured foundation for operationalizing evaluation practices. This framework motivates systematic investigation into how granularity choices shape detection capability, how reference availability constrains verification, and how automation levels influence practical deployment. Collectively, these factors expose fundamental trade-offs between precision and coverage, computational cost and accuracy, and scalability and reliability [3], [5]. Beyond these dimensions, evaluation design is further shaped by task-specific goals, modality constraints, hallucination typologies, and domain-specific requirements, as outlined in Section III.

Accordingly, the following sections first examine three operational dimensions—Granularity, Reference Availability, and Automation–Human Balance—before turning to the core dimensions that define what properties should be measured. Framing the methodological design space in this way establishes a foundation for evaluation frameworks that are rigorous, interpretable, and practically deployable. A closer inspection of relevant metrics further emphasizes that robust assessment depends on explicit methodological choices that balance precision with coverage, scalability with interpretability, and automation with human oversight.

1) Granularity: Among evaluation dimensions, granularity plays a central role in shaping hallucination evaluation outcomes [5]. Hallucination evaluation can be conducted at multiple levels of granularity, each offering distinct advantages and limitations in terms of diagnostic precision, scalability, and interpretability. Existing approaches can be broadly classified into four complementary levels:

- (i) *Token-level*, which pinpoints minimal factual errors at the word or subword level. This allows precise localization of inaccuracies but requires careful contextual interpretation.
- (ii) *Sentence- and claim-level*, which verifies the factuality of self-contained statements or atomic facts. This level is central to recent benchmarks that adopt claim decomposition for fine-grained factuality assessment.
- (iii) *Dialogue- and document-level*, which captures long-range dependencies such as coherence across dialogue turns, cross-paragraph consistency, and global factual reliability, making them indispensable for summarization and multi-turn QA.
- (iv) *Specialized units (e.g., entities and tool-calls)*, which address application-specific failure modes, such as entity grounding, relation verification, or correctness of API/tool invocations, increasingly relevant in retrieval- and tool-augmented systems [5].

The choice of granularity reflects a fundamental trade-off between precision and practicality, as comprehensively surveyed by Qi et al. [5]. Fine-grained evaluation methods (e.g., token- or entity-level) capture subtle errors but require extensive annotation effort and computational resources, while coarse-grained methods (e.g., document-level) provide efficiency but risk overlooking localized inaccuracies and missing critical details. Empirical analyses reveal that early approaches tended to emphasize entity-level grounding, while the rise of LLMs popularized holistic, direct-response judgments [5]. More recently, frameworks have shifted toward finer-grained atomic claim-level decomposition for more precise fact com-

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parison, which enables structured verification against source material with improved sensitivity to subtle inconsistencies.

Importantly, as Qi et al. [5] emphasize, identifying a universally optimal level of granularity remains an open challenge, heavily dependent on the specific task and application context. For example, summarization requires multi-sentence coherence checks, whereas open-domain QA benefits from fine-grained claim verification. The subsequent discussion examines these granularities in more detail, outlining representative methodologies, inherent challenges, and potential directions for improvement.

a) Token-level Evaluation: At the finest granularity, token-level evaluation examines individual word or subword units for factual accuracy. The HADES (HAllucination DEtection dataSet) benchmark [79] pioneered this approach by introducing a reference-free framework for hallucination detection at the token level. HADES addresses the challenge that coarser-grained evaluations miss subtle factual errors embedded within otherwise coherent text [79]. The dataset, constructed by perturbing Wikipedia segments and validating through crowd-sourcing, enables models to identify specific hallucinated tokens within broader contexts [79]. Using BERT-based architectures, HADES reported Not-Hallucination F1 scores ranging from 72.84% to 74.41%, demonstrating the feasibility of token-level annotation for evaluation. Nonetheless, the approach faces a fundamental limitation: the factuality of an isolated token is often context-dependent, requiring interpretation within surrounding discourse and background knowledge [79].

b) Sentence and Claim-level Evaluation: Sentence-level evaluation has become the dominant paradigm, balancing detection precision with practical utility. SelfCheckGPT [92] operates at this level, leveraging the insight that factual statements from knowledgeable models exhibit consistency across multiple generations, while hallucinated content shows high variance. This zero-resource approach eliminates the need for external databases, achieving the AUC-PR for hallucination detection increasing from 72.96 to 83.21. [92].

Claim-level evaluation bridges the gap between surface-level assessment and semantic understanding [93]. FActScore (Factual precision in Atomicity Score) [93] exemplifies this approach by decomposing generated text into atomic claims: individual verifiable facts that can be independently assessed. FActScore addresses the challenge that long-form generations often contain mixtures of accurate and inaccurate information [93]. By breaking down biographies into atomic facts (averaging 180 facts per generation), the method revealed that even advanced models like ChatGPT achieve only 58% factual precision on biographical generation [93].The analysis reveals systematic failure patterns: models particularly struggle with specific dates, lesser-known facts, and contextual relationships, suggesting that hallucinations concentrate in areas of sparse training data or complex relational reasoning [93].

c) Dialogue and Document-level Evaluation: Dialogue-level evaluation captures hallucinations that emerge through conversational dynamics. HalluDial [16], comprising 4,094 dialogues with 146,856 samples, addresses the unique challenges of multi-turn interactions where hallucinations can propagate and compound across exchanges [16]. The benchmark distinguishes between spontaneous hallucinations (naturally occurring errors) and induced hallucinations (deliberately triggered through adversarial prompts), achieving detection accuracies of 86.48% respectively using their HalluJudge model [16]. DiaHalu [17] complements this by focusing on faithfulness hallucinations in task-oriented dialogues, where models must maintain consistency with both factual knowledge and conversational history [16], [17].

d) Entity and Tool-call Level: Specialized granularities emerge from domain-specific needs [84], [95]. Entity-level evaluation focuses on named entity accuracy for people, organizations, and locations, where performance typically degrades when evaluating entity relationships compared to entity existence, revealing the challenge of compositional understanding. Tool-call level evaluation examines whether models correctly invoke external tools and accurately incorporate returned information. Recent benchmarks like ToolLLM [84] and ToolBeHonest [95] reveal that tool-augmented generation introduces new failure modes: State-of-the-art models achieve less than 50% accuracy when coordinating multiple API calls, suggesting that augmentation may create as many problems as it solves [84], [95].

2) Reference Availability: The availability of ground truth fundamentally shapes the methodology of hallucination evaluation. This dimension highlights a core tension: the desire for objective verification versus the reality that many generative tasks lack definitive references. Evaluation approaches differ by whether gold references are available and can thus be categorized into three main groups — reference-based, reference-free, and hybrid/knowledge grounded — each carrying unique advantages and limitations.

- *Reference-based* metrics compare model outputs to human-written or curated references, ranging from surface-level overlap to semantic and claim-level checks.
- *Reference-free* methods dispense with external corpora and instead rely on model-internal signals such as semantic uncertainty estimation or response consistency.
- *Hybrid and knowledge-grounded* approaches integrate retrieved evidence or structured knowledge to verify claims, thereby retaining flexibility across different tasks or domains while introducing additional engineering overhead.

This axis directly influences the objectivity of evaluation outcomes, the breadth of domain coverage that can be achieved, and the technical overhead required for implementation. The subsequent discussion elaborates on these categories in greater depth.

a) Reference-based Metrics: Traditional reference-based metrics, originally designed for machine translation and summarization, provide established baselines for hallucination detection. *ROUGE* [96] and *BLEU* [97] scores, while originally designed for surface-level similarity, have been repurposed to detect factual inconsistencies by comparing generated content against gold-standard references. However, these metrics show

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weak correlation with human hallucination judgments [98], as they prioritize lexical overlap over semantic accuracy.

BERTScore [99] improves upon surface metrics by leveraging contextual embeddings, achieving better correlations with human annotations compared to traditional metrics. Yet it still struggles with paraphrases and fails to distinguish between semantically similar but factually different statements. *FActScore* [93] represents the evolution toward semantic reference-based evaluation, using Wikipedia as a knowledge source to verify atomic facts. This approach considers factual precision in automated fact-checking but requires extensive retrieval infrastructure and assumes reference completeness—problematic for emerging or specialized domains [93], [99].

b) Reference-free Approaches: Reference-free methods address scenarios where gold-standard references are unavailable or impractical. These methods typically exploit intrinsic model signals. Uncertainty-based approaches leverage model confidence as a hallucination signal [94]. Semantic Entropy [94], published in Nature 2024, computes uncertainty at the meaning level rather than the token level, grouping semantically equivalent outputs before calculating entropy. This method achieves 0.78–0.81 AUROC across diverse tasks without task-specific training, though it requires multiple model samples per query, increasing computational cost [94].

SelfCheckGPT [92] exploits response consistency without external references, sampling multiple responses and measuring their agreement. The method achieves comparable performance to reference-based approaches (72.96-83.21 AUC-PR) while remaining fully self-contained [92]. This assumption reveals a fundamental limitation: it presupposes that factual knowledge produces consistent outputs, breaking down for ambiguous or context-dependent information where multiple valid interpretations exist [92].

c) Hybrid and Knowledge-Grounded Approach: Hybrid approaches combine reference-based precision with reference-free flexibility—for example, vision-grounded alignment that checks phrases against image regions (DBD [68]) and retrieval-grounded span labeling that judges support at the word/span level (RAGTruth [76]). Despite their promise, these methods face notable challenges. RAGTruth, for example, struggles to distinguish genuine performance improvements from benchmark overfitting, and its reliance on continuous human annotation complicates scalability and long-term maintenance [76]. Knowledge-grounded metrics integrate structured knowledge graphs with neural evaluation for improved factual verification while providing interpretable reasoning paths, as in automatically verifiable entity–relationship triples (ERBench [100]) and graph-based assessments of hallucination/omission during knowledge-graph construction [101]. These methods can construct dynamic knowledge graphs from both references and generated content, detecting explicit contradictions and implicit inconsistencies; in practice, pipeline tools also couple evidence building with evaluation to stabilize judgments (THaMES [71]), and hybrid reference construction under retrieval further reduces judge hallucinations on in-the-wild queries (HaluEval-Wild [42]).

d) Graph-based Evaluation: A specialized subcategory of Hybrid and Knowledge-Grounded Approach is *Graph-based Evaluation*, which verifies claims against structured external evidence - such as knowledge graphs built from curated corpora or dynamically retrieved sources - while optionally incorporating model-internal signals or automatic extraction from generated text. Unlike reference-free methods, graph-based evaluation necessarily relies on external evidence encoded in graph structures, as the approach explicitly models relationships among claims, references, and retrieved evidence using techniques such as graph structures and message passing for scientific text evaluation [24], [100], [101] At the same time, the approach is not purely reference-based, because the "reference" isn't a fixed gold text but a structured and often dynamically built graph, enabling reasoning beyond surface-level string or semantic overlap. Accordingly, graph-based evaluation is best understood as hybrid: leaning reference-based when fixed knowledge graphs are employed, but increasingly hybrid when graphs are dynamically constructed through retrieval and information extraction.

While offering strong interpretability and fine-grained relational reasoning, graph-based evaluation introduces significant challenges. Graph construction remains computationally expensive—often slower than purely neural evaluation pipelines—and may require domain-specific relationship extraction models [100], [101]. These constraints limit its general-purpose applicability. Nonetheless, graph-based approaches have proven especially effective in scientific and technical domains, where accurate modeling of entity relations, logical consistency, and evidence integration is critical [24], [100], [101].

3) Automation vs. Human-in-the-Loop: The spectrum from automation to human involvement reflects the trade-off between scalability and the irreplaceable nuance of human judgment. This dimension determines not just evaluation cost but also the types of hallucinations that can be reliably detected. Broadly, three operational modes can be distinguished: *fully automatic methods*, which maximize scalability and repeatability but may overlook nuanced errors [5]; *human annotation frameworks*, which incorporate expert or crowd validators for adjudication of difficult cases, improving reliability at a higher cost; and *hybrid strategies*, which combine automation screening with targeted human review to optimize efficiency and accuracy, often yielding the best cost–quality trade-off for high-stakes applications.

a) Fully Automatic Methods: Fully automatic evaluation enables large-scale, repeatable assessment without human intervention [5]. For example, the HADES framework exemplifies token-level automatic detection using masked language models [79], while ReEval [26] introduces adversarial automatic evaluation that generates challenging test cases to expose model hallucinations.

HalluDial [16] represents the first large-scale benchmark for automatic dialogue-level hallucination evaluation, addressing key limitations in existing benchmarks. Both HalluDial and

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HalluJudge are utilized to conduct an automatic evaluation of dialogue-level hallucination, which covers both factuality and faithfulness hallucinations and employs a two-scenario framework: a spontaneous hallucination scenario using a diverse dialogue sampling and GPT-4 annotation pipeline, and an induced hallucination scenario where GPT-4 directly generates hallucinated samples with specific instructions [16]. This comprehensive dataset enables rigorous meta-evaluation of LLMs' capabilities and supports the development of HalluJudge, a specialized judge model that demonstrates superior performance in hallucination detection, localization, and rationale provision, achieving a 90.2% inter-annotator agreement with human evaluation.

b) Human Annotation Frameworks: Human evaluation remains the gold standard for detecting complex or domain-sensitive hallucinations. LLMAuditor [31] exemplifies this approach by providing a structured rubric-based framework in which experts assess factuality, consistency, and relevance.

Domain expertise proves especially valuable: for example, the MedHallBench annotation process [9] demonstrates that employing board-certified physicians for the evaluation of specialized medical content, the framework assigns a weight of 0.7 to expert evaluations compared to 0.3 for general annotators, underscoring the indispensable role of expert knowledge in high-stakes domains.

c) Hybrid Strategies: Hybrid approaches balance automation's efficiency with human judgment's accuracy. Combining automation with selective human review provides an effective compromise. The LLMAuditor framework [31] employs a human-in-the-loop process to iteratively refine evaluation criteria. Among the two methods of dataset construction that can automatically annotate data, one method utilizes active learning, while the other leverages Reinforcement Learning with Human Feedback (RLHF) for automatic annotation [31]. This method significantly improves annotator agreement, with Cohen's kappa scores increasing from 0.46 to 0.69 for relevance and from 0.62 to 0.90 for diversity, thereby enhancing the reliability of hallucination assessments [31].The framework further develops standardized probe templates that enable consistent evaluation across diverse models and domains, demonstrating particular utility in high-stakes contexts [31]. Active learning further optimizes resource use by focusing annotation efforts on the most uncertain or high-risk samples.

4) Core Dimensions: Having established the operational scaffolds of hallucination evaluation—granularity, reference availability, and the balance between automation and human oversight—we now turn to the Core Dimensions that operationalize these foundations into concrete evaluative practices. These dimensions collectively provide the basis for a thorough and reliable assessment of hallucinations.

We organize the Core Dimensions into four progressive clusters that collectively capture the essential properties of hallucination evaluation. Specifically, the dimensions listed below reflect the design principles outlined in Section I-D. While the sequence in Section I-D emphasizes their relative significance, the ordering presented here instead follows the structural logic of a comprehensive evaluation framework, representing the foundational components that reasonable metrics should incorporate.

- Grounding and Alignment
  - *Factuality*: accuracy of content against external world knowledge.
  - *Faithfulness*: adherence of outputs to the provided source material.
  - *Safety Likelihood and Risk Assessment*: quantifying potential harm beyond correctness.
- Reasoning Integrity
  - *Vertical Reasoning Traceability*: layer-by-layer localization of error origins.
  - *Causal Reasoning Capabilities*: ability to distinguish causation from correlation and identify root causes.
  - *Interpretability*: transparency and auditability of evaluation judgments themselves.
- Systemic Reliability
  - *Self-Consistency*: stability of outputs under paraphrasing or rephrasing.
  - *Coherence and Consistency*: logical harmony within single responses.
  - *Reliability and Consistency*: stability of behavior across time, contexts, and user populations.
  - *Robustness*: resilience against adversarial perturbations and edge cases.
- Practical Constraints
  - *Tool Appropriateness*
  - *Computational Cost and Efficiency Trade-offs*: balancing diagnostic quality with feasibility and scalability.

Together, these dimensions capture the foundational qualities that distinguish accurate from hallucinatory outputs, and extend evaluation beyond surface-level correctness to encompass reasoning integrity, discourse-level coherence, interpretability, and risk-awareness. By clustering them into progressive layers, the framework supports both diagnostic depth and practical applicability—beginning with factual grounding, advancing through reasoning quality, scaling to systemic reliability, and concluding with pragmatic trade-offs. By structuring the analysis around these dimensions, we create a diagnostic lens that is both comprehensive in scope and actionable in practice.

The selection of these dimensions is deliberately grounded in typologies of hallucinations articulated in contemporary understanding of hallucination typologies articulated in contemporary research [78], [87]. This foundation ensures that the evaluation framework captures the full spectrum of failure modes—from intrinsic reasoning errors, such as logical fallacies and factual inaccuracies, to extrinsic challenges stemming from contextual omissions or reliance on outdated knowledge—thereby enabling a holistic and rigorous assessment of model reliability.

Recent empirical findings further underscore the necessity of broad evaluation dimensions. For instance, Novikova et al.

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demonstrate that models may score highly on factual alignment while failing to maintain logical consistency or avoid contradictions across contexts [102]. Likewise, Asgari et al. demonstrate that domain-specific risks (e.g., in clinical applications) are overlooked by surface-level metrics like ROUGE or BLEU [103]. Their proposed framework, which incorporates coherence, factual consistency, comprehensiveness, and potential harm, highlights that evaluation must move beyond superficial overlap to capture medical concept correctness and assess downstream risks [103]. Evaluation efforts that remain narrowly focused on accuracy or surface similarity risk overlooking deeper and often more consequential sources of error, including flawed causal reasoning, internal contradictions, and domain-specific safety hazards [102], [103]. These insights highlight the necessity of extending hallucination evaluation beyond superficial measures to capture the foundational qualities that distinguish accurate from hallucinatory content.

The proposed core dimensions function as complementary analytical lenses that collectively provide an "MRI-like" diagnostic capability, enabling the hierarchical tracing of reasoning processes across multiple levels of abstraction. This framework offers both breadth—capturing diverse failure modes such as factual inaccuracies, misleading causal inferences, stylistic misalignments, and safety risks—and depth, by elucidating how these errors originate and propagate through reasoning chains and contextual dependencies.

By organizing the core dimensions into this structured progression—from factual grounding and reasoning quality to systemic reliability and practical constraints, such framework balances breadth (capturing diverse failure modes such as factual errors, causal mis-attribution, and safety risks) with depth (tracing how errors originate and propagate) and supports both diagnostic depth and practical applicability.

In the subsections that follow, each dimension is defined, paired with representative metrics for its assessment, and discussed in terms of trade-offs and open challenges that remain in its implementation.

a) Factuality: Factuality Assessment remains a foundational dimension of hallucination evaluation, as it verifies whether generated content aligns with objective truth and reliable external knowledge sources. Benchmarks such as FactScore introduced atomic-level evaluation, revealing significant variation in factuality depending on the type of information being assessed [93]. Modern approaches further extend this granularity by evaluating factuality at both the node level (atomic claims) and the path level (composite inferences) to support vertical traceability across reasoning steps.

Recent studies on *temporal factuality* have further highlighted the limitations of static corpora and knowledge cutoffs. To address this limitation, dynamic fact-checking systems have been proposed, integrating temporal knowledge graphs and adaptive reasoning modules.

When hallucination analysis is extended into the temporal domain, recent long-video benchmarks report that hallucination risk tends to grow with input size and semantic complexity—particularly across multi-event videos. An additional phenomenon, termed *temporal drift*, illustrates the accumulative nature of hallucinations in time-dependent tasks and leads to temporal hallucination [104]–[106].

This *compounding effect* highlights the fragility of temporal reasoning in multimodal models, where extended inputs exacerbate inconsistencies and undermine reliability. Such evidence suggests that temporal drift not only challenges evaluation methodologies but also necessitates the design of more robust benchmarks that account for sequence length, accumulation effects, and dynamic event consistency.

For Large Vision–Language Models (LVLMs), factuality requires domain-specific definitions that extend beyond text-based accuracy to encompass multimodal alignment. For example, Accuracy and precision metrics, and F1 scores are traditional metrics that provide an incomplete picture of hallucination behavior [6]. POPE quantified hallucinations in terms of object existence, reporting error rates of 30–40% [25] while H-POPE revealed that attribute hallucinations—cases where models correctly identify objects but fabricate their properties—are substantially higher, reaching 67% [107].

b) Faithfulness: Faithfulness evaluation examines whether model outputs accurately reflect the information contained in the provided source material. This dimension is especially critical for retrieval-augmented systems, where hallucinations often arise as unfaithful extrapolations from retrieved context. A persistent challenge is confident confabulation, in which models produce plausible-sounding but fabricated citations or facts that can pass superficial checks. Empirical audits have quantified this behavior: for instance, when prompted to write literature reviews, GPT-3.5 fabricated 55% of citations and GPT-4 fabricated 18%; 43% (GPT-3.5) and 24% (GPT-4) contained substantive errors [108]. Complementary verifiability audits of generative search engines report that only 51.5% of generated sentences are fully supported by citations and that 25.5% of citations do not support the sentence they accompany [109]—indicating that many responses that look "grounded" still contain unsupported or distorted content.

c) Safety Likelihood and Risk Assessment: Safety-likelihood evaluation quantifies the probability that a model will hallucinate under varying conditions, enabling crucial risk-aware deployment decisions. Rather than a binary judgment, this perspective adopts probabilistic frameworks that capture the stochastic nature of error generation. Studies on epistemic and aleatoric uncertainty enable hallucination estimation for single- and multi-answer settings [110], and work on semantic entropy predicts the likelihood of confabulations in free-form text generation across models and domains without prior domain knowledge [94]. Thus, hallucination propensity can be modeled via uncertainty-based estimators (e.g., semantic entropy) and internal-state probes, with risk patterns that track identifiable factors such as query difficulty, domain specificity, and model confidence.

Statistical risk modeling—via multivariate analyses—characterizes hallucination likelihood as a function of identifiable risk factors (e.g., query complexity, domain

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specificity, model confidence), decomposes their contributions to overall risk, and isolates the primary drivers while controlling for confounders.

CausalBench evaluates safety from a complementary angle by probing an LLM's causal-reasoning ability and examining its relationship to hallucination tendencies [52]. Moreover, models with stronger causal-reasoning performance on CausalBench are observed to have lower hallucination rates on external evaluations, suggesting—but not proving—a link between causal understanding and factual faithfulness [52]. The CausalBench framework advances safety assessment through causal inference techniques that isolate the contribution of individual risk factors [52]. This causal understanding enables targeted mitigation: addressing root causes rather than symptoms reduces hallucination rates [52].

d) Vertical Reasoning Traceability: Vertical reasoning traceability enables layer-by-layer dissection of inference chains and emphasizes how reasoning chains unfold and where they fail. It operationalizes the "vertical assessment" principle as described in Table I for the LENS framework (Section I-F). Through tree-structured decomposition, complex queries are broken into atomic verification units (nodes) linked by logical dependencies (edges). This structured dissection enables evaluators to localize the origin of hallucinations—whether stemming from a faulty premise, flawed synthesis, or cascading error propagation—thereby enhancing diagnostic resolution and informing targeted mitigation strategies.

Representative metrics include:

- *Node-level Factuality*: Atomic claim verification rate (Section III-B4a)
- *Error Propagation Index*: Measures cascading failures across dependent nodes and quantifies the degree to which failures in upstream nodes affect downstream reasoning.
- *Path Consistency*: Logical coherence across multiple reasoning branches

By systematically tracing inference paths, i.e., tracking how reasoning processes unfold and pinpointing where they break down, the vertical reasoning traceability dimension not only enhances diagnostic resolution but also provides interpretable signals that inform targeted mitigation strategies. Rather than treating hallucination as a surface-level misalignment, this dimension highlights its structural embedding in reasoning chains, making error localization and remediation both more precise and more actionable.

e) Causal Reasoning Capabilities: Causal reasoning constitutes a distinct and indispensable dimension of hallucination evaluation. It assesses whether a model can correctly infer causal relationships, differentiate them from mere spurious correlations, uncover underlying generative mechanisms, and simulate the consequences of hypothetical interventions [111], [112]. Mastery of causal reasoning is especially vital in high-stakes domains such as medicine, law, or policy-making, where misattributing correlation as causation can lead to erroneous conclusions and harmful downstream effects [36].

Rigorous evaluation of causal reasoning often requires mapping attribution to specific nodes within reasoning trees. This approach identifies the root causes—such as distinguishing between errors arising from reliance on spurious correlations, knowledge cutoffs, or structural gaps in reasoning versus those due to inherent knowledge boundaries—to enable targeted mitigation strategies, as aligned with Section I-F's Contribution. In doing so, it complements vertical reasoning traceability: while traceability shows where reasoning breaks down, causal reasoning establishes why those failures occur and provides a structured account of how specific inputs lead to particular outputs.

Empirical studies further underscore its importance. By systematically testing models on causal inference tasks, researchers have shown that causal attribution improves interpretability, enables error correction targeted at root causes, and fosters greater user trust in model outputs [112]. Integrating causal reasoning into evaluation frameworks thus ensures not only structural transparency but also conceptual soundness in reasoning processes.

f) Interpretability: As discussed in Section I-E, interpretability is an indispensable core dimension of hallucination evaluation. Unlike factuality, faithfulness, or reasoning-oriented dimensions that focus on the quality of model outputs, interpretability emphasizes the transparency, accountability, and auditability of the evaluation process itself.

While Vertical Reasoning Traceability (Section III-B4d) provides structural transparency by showing where errors occur, and Causal Reasoning Capabilities (Section III-B4e) provide conceptual transparency by explaining why they arise, interpretability uniquely addresses how evaluation judgments are communicated and made accountable. It requires that evaluation frameworks be auditable, reproducible, and trustworthy—not only for researchers diagnosing model failures, but also for practitioners, auditors, and policymakers overseeing real-world deployments.

In this sense, interpretability complements other dimensions by ensuring that hallucination evaluation is not only rigorous, but also communicable, reproducible, and deployable in high-stakes applications.

g) Self-Consistency: As discussed in Section I-D, the *Consistency* principle requires that models produce reproducible and reliable outputs across *runs*, *time*, *environments*, *configurations*, and *query variations*. This requirement is essential for both scientific validity and real-world deployment, where users expect stable answers regardless of phrasing or timing. While consistency issues can be examined from several perspectives, three aspects deserve particular attention: self-consistency, coherence consistency, and reliability consistency.

Self-consistency addresses the question of whether a model provides stable answers when semantically equivalent queries are rephrased or lightly perturbed [102], [113]. Perturbations include paraphrasing prompts, reordering clauses, or substituting synonyms, as well as comparing outputs sampled under fixed settings. Self-consistency metrics leverage the principle that true knowledge should remain stable across perturbations.

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Such instability undermines user trust and directly challenges the Consistency principle. Empirically, self-consistency tests that apply semantic-preserving perturbations reveal notable degradation under adversarial rephrasings, indicating fragile understanding rather than robust knowledge [30], [102], [113].

Typical metrics include agreement rates across paraphrases, pairwise similarity metrics, contradiction for pairs of outputs, variance across answers, consistency-accuracy, and semantic-equivalence clustering stability [102], [113]. These metrics capture whether a model's answers remain robust to superficial changes in wording (e.g., when users ask the same question in different ways) [102], [113].

Self-consistency evaluation thus targets a critical facet of hallucination behavior: a model's propensity to generate contradictory outputs for equivalent queries. Ultimately, self-consistency evaluation provides a diagnostic lens that bridges factual correctness with behavioral reliability, revealing when models fail not only in accuracy but also in stability.

To address this challenge, evaluation frameworks must go beyond surface comparison. Standardized protocols, seed-controlled sampling, and rigorous replication practices safeguard against randomness and variability while ensuring comparability across systems [29].

h) Coherence and Consistency: Coherence and consistency focuses on whether a model maintains internal logical alignment across its outputs. Unlike self-consistency, which probes stability across query variations, coherence evaluation examines logical structure and internal consistency, i.e., whether claims made within the same response (or across related responses) are logically compatible with each other. Failures of coherence often manifest as contradictions between otherwise factual statements, or as multi-hop inconsistencies where individual claims may be correct but their combination forms an invalid reasoning chain.

Recent work on consistency evaluation shows that surface fluency is a poor proxy for logical soundness: a substantial proportion of seemingly coherent responses contain subtle contradictions, and models frequently fail to maintain consistent claims across paraphrased contexts, with significant portions of seemingly coherent outputs containing internal contradictions [102], [113]. Perturbation-based approaches—such as paraphrased contexts or reordered premises—are often applied to stress-test coherence. Coherence evaluation, therefore, plays a critical role in diagnosing structural hallucinations that undermine reasoning validity.

i) Reliability and Consistency: Reliability and Consistency extends the notion of consistency beyond single responses to characterize stability of model behavior across *time*, *contexts*, and *user populations*. Reliability is indispensable for deployment: models that appear accurate under controlled benchmarks may exhibit substantial instability in practice. Whereas self-consistency asks whether rephrasings of the same query yield the same answer, and coherence assesses agreement among statements within a response, reliability evaluation assesses the stability and predictability of model behavior across repeated trials and varying conditions [21], [102], [113]. Key challenges include *temporal drift*—identical queries issued at different times returning divergent answers—*contextual variability* across domains (e.g., academic vs. conversational tasks), and *demographic influences* in which performance varies systematically across user groups.

Capturing and quantifying this variability provides essential confidence bounds for scientific evaluation and risk-aware deployment scenarios. Unlike accuracy metrics that measure correctness at a single point, reliability metrics, such as Test-retest Reliability, Parallel-forms Reliability, capture the consistency of model performance over time and across contexts [21], [102], [113].

Cross-context reliability evaluation examines consistency across different application domains and user populations.

j) Robustness: Robustness quantifies hallucination stability across conditions. Adversarial robustness testing shows that hallucination rates increase significantly under targeted attacks, with models showing particular vulnerability to authority bias and other manipulation techniques [114], [115].

Paradoxically, calibration often degrades with model scale: larger models become more confidently wrong, necessitating explicit uncertainty training and challenging assumptions about the relationship between capability and reliability [94].

k) Tool Appropriateness: We further include an explicit assessment of Tool Appropriateness, comprising two facets: *Tool Necessity Detection Accuracy*, which measures whether models recognize when external verification is required, and *Tool Selection Adequacy*, which assesses the appropriateness of chosen verification sources.

Recent diagnostic work on tool-augmented LLMs formalizes related capabilities—such as solvability detection and multi-step tool planning—and shows that even advanced models struggle on these axes, underscoring the need for targeted evaluation.

l) Computational Cost and Efficiency Trade-offs: Computational Cost and Efficiency Trade-offs analysis for hallucination evaluation reveals fundamental tensions between detection quality and practical feasibility, with comprehensive evaluation requiring orders of magnitude more computation than standard inference. This analysis includes cost considerations for User-Centric Benchmark Construction (as outlined in Section I-F's Contribution 3), which emphasizes balancing evaluation rigor with accessibility. The economic dimension of evaluation directly impacts deployment decisions: full-spectrum hallucination detection significanly increases inference costs, making real-time evaluation prohibitively expensive for many applications.

The high Large language model evaluation costs that scale with model size create evaluation inequality where only well-resourced organizations can afford rigorous testing, potentially leading to deployment of inadequately evaluated models by smaller entities. Shared evaluation infrastructure and benchmark pooling initiatives attempt to democratize access, though coordination challenges and competitive concerns limit adoption [116].

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Efficiency optimization strategies attempt to balance cost and quality through selective evaluation. Cascaded evaluation pipelines employ lightweight filters to screen obvious cases, reserving expensive deep evaluation for ambiguous instances, achieving improved detection reliability and maintaining assessment integrity through tool-augmented frameworks like UNIHD [2].

5) Emerging Metrics: As models grow more sophisticated, traditional metrics prove insufficient. Emerging approaches leverage advances in interpretability, uncertainty quantification, and cross-modal understanding to detect increasingly subtle forms of hallucination.

a) Vision-Aware and Multimodal Metrics: Vision-Aware and Multimodal Metrics need to address the unique challenges of multimodal hallucination and assess alignment between visual inputs and textual outputs. Typically, the hallucination evaluation for Multimodal-LLMs suffers from insufficient evaluation dimensions, high costs, and over-reliance on language priors rather than visual input [117]. For example, CHAIR (Caption Hallucination Assessment with Image Relevance) [59] evaluates object hallucination in image captions but only captures existence, not attributes or relationships.

Newer frameworks like AMBER (An LLM-free Multi-dimensional Benchmark) [118], provide scalable, low-cost alternatives to human annotation with multi-dimensional analysis across object existence, attributes, and relations. AMBER is an LLM-free multi-dimensional benchmark for the hallucination evaluation of MLLMs and evaluates across various tasks without relying on additional LLMs [118]. The framework's multi-dimensional assessment spans both generative tasks such as image captioning and discriminative tasks including visual question answering, capturing the full spectrum of multimodal reasoning capabilities [30].

The AMBER benchmark [30] represents a paradigm shift in multimodal hallucination evaluation by eliminating dependence on expensive LLM judges or human annotators while maintaining evaluation quality. This framework addresses a critical gap where previous evaluation methods either required costly GPT-4 API calls (with OpenAI's official pricing at $0.03 per 1,000 prompt tokens and $0.06 per 1,000 completion tokens [119]) or expensive human annotation processes, making large-scale evaluation economically infeasible [30].AMBER's automated pipeline achieves comparable accuracy to human evaluation at less than 1% of the cost, democratizing rigorous multimodal evaluation for the broader research community [30].This granular decomposition reveals that models like GPT-4V achieve 94.5% accuracy on object existence but only 77.1% on relational reasoning, exposing systematic weaknesses in compositional understanding [30].*LVLM-eHub* [65] unified six evaluation dimensions, revealing cross-task correlations: models excelling at object detection (higher than 80% accuracy) still fail at visual reasoning (45% accuracy), indicating task-specific competencies don't transfer [65].

While fine-grained visual grounding metrics trace generated text spans to image regions, visual hallucinations often stem from over-reliance on language priors rather than visual input [117]. *PhD* [120] introduces adversarial multimodal conflicts, showing models default to textual information 76% of the time when modalities disagree [120].

Temporal consistency metrics for video understanding show that hallucination rates increase substantially for events spanning multiple frames, highlighting the challenge of temporal reasoning in multimodal models [104].

b) Causal Attribution and Mechanistic Analysis: Causal Attribution and Mechanistic Analysis has attracted substantial attention as a lens on hallucination. This line of work treats hallucinations as phenomena inherently induced by the representation of internal states and traces hallucinations via attention analysis and activation patching.

Recent mechanistic interpretability research [121] has revealed that different attention heads and layers play distinct roles in factual processing, with middle layers being particularly critical for maintaining factual accuracy. Mechanistic interpretability approaches [122] focus on identifying and understanding the internal representations responsible for generating factual versus hallucinated content. The identification of potential hallucinations can also be conducted by analyzing hidden layer activation at the last token of generated texts [24]. These methods enable researchers to pinpoint specific subnetworks and circuits that contribute to different types of errors, providing insights that can inform targeted mitigation strategies and improve model reliability.

Intervention studies using techniques such as causal tracing [123] demonstrate that targeted modifications to specific model components can reduce hallucination rates while preserving overall model performance. For example, attention attribution and activation patching identify circuits linked to factual errors, enabling targeted interventions that reduce hallucination rates without major performance loss [24], [124].

c) Entropy-based and Uncertainty-aware Scoring: Semantic Entropy [94] advances beyond token-level uncertainty by clustering semantically equivalent outputs before computing entropy, thereby measuring uncertainty at the level of meaning rather than specific word sequences. The method operates across datasets and tasks without task-specific supervision or prior task knowledge and generalizes to unseen tasks [94]. By flagging when a prompt is likely to trigger confabulations, it indicates when extra caution is warranted and enables users otherwise limited by model unreliability [94]. This semantic-aware approach achieves an AUROC of 0.8 without task-specific training, though computational requirements limit real-time application [94].

d) Risk-weighted and Severity-aware Metrics: Context-aware severity assessment incorporates user expertise and application domain, adjusting evaluation thresholds accordingly. For example, Medical hallucination severity scoring weights errors by potential patient harm, finding that while only a small portion of hallucinations are high-severity, they account for the majority of potential adverse outcomes. Financial risk metrics similarly weight hallucinations by potential monetary impact, revealing that seemingly minor numerical errors can cascade into million-dollar miscalculations. Educational applications

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tolerate high hallucination rates for exploratory learning but require low tolerance for examination systems, demonstrating the need for adaptive evaluation standards. Risk-weighted evaluation acknowledges that not all hallucinations are equally harmful. 0A risk-weighted (e.g., weighted-sum) evaluation metric helps improve model performance in safety-critical applications where hallucinations can have a significant consequence and provides transparent, interpretable trade-offs [125].

6) Cross-Cutting Challenges and Future Directions: The evaluation dimensions presented above reveal fundamental challenges that transcend individual methodologies, exposing critical gaps in our ability to comprehensively assess hallucination phenomena. Current evaluation frameworks suffer from insufficient coverage across multiple dimensions: they fail to capture the full spectrum of task types (from factual QA to creative generation), overlook rare but important hallucination categories, and systematically miss errors in less common objects or concepts [30]. Thus, models appear more reliable than they actually are, creating a dangerous blind spot [30]

These challenges align with and expand upon those outlined in the introduction (Section I-E), including *Compositional Complexity*, *Ground Truth Ambiguity*, *Computational Efficiency*, *Cross-Modal Consistency*, and the need to handle *Dynamic Knowledge*.

The compositional nature of hallucinations further compounds these challenges, where multiple small errors cascade into major falsehoods through error propagation mechanisms that current evaluation approaches fail to capture. The precision for atomic units for individual facts is much higher than for compositional hallucinations involving multiple interrelated claims [30], [93].

This compositional challenge becomes particularly acute in multi-hop reasoning tasks, where a single incorrect intermediate step can invalidate an entire chain of reasoning, yet current metrics often award partial credit that obscures the severity of such failures.

A deeper concern emerges from Goodhart's law: when a measure becomes a target, it ceases to be a good measure. Models trained to optimize specific metrics often develop new hallucination types undetected by those same metrics [2]. The "evaluation overfitting" phenomenon shows that models that improves on HaluEval actually perform worse on held-out real-world data [7], [42], highlighting the need for dynamic, evolving benchmarks.

The Western-centric design of current benchmarks creates systematic blind spots in evaluation. Hallucination detection accuracy drops for non-English languages and for culturally-specific content, reflecting the Western-centric nature of most benchmarks [43], [63].

Future work must develop culturally aware evaluation frameworks that recognize legitimate variation in knowledge and perspective versus genuine hallucinations [49].

Perhaps most troublingly, a fundamental tension exists between model capability and reliability: more capable models often hallucinate more confidently and creatively, making detection paradoxically harder as models improve [13], [47]. This suggests that hallucination may not be a bug to be fixed but an inherent property of the probabilistic reasoning that enables these models' impressive capabilities. Resolving this paradox requires rethinking evaluation from binary detection toward nuanced understanding of model knowledge boundaries and uncertainty [94].

a) Full-Spectrum Evaluation Framework: The robustness and reliability of evaluation frameworks require a full-spectrum approach that transcends traditional metrics. Such a comprehensive paradigm systematically examines model behavior across multiple orthogonal dimensions, including accuracy under different granularity, self-consistency across repeated queries, resilience to adversarial attacks, faithfulness to source material, computational efficiency, and inference capabilities under knowledge constraints. Each dimension exposes distinct failure modes that remain hidden when evaluation is limited to surface-level correctness, underscoring the necessity of a holistic framework capable of capturing the full complexity of hallucination phenomena. The items listed below illustrate typical combinations of evaluation metrics, though they are not exhaustive and do not capture all possible combinations.

b) Factuality at Different Granularity: Traditional accuracy metrics should be discussed at different granular levels. For example, in *Token-level* methods, models achieve 56.31%–71.05% Hallucination F1 Scores and 66.86%-75.37% Not Hallucination F1 Scores [79]; In *Sentence-level* approaches, factual sentences can be identified with an AUC-PR of 53.97%, balancing practicality and interpretability [92].; Besides, claim-level decomposition shows that even advanced models achieve only 58% factual precision [93]. These results demonstrate the significance of structured exploration on how granularity choices affect detection capability.

c) Faithfulness, Source Attribution, and Task-Specific Definition: Faithfulness in this broader sense encompasses not only whether generated content reflects the given source material but also how models handle attribution and task-specific requirements. Evaluating faithfulness is therefore essential for assessing whether large language models (LLMs) remain grounded in context without introducing unsupported or contradictory information.

A structured taxonomy of faithfulness violations provides an analytical lens for diagnosing different failure modes in context adherence. This categorization enables targeted evaluation and mitigation strategies, revealing that different model architectures exhibit distinct faithfulness failure patterns and highlighting the importance of model-specific evaluation.

Recent advances in attribution analysis integrated with vertical tracing highlight the scale of this issue: approximately 40% of responses that appear to be grounded contain subtle distortions or unsupported inferences that escape detection by traditional metrics [95].

For example, FaithDial [126] can be used to train both high-quality dialogue generation models and a hallucination critic that classifies utterances as faithful or unfaithful. Models trained on FaithDial achieve improved performance on

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the BEGIN benchmark, outperforming counterparts trained on existing dialogue-coherence datasets.FaithDial benchmarks contemporary dialogue models and introduces an auxiliary contrastive learning objective. According to multiple automatic metrics, this objective yields highest faithfulness and increased abstractiveness [126]. FaithDial benchmarks state-of-the-art models and proposes an auxiliary contrastive objective that achieves the highest level of faithfulness and abstractiveness based on several automated metrics [126]. In human evaluations, models trained on FAITHDIAL produce responses rated as more interpretable, cooperative, and engaging, and improve faithfulness and abstractiveness, both in domain and during zero-shot transfer to other datasets, such as TopicalChat and CMU-DoG [126].

d) Inference Capability and Knowledge Boundaries: Inference capability evaluation probes the model's ability to recognize and respect its knowledge boundaries, distinguishing between legitimate reasoning and hallucinated extrapolation. Many model errors stem not from knowledge gaps but from deficiencies in causal reasoning and failure to recognize knowledge boundaries. Moreover, the ability to generalize causal knowledge across contexts strengthens model adaptability to novel scenarios and distribution shifts [127].The integration of causal reasoning enhances both transparency and robustness in AI systems.

Ultimately, robustness-centered evaluation highlights the importance of moving beyond static measures of factual accuracy toward a more dynamic and adversarially aware understanding of hallucinations. By capturing both reasoning stability and retrieval fidelity, this dimension ensures that large language models are not only accurate in controlled conditions but also trustworthy when confronted with real-world variability and adversarial challenges. Such evaluations will be indispensable in advancing AI systems that can maintain reliability, transparency, and safety across diverse operational contexts.

e) Self-Consistency across Tasks and Temporal Stability: Empirical evidence confirms that self-consistency remains a pervasive weakness. Models exhibit varying degrees of self-consistency across different tasks and different temperatures.

Longitudinal analyses indicate that GPT-4's answers can vary when semantically equivalent questions are rephrased or asked at different times, with the variability becoming more pronounced on complex reasoning tasks [30]. These findings suggest that such inconsistency is not mere noise but reflects structural fragility in reasoning and contextual grounding.

Temperature-dependent consistency analyses reveal a non-linear relationship: near-deterministic settings with very low temperatures tend to produce repetitive yet consistently incorrect outputs, whereas moderately stochastic settings with moderate temperatures introduce greater diversity and can occasionally yield correct responses [46].

These findings indicate that inconsistency is not simply noise but reflects structural fragility in reasoning and contextual grounding.

Furthermore, it exacerbates the *Ground Truth Ambiguity* challenge (Section I-E) by making definitive verification even more difficult.

f) Adversarial Robustness: Adversarial evaluation reveals critical vulnerabilities in hallucination detection and mitigation strategies. Targeted perturbations that preserve semantic meaning but alter surface forms can increase hallucination rates, with models confidently generating false information when prompted with carefully crafted adversarial inputs. Universal adversarial triggers—short phrases that reliably induce hallucinations across diverse contexts—demonstrate that models possess systematic weaknesses exploitable through minimal input modifications [6].

These findings directly test the *Robustness* design principle (Section I-D), which demands resilience to edge cases and adversarial inputs. The ease with which hallucinations can be induced underscores the critical need for evaluation frameworks to incorporate rigorous adversarial testing, as envisioned in the introduction's call for comprehensive assessment.

g) Robustness to Distribution Shift: Distribution shift evaluation examines how hallucination patterns change when models encounter out-of-distribution inputs, revealing dramatic degradation in reliability. In cross-domain transfer, models trained on general text produce substantially more hallucinations in specialized settings—particularly in medical and legal applications [1]. Distribution shift analysis reveals that models trained on Wikipedia exhibit elevated hallucination rates when applied to social media, highlighting generalization challenges [63]. Geographic and cultural shifts further expose Western-centric biases, with non-Western contexts showing higher hallucination rates even for factual queries [6]. This sensitivity to distribution shifts exemplifies the *Dynamic Knowledge* and *Cross-Modal Consistency* challenges described in Section I-E and underscores the necessity for evaluation frameworks, as called for in Section I-F, to be adaptable and capable of handling evolving knowledge and diverse contextual deployments beyond static training distributions.

h) Computational Efficiency and Scalability: The computational demands of comprehensive hallucination evaluation pose practical barriers to deployment: full-spectrum assessments typically require substantially more resources than standard accuracy testing [92]. Sampling-based approaches that score only a subset of outputs can approximate exhaustive evaluations at much lower computational cost, but they risk missing rare but catastrophic failures [92]. In reality, strict real-time constraints necessitate lightweight detectors and force trade-offs between detection accuracy and latency [2]. The economic dimension of evaluation—where comprehensive testing of large models costs thousands of dollars per evaluation cycle—creates inequities where only well-resourced organizations can afford rigorous hallucination assessment [30].

This challenge of *Computational Efficiency* was explicitly foreseen in Section I-E. Overcoming these barriers is essential for realizing the practical *User-Centric Benchmark Construction* and widespread applicability envisioned for the LENS

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framework in Section I-F, ensuring that rigorous evaluation is not just theoretically sound but also practically feasible.

i) Future Directions: Toward Unified and Adaptive Evaluation: The path forward requires fundamental reconceptualization of hallucination evaluation from fragmented metrics toward unified, adaptive frameworks that evolve with model capabilities and deployment contexts. This vision directly addresses the gaps and limitations identified in the introduction's survey of the current evaluation landscape (Section I-B) and its outlined challenges (Section I-E), particularly the need to handle *Compositional Complexity* and *Dynamic Knowledge*. Dynamic benchmark generation using procedural content creation and adversarial techniques can produce continuously novel test cases, preventing evaluation overfitting while maintaining consistent difficulty calibration [116]. These adaptive systems must incorporate feedback loops that learn from deployment failures, automatically generating new test cases that probe discovered vulnerabilities.

Multi-stakeholder evaluation frameworks must balance competing objectives: researchers need fine-grained diagnostic information, practitioners require actionable deployment guidance, and end-users demand interpretable reliability indicators. This aligns with the LENS framework's goal of providing *Actionable Insights* (Section I-F) and adheres to the *Interpretability* and *Faithfulness* design principles (Section I-D) by making evaluation results transparent and useful for different audiences. Hierarchical evaluation protocols that provide different levels of abstraction—from detailed mechanistic analysis to simple trust scores—can serve these diverse needs while maintaining consistency across levels. The development of "evaluation APIs" that allow domain experts to contribute specialized test cases without deep ML expertise will democratize benchmark creation and ensure coverage of long-tail use cases.

Cross-modal and cross-lingual evaluation represents a critical frontier, as current English-centric, text-focused frameworks fail to capture the full spectrum of hallucination phenomena in multilingual, multimodal systems. This directly tackles the *Cross-Modal Consistency* challenge from Section I-E and is a necessary extension for any framework aiming to be truly comprehensive. Universal hallucination taxonomies that transcend modality boundaries—defining hallucination in terms of information-theoretic divergence rather than surface-level errors—could enable consistent evaluation across diverse model types. Cultural adaptation layers that distinguish legitimate knowledge variation from genuine hallucinations will be essential for global deployment, requiring collaboration with diverse communities to establish culturally-aware ground truth. This effort is crucial for resolving the *Ground Truth Ambiguity* challenge in culturally nuanced contexts, moving beyond Western-centric benchmarks mentioned in the introduction.

The integration of human-in-the-loop evaluation with automated metrics promises to combine scalability with nuanced judgment. Active learning approaches that strategically select instances for human review—focusing on high-uncertainty or high-impact cases—can maximize the value of limited human evaluation resources. Crowdsourced evaluation platforms that aggregate judgments from diverse evaluators while accounting for expertise and cultural background can provide more robust assessments than single-annotator ground truth.

Causal evaluation frameworks that move beyond correlation to establish mechanistic links between model properties and hallucination behaviors will enable targeted improvements. This deeper analysis is a key step toward fulfilling the promise of *Actionable Insights* and moving from mere detection to understanding *why* hallucinations occur, as advocated in Section I-F. Interventional studies that systematically modify model components, training procedures, or inference strategies can identify causal factors driving hallucinations. Counterfactual evaluation that assesses how models would behave under alternative conditions provides insights into the robustness and generalizability of hallucination patterns.

Finally, the development of metacognitive evaluation that assesses models' awareness of their own limitations represents perhaps the most promising direction. Models that can accurately express uncertainty, refuse to answer beyond their knowledge, and actively seek clarification when faced with ambiguous queries would fundamentally change the hallucination landscape. This direction resonates strongly with the introduction's ultimate goal (Section I-G) of transforming LLMs into "trustworthy partners" and addresses the core issue of models failing to recognize their own knowledge boundaries, a source of many hallucinations. Evaluation frameworks must therefore evolve to reward appropriate uncertainty expression and penalize confident confabulation, shifting the focus from eliminating hallucinations to managing them transparently.

The ultimate goal is not perfect factuality—an impossible standard for systems trained on imperfect data—but reliable, interpretable, and controllable behavior that allows users to make informed decisions about when and how to trust model outputs. This conclusion perfectly encapsulates the overarching argument of the introduction: that systematic evaluation is paramount for "ensuring safe and trustworthy AI deployment," as mentioned in Section I and achieving the *Accuracy*, *Faithfulness*, and *Interpretability* demanded by the design principles in Section I-D.

C. Mechanistic Understanding for Decision-Making

Decision-making evaluation plays a critical role in understanding and mitigating hallucinations in LLMs and LVLMs, yet the field remains fragmented by methodological inconsistencies that obscure causal mechanisms. Current studies often employ heterogeneous experimental setups that conflate multiple sources of error, making it impossible to isolate specific failure modes or design targeted interventions [49].

The evaluation landscape is characterized by a proliferation of datasets, from factual QA benchmarks like TruthfulQA [4] and HaluEval [7] to domain-specific collections like MedHallBench [9] and FinanceBench [128], each capturing different aspects of hallucination phenomena but lacking unified evaluation protocols. Similarly, metrics range from token-level accuracy and entity-level F1 scores to semantic similarity measures

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and human judgment proxies, creating an evaluation Tower of Babel where results cannot be meaningfully compared across studies [5]. This methodological fragmentation not only impedes scientific progress but also renders deployment decisions essentially arbitrary, as practitioners lack principled frameworks for selecting appropriate evaluation strategies for their specific use cases.

Constructing benchmarks and designing targeted interventions requires a systematic separation of hallucination causes—distinguishing knowledge gaps from reasoning failures, parametric memory conflicts from retrieval errors, and linguistic biases from genuine misunderstanding—to ensure reliable diagnosis and effective mitigation [49].

Interpretability and intervention have emerged as two central perspectives in this line of work. On the interpretability side, investigations focus on understanding the internal thought process of LLMs that give rise to hallucinations, while intervention-based approaches directly modify model components to suppress undesired outputs. Results demonstrate that the success of interventions is highly component-dependent, with modifications to attention blocks proving effective, whereas adjustments to the residual stream are detrimental to language modeling performance [49].

Beyond the architectural and algorithmic considerations, the manifestation and detection of hallucinations are profoundly influenced by a constellation of contextual factors that shape model behavior at inference time. These impact factors—ranging from the granularity of evaluation units to the subtle interplay between linguistic priors and prompting strategies—determine not only whether hallucinations occur but also their severity, persistence, and detectability. Understanding these factors is essential for developing robust evaluation frameworks that can generalize across diverse deployment scenarios and provide actionable insights for mitigation strategies. Recent empirical studies have identified three primary dimensions along which these factors operate: the granularity at which hallucinations are measured, the influence of preexisting language biases that override factual grounding, and the critical role of prompt engineering in either amplifying or suppressing hallucinatory behaviors [5], [45]. Each dimension reveals distinct failure modes and suggests targeted intervention strategies, yet their interactions remain poorly understood, creating blind spots in current evaluation methodologies.

1) Granularity: The granularity at which hallucinations are evaluated fundamentally shapes both detection capabilities and intervention strategies, yet no consensus exists on optimal evaluation units. Hallucinations manifest across a spectrum of granularities—from individual tokens and named entities to complex propositions, multisentence passages, and entire document structures [5]. Token-level evaluation, exemplified by methods like SelfCheckGPT [92], enables precise localization of factual errors but suffers from exponential computational costs and difficulty in capturing semantic-level inconsistencies. Entity-level approaches, widely adopted in early fact-checking systems, excel at identifying specific factual inaccuracies—particularly for named entities, dates, and quantities—but fail to detect more subtle forms of hallucination such as relational errors or contextual misrepresentations [5].

At coarser granularities, sentence- and claim-level evaluations have gained prominence with the advent of neural factuality scoring methods like FactScore [93], which decompose responses into atomic claims for independent verification. These approaches strike a balance between computational tractability and semantic completeness, enabling evaluation of complex factual relationships while maintaining reasonable processing requirements. Document-level evaluation, while computationally efficient for large-scale assessment, obscures fine-grained error patterns and provides limited diagnostic value for targeted improvements.

The evolution of granularity preferences reflects deeper shifts in the field: early retrieval-based methods favored entity-level precision, the rise of generative models drove adoption of sentence-level metrics, and current multimodal systems demand hierarchical evaluation spanning multiple granularities simultaneously [5]. Empirical studies reveal that optimal granularity is inherently task-dependent—medical applications require token-level precision for drug names and dosages, while conversational systems benefit from discourse-level coherence checks [9]. This granularity–performance trade-off remains a fundamental challenge, with recent work exploring adaptive granularity selection that dynamically adjusts evaluation resolution based on detected error patterns and computational constraints.

2) Language Priors and Bias: The pernicious influence of language priors and learned biases represents one of the most insidious sources of hallucination in modern LLMs and LVLMs, where statistical regularities in training data override factual grounding and sensory evidence. These priors manifest as systematic preferences for plausible-sounding but incorrect completions, creating a fundamental tension between fluency and factuality that current architectures struggle to resolve [45]. The problem is particularly acute in multimodal systems, where language priors can completely dominate visual or auditory signals, leading to descriptions that are linguistically coherent but entirely disconnected from the actual input.

EventHallusion [61] provides compelling evidence of this phenomenon in video-based LVLMs, demonstrating that models consistently hallucinate temporal sequences based on learned event scripts rather than observed visual evidence. When presented with videos showing events in noncanonical order (e.g., putting on shoes before socks), models overwhelmingly generate descriptions following typical temporal patterns, revealing a catastrophic failure to ground language generation in actual sensory input [61]. The proposed Temporal Cue Disentanglement (TCD) method addresses this by systematically manipulating temporal markers, forcing models to rely on visual information rather than defaulting to linguistic shortcuts [61].

Beyond temporal biases, language priors manifest across multiple dimensions: frequency biases lead models to favor

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common entities over rare but correct ones, co-occurrence statistics drive spurious associations (e.g., automatically mentioning "Eiffel Tower" when discussing Paris, even when irrelevant), and syntactic preferences can override semantic accuracy [45]. Cultural and demographic biases embedded in training corpora further compound these issues, causing models to hallucinate stereotypical attributes or relationships that reflect societal biases rather than factual reality [49].

Empirical evaluations on the EventHallusion benchmark reveal striking disparities in model performance: open-source VideoLLMs demonstrate significantly higher susceptibility to prior-driven event hallucinations compared to their closed-source counterparts. Furthermore, the proposed Temporal Contrastive Decoding (TCD) method proves effective as a training-free approach to mitigate such hallucinations across various model architectures. [61].

3) Prompting Strategy: Prompting strategy emerges as a critical yet underappreciated determinant of hallucination rates, with subtle variations in prompt construction producing dramatic differences in model reliability and factual accuracy. The relationship between prompt design and hallucination is far more nuanced than initially understood, involving complex interactions between linguistic properties, structural patterns, and cognitive scaffolding that either amplify or suppress the model's tendency to confabulate [46].

Counterintuitively, empirical evidence reveals that simpler prompting strategies consistently outperform their more complex counterparts in reducing hallucinations [45]. Zero-shot prompts with clear, direct instructions produce fewer hallucinations than elaborate few-shot examples that may introduce spurious patterns or conflicting signals. This *complexity penalty* becomes particularly pronounced when LLMs are augmented with external tools or multistep reasoning chains—each additional component introduces new failure modes and compounds the probability of hallucination [45].

The impact of readability proves more complex and context-dependent, suggesting intricate interactions with other linguistic features and task characteristics.

A groundbreaking empirical investigation into the linguistic properties of prompts reveals three critical dimensions that govern hallucination rates: formality, concreteness, and readability [45].Prompts exhibiting higher formality—characterized by precise technical language, explicit constraints, and structured formatting—demonstrate a marked reduction in hallucination rates across diverse model architectures.

Concreteness, measured through specificity of references and grounding in verifiable facts, shows even stronger effects, with highly concrete prompts significantly reducing factual errors in categories like numerical values and proper nounss [45].

The impact of readability proves more complex and context-dependent, suggesting intricate interactions with other linguistic features and task characteristics.

Advanced prompting techniques reveal distinct tradeoffs in their effectiveness against hallucinations. Chain-of-Thought (CoT) prompting [23], while improving reasoning transparency, can paradoxically increase hallucinations when intermediate steps contain errors that propagate through the reasoning chain [46]. Self-consistency approaches that sample multiple reasoning paths show promise in identifying and filtering hallucinations through divergence detection, but at substantial computational cost [46]. Multiagent debate frameworks demonstrate impressive hallucination reduction through adversarial verification, yet their complexity makes them impractical for real-time applications [46].

The effectiveness of prompting strategies is further modulated by the linguistic properties of the prompt itself. Empirical findings indicate that prompts characterized by greater formality and concreteness tend to result in reduced hallucination across various categories (Person, Location, Number, Acronym) [46]. However, the impact of readability on hallucination rates shows a more mixed and inconclusive pattern. These relationships between prompt design and model output fidelity are observed to be more pronounced in advanced LLMs such as GPT-4 and OPT [46].

Recent work has begun exploring adaptive prompting strategies that dynamically adjust based on detected uncertainty or contradiction signals, showing promising early results in reducing hallucinations while maintaining output quality [45]. These approaches leverage metacognitive signals—such as token-level confidence scores, attention pattern anomalies, and semantic coherence metrics—to identify potential hallucination points and trigger more conservative generation strategies or explicit uncertainty expression [46].

4) Causal Attribution and Mechanistic Insights: Understanding the causal mechanisms underlying hallucinations in decision-making contexts requires moving beyond correlational observations to establish direct causal links between model components and hallucinatory outputs.

Particularly revealing are findings about the internal mechanisms transformers employ for multi-step reasoning. Mechanistic analysis of a 6-layer model trained on a synthetic path-finding task reveals that it implements a backward chaining algorithm. This algorithm operates in a layer-wise manner, with early layers aggregating edge information, and subsequent deduction heads in middle layers (e.g., layers 2-4) responsible for recursively climbing the tree structure one step per layer. Perturbing these critical components via causal scrubbing techniques disproportionately disrupts the model's reasoning capability, demonstrating their causal necessity for correct output generation [57].

This "factuality bottleneck" suggests that hallucinations may result from insufficient information flow through these critical layers, potentially addressable through architectural modifications or targeted fine-tuning strategies [57].

5) Evaluation Benchmark Design and Validation: The proliferation of hallucination benchmarks without systematic validation criteria has created a paradox: we have numerous ways to measure hallucinations but limited confidence in what these measurements actually capture. Effective benchmark design for decision-making evaluation must address several funda-

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mental challenges that current approaches largely ignore [116], [118].

First, the temporal validity problem: benchmarks quickly become obsolete as models are trained on test data, either directly through data contamination or indirectly through similar distributions appearing in training corpora. Dynamic benchmark generation, as proposed in recent work [116], offers a promising solution by creating novel test instances from verified knowledge sources, though this introduces new challenges in ensuring consistent difficulty and coverage.

Second, the domain transfer challenge: benchmarks developed for general factual accuracy often fail catastrophically when applied to specialized domains like medicine, law, or finance, where hallucinations carry vastly different risks and manifestations [9], [128]. Domain-specific benchmarks must capture not just factual accuracy but also appropriate uncertainty expression, regulatory compliance, and professional standards that vary dramatically across fields.

Third, the compositionality gap: current benchmarks primarily test atomic facts or simple relationships, failing to evaluate complex multistep reasoning where hallucinations can compound through inference chains. Hierarchical benchmarks that decompose complex queries into verifiable subcomponents while maintaining logical consistency represent a critical frontier for evaluation development [129].

6) Cross-Modal Hallucination Dynamics: The emergence of vision–language models introduces qualitatively new forms of hallucination that transcend traditional text-based evaluation paradigms. Cross-modal hallucinations arise from fundamental misalignments between visual encoders and language decoders, creating failure modes that cannot be detected through unimodal evaluation [61], [130].

Visual grounding failures manifest in multiple ways: object hallucination (describing nonexistent entities), attribute confusion (misattributing properties across objects), spatial relationship errors (incorrect relative positions), and temporal sequence violations (misordering events in video) [130]. Each category requires distinct evaluation strategies—pixel-level verification for object presence, scene graph matching for relationships, and temporal logic checking for event sequences.

The modality imbalance problem proves particularly pernicious: language models' superior training on text leads to linguistic dominance, where plausible textual completions override visual evidence. This creates a "hallucination gradient" where errors increase with the semantic distance between visual input and textual prior knowledge. Models consistently hallucinate typical attributes for recognized objects (e.g., describing a white crow as black) and generate stereotypical scene descriptions even when contradicted by visual input [61].

Recent approaches to cross-modal evaluation employ adversarial visual inputs, counterfactual image-text pairs, and synthetic scene modifications to stress-test grounding mechanisms. These reveal that current VLMs often rely on superficial visual features and textual shortcuts rather than deep scene understanding, suggesting fundamental architectural limitations in how visual and linguistic information are integrated [130].

7) Implications for Real-World Deployment: The translation of hallucination evaluation from research settings to production environments reveals critical gaps between academic metrics and operational requirements. Real-world deployment demands not just detection but also graceful handling of hallucinations, requiring evaluation frameworks that assess both failure rates and failure modes [18].

Risk-stratified evaluation becomes essential in deployment contexts: a hallucinated drug dosage in medical applications carries vastly different consequences than a hallucinated historical date in educational content. Current evaluation frameworks fail to incorporate risk weighting, treating all hallucinations as equally problematic. Operational metrics must distinguish between categories of harm—safety-critical errors, legally consequential mistakes, reputation-damaging inaccuracies, and minor factual errors, with evaluation protocols tailored to each risk level.

The challenge of evaluation under distribution shift proves particularly acute in production settings, where models encounter inputs far outside their training distribution. Robustness evaluation must assess not just average-case performance but worst-case behavior under adversarial or out-of-distribution inputs. This requires moving beyond static benchmarks to dynamic evaluation protocols that continuously probe model boundaries and identify emerging failure modes [118].

Furthermore, the interaction between hallucination evaluation and user behavior creates feedback loops that current frameworks ignore. Users adapt to model limitations, developing workarounds that may mask or exacerbate hallucination problems. Evaluation must therefore consider the full sociotechnical system, including user strategies, interface design, and organizational processes that mediate model outputs [18].

Together, these studies suggest that illusions in decision-making are multifaceted and multilayered, from low-level perceptual inaccuracies to domain-critical reasoning errors. Although specialized benchmarks can capture the subtleties of decision hallucinations, a unified framework ensures intertask comparability and scalability. A balanced integration of these two approaches is crucial to ensure comprehensive coverage without sacrificing contextual depth. Future research must prioritize the integration of benchmark development with real-world decision-making scenarios, as hallucinations can not only be costly in the form of technical failures but also pose potential risks to safety, fairness, and trustworthiness.

D. Evaluation Methodologies (Implementation Approaches)

Evaluation methodologies operationalize the theoretical frameworks and dimensional choices into concrete implementation strategies, determining not just what we measure but how we execute these measurements at scale. The field has evolved from simple output comparison to sophisticated approaches that probe model internals, manipulate causal pathways, and leverage uncertainty quantification. Pioneering methodologies include SelfCheckGPT's [92] self-consistency

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paradigm achieving reference-free detection, ReEval's [26] adversarial perturbation framework exposing model vulnerabilities, and TruthPrint's [27] latent space interventions identifying universal truthful directions. These diverse approaches reflect a fundamental tension: while automated methods enable scalability, they often miss nuanced errors that human judgment catches [92]; while internal state analysis provides mechanistic insights, it remains model-specific [27]; while adversarial testing reveals vulnerabilities, it may not reflect real-world usage patterns [26].

The methodological landscape has crystallized around six complementary paradigms, each addressing different aspects of the hallucination challenge. Probe-based and prompt-based methods leverage external tools or crafted inputs to elicit inconsistencies, balancing flexibility with computational overhead. Adversarial testing actively perturbs inputs to stress-test model robustness under controlled conditions. Causal intervention techniques isolate failure mechanisms through counterfactual analysis. Uncertainty-guided approaches exploit intrinsic model properties for reference-free detection. Internal state analysis traces hallucination origins through attention patterns and activation dynamics. Online evaluation captures real-world performance through user-facing deployments. This diversity reflects the multifaceted nature of hallucinations—no single methodology suffices; effective evaluation requires strategic combination based on specific use cases, resource constraints, and reliability requirements.

1) Probe-based and Prompt-based Evaluation: Probe-based and prompt-based evaluation methods represent a prominent category in hallucination detection and probability estimation for LLMs. The fundamental innovation lies in their dual approach: probes leverage external tools, metrics, auxiliary models, or structured annotations to scrutinize outputs for inconsistencies, while prompt-based methods craft specific inputs to elicit self-assessment or provoke hallucinations for analysis. A seminal work in this category is POPE (Polling-based Object Probing Evaluation) [25], which pioneered systematic object hallucination assessment in VLMs by converting detection into binary questions (e.g., "Is there a [object] in the image?"). POPE's novelty lies in revealing that VLMs exhibit systematic biases toward common and co-occurring objects, with overall accuracy around 70-77% under various sampling strategies, exposing a fundamental tendency to hallucinate frequently appearing items [25].

In the culturally sensitive test framework, prompt-based questioning across multiple query categories is used to provoke hallucinations in GPT models, evaluated through a human-centric probe-based comparison that modifies Turing's Imitation Game to highlight cultural nuances and societal contexts, outperforming purely algorithmic detection by emphasizing non-binary judgments [49]. THaMES [71], on the other hand, uses a hybrid approach that combines prompt-based test set generation through weighted sampling, batch processing, and various question types, such as counterfactual validation, to generate QA pairs from corpora like Wikipedia and academic papers [71]. Probe-based benchmarking uses multifaceted metrics for hallucination generation and identification across tasks like text generation and binary classification. This results in an end-to-end pipeline that incorporates mitigation strategies, such as RAG, ICL, and PEFT and outperforms isolated prompts by achieving cost-effective diversity [71].

Using prompt-based classifiers for atomic fact evaluation and context reasoning, OnionEval [58] presents a layered probe-based metric called context-influence score (CI) to quantify fact-conflicting hallucinations in SLLMs across nested contexts [58]. This reveals that SLLMs are better at factual compression than they are at context comprehension when compared to larger models, allowing for targeted mitigations like chain-of-thought that improve performance without incurring high computational costs [58].

In contrast to synthetic prompt-induced datasets, RAGTruth [76] highlights fine-tuning's advantage over RLHF in sparse evaluations. RAGTruth focuses on word-level probe-based annotations in a corpus of 18,000 natural responses from diverse LLMs in RAG settings [76], benchmarking detection via fine-tuned models that surpass prompt-based self-checking or intrinsic state probes by addressing both benign and harmful hallucinations [76].

Med-HALT's prompt-based tests, such as False Confidence Test (FCT), None-of-the-Above (NOTA), and Fake Questions Test (FQT), evaluate reasoning hallucinations, while probe-based memory tests assess retrieval accuracy in medical LLMs using multinational datasets [1].

In contrast to English-centric probes like TruthfulQA, HalluQA [70] for Chinese LLMs uses GPT-4 as a prompt-based automated evaluator on 450 adversarial questions across multiple domains. Probe-based critiques highlight cultural and alignment factors, demonstrating that alignment reduces imitative issues more effectively than scaling and promoting human intervention in culture-dependent hallucinations [70].

HaDes focuses on reference-free, token-level probing through binary classification tasks applied to modified Wikipedia texts [33]. It utilizes prompt-based labeling combined with iterative model-in-loop approaches to ensure data balance. This facilitates real-time detection during the decoding process, surpassing the effectiveness of sentence-level prompting by identifying detailed, subtle signals that allow immediate intervention in open-ended text generation [33].

EventHallusion detects event hallucinations in video-based large language models by utilizing GPT-assessed prompts that focus on temporal discrepancies and vision-language inconsistencies. It employs a probe-driven approach known as Temporal Contrastive Decoding (TCD) to minimize interference without requiring additional training [61]. This method tackles challenges unique to video data, such as the accurate representation of chronological sequences, which are often neglected by static image-based prompts. Furthermore, EventHallusion highlights the weaknesses of open-source models when compared to their closed-source counterparts [61].

Bias and Interference Challenges in Visual Language Models (Bingo)'s probe-based evaluation framework carefully selects instances of model failures to evaluate various bi-

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ases [39], such as regional and OCR-related biases, and interference effects in GPT-4V. The assessment employs prompt-driven leading questions and multi-image test cases, incorporating self-correction mechanisms that reduce hallucinations [39]. This benchmark reveals underlying multimodal weaknesses, including a noticeable Western bias, which goes beyond the limitations observed in text-only inputs used in models [39].

The evolution from simple probing (POPE) to sophisticated frameworks (THaMES, EventHallusion) demonstrates the field's progression toward comprehensive evaluation [25]. EventHallusion's temporal probing for video understanding represents the frontier, detecting chronological inconsistencies that static methods miss entirely [61]. These methodologies collectively reveal that effective probing requires domain-specific adaptation: medical contexts demand specialized knowledge probes (Med-HALT), cultural evaluation needs modified Turing tests, and multimodal scenarios require cross-modal consistency checks.

2) Adversarial and Stress Testing: Adversarial and stress testing plays a crucial role in evaluating the robustness of large language models (LLMs) and large vision–language models (LVLMs), particularly under conditions that intentionally expose weaknesses not captured by standard benchmarks. Unlike conventional evaluation, which often measures performance on well-structured datasets, adversarial testing deliberately perturbs inputs or simulates edge cases to uncover systematic vulnerabilities. Stress testing further extends this paradigm by assessing model behavior under complex, noisy, or ecologically valid scenarios that better reflect real-world use cases.

To capture the breadth of current approaches, this section reviews several representative methodologies within this category. Specifically, we discuss adversarial methodologies, which generate controlled perturbations to reveal fragility in model reasoning; vision–language stress testing, which adapts adversarial principles to multimodal systems to evaluate cross-modal consistency and resilience; and real-world adversarial queries, which focus on user-derived or naturally occurring prompts to assess ecological validity. Collectively, these methods provide complementary perspectives on robustness, scalability, and practical reliability, highlighting both the strengths and limitations of adversarial evaluation paradigms.

a) Adversarial Methodologies: Adversarial methodologies actively perturb inputs to expose model vulnerabilities under controlled stress conditions. ReEval [26] pioneers a transferable framework for retrieval-augmented LLMs by dynamically generating adversarial test cases through *evidence perturbation*. Its key innovation employs two strategies: *answer swapping* - replacing correct answers with valid alternatives and *context enriching* - adding relevant information, both synthesized via LLM-based prompt chaining. Empirical results show that ReEval causes GPT-4 accuracy to drop from 83% to 47% on perturbed queries, while ChatGPT's performance degrades from 76% to 42% [26]. Strikingly, the attack data generated by Alpaca-7B has a relatively small impact on the accuracy of GPT-4 (with an accuracy rate of 89%), indicating low attack effectiveness. [26].

b) Vision–Language Stress Testing: Extending these principles to multimodal contexts, vision–language stress testing leverages automated adversarial construction. AutoHallusion [20] automates adversarial benchmark generation for LVLMs by reverse-engineering *schema-based conflicts*. Inspired by cognitive dissonance theory, it manipulates images via three strategies: (1) *abnormal object insertion*, (2) *paired object insertion*, and (3) *correlated object removal*, creating "contextual traps" that exploit language priors.

The interventions induce hallucinations at rates exceeding 97% on models such as GPT-4V and Claude 3, thereby uncovering systematic biases in cross-modal reasoning while substantially reducing the reliance on manual benchmark construction and human curation costs [20].

c) Real-World Adversarial Queries: From another perspective, Real-world adversarial queries prioritize ecological validity by capturing naturally challenging user prompts. HalluEval-Wild [42] curates challenging user interactions from ShareGPT, filtered adversarially against Alpaca to target five hallucination-prone query types (e.g., out-of-scope information, beyond-modality requests). Using retrieval-augmented generation (RAG) to provide reference answers, this benchmark highlights heightened hallucination risks in knowledge-distilled models, which often perform well on alignment benchmarks yet falter under real-world stress conditions.

Furthermore, *Collective limitations* persist across adversarial evaluation methodologies. ReEval's concentration on short-answer QA restricts its coverage of complex tasks. AutoHallusion's primitive image stitching compromises perceptual naturalness. HalluEval-Wild's static benchmark struggles with evolving LLM capabilities. These methodologies collectively underscore trade-offs between *automation scalability*, *ecological validity*, and *granular controllability* in stress-testing paradigms.

3) Causal Intervention and Counterfactual Analysis: Causal intervention and counterfactual analysis represent a rapidly growing area of research that seeks to move beyond surface-level performance metrics toward uncovering the mechanisms that drive hallucinations in LLMs and LVLMs. By explicitly modeling cause–and–effect relationships and simulating controlled counterfactual scenarios, these approaches provide deeper insight into how hallucinations emerge and how they can be effectively mitigated. This section is organized into four complementary perspectives: (1) Counterfactual Probing; (2) Causal Intervention based on Internal States, which directly manipulates internal states to probe model reliability; (3) Causal Probing based on Causality Graph, which constructs structured causal graphs to identify spurious correlations and hidden dependencies; and (4) Comprehensive Interventions Strategy, which integrates multiple intervention levels to guide principled decision-making about when, where, and how to apply corrections. Together, these approaches illustrate the potential of causal frameworks to transform hallucination evaluation from ad hoc fixes into systematic, mechanism-driven methodologies.

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a) Counterfactual Probing: Counterfactual probing is a novel approach that helps detect the causal relationships in LLM structures and to detect and mitigate hallucinations [131]. Counterfactual Probing [131] hypothesizes that robust knowledge should demonstrate consistency when confronted with plausible alternatives. The method evaluates the model's sensitivity to different counterfactual perturbations [131] and achieves superior detection performance compared to baseline methods. There are four different probe types: *Factual Probes* that alter key entities, relationships, or attributes while maintaining semantic plausibility; *Temporal Probes* that modify time-related information, such as dates, durations, or temporal sequences; *Quantitative Probes* that perturb numerical values, statistics, or measurements; and *Logical Probes* that introduce logical inconsistencies or invalid causal relationships [131]. Recent studies reveal that logical probes effectively identify reasoning errors (F1: 0.771), factual probes are most effective for entity-related hallucinations (F1: 0.834), while temporal probes excel at detecting chronological errors (F1: 0.798) and quantitative probes show strong performance on numerical claims (F1: 0.812) [131].

b) Causal Intervention based on Internal States: Causal probing can help interpret the behavior of an existing pre-trained model [132]. Causal Intervention based on internal states methodologies provides a principled framework for probing the underlying mechanisms of hallucination by simulating controlled counterfactual scenarios. The TruthPrint framework [27] pioneers a latent space approach with remarkable universality, identifying "truthful directions" shared across diverse LVLMs. Its core innovation lies in discovering that truthfulness manifests as consistent geometric patterns in latent space, with a common subspace capturing transferable hallucination features. By projecting internal states and applying pre-intervention during decoding, TruthPrint achieves significant hallucination reduction on benchmarks like CHAIR and POPE without any model retraining, while maintaining high performance on standard benchmarks [27].

c) Causal Probing based on Causality Graph: Complementary to this, *Causal Probing* [133] constructs multimodal causality graphs revealing that object hallucinations primarily stem from learned co-occurrence patterns rather than visual misinterpretation, as evidenced by frequent hallucinatory-inducing words like 'tree' or 'grass' [133]. The method exposes how benign contextual objects (e.g., grassy fields) induce spurious generations (e.g., frisbees) with measurable causal effects, though no specific correlation strength is quantified [133]. Its three-tier intervention strategy achieves consistent improvements: image-pasting (e.g., adding a small object) reduces hallucination metrics such as CHAIR scores, semantic prompt restructuring (e.g., foreground-background prompting) enhances coverage while mitigating hallucinations, and embedding manipulation alters latent representations to decrease hallucinations, with combined approaches showing additive benefits without compromising task performance metrics like coverage rates [133]. Remarkably, these simple input-space modifications outperform complex fine-tuning approaches that require thousands of human-annotated samples [133].

d) Comprehensive Interventions Strategy: Beyond individual interventions, recent research has shifted toward optimizing the decision-making process surrounding when, where, and how to apply interventions. Rather than focusing solely on the design of isolated techniques, this perspective emphasizes strategic deployment: identifying the most critical model components to target, determining the optimal timing for intervention, and calibrating intervention intensity to balance effectiveness with efficiency [18].

Such an approach reframes intervention design as a dynamic and context-aware process, moving the field from ad hoc experimentation toward principled, systematic methodologies [18]. Simhi et al. [18] argue that disparate research setups obscure the true causes of hallucinations, leading to misguided interventions. They constructed the *WACK* [18] benchmark to systematically evaluate interventions in both open-book and closed-book QA settings. Their investigation reveals that intervention success is highly dependent on the targeted component (e.g., attention blocks vs. residual stream) and the timing of using representative vectors (collecting them *before* a hallucination occurs proves more beneficial than after) [18]. Most notably, they introduce a *Dynamic Intervention* strategy that triggers only when necessary, enhancing robustness compared to static intervention paradigms [18]. This research provides a crucial framework for making informed decisions in deploying causal interventions, moving from ad-hoc applications to a principled methodology [18].

Collectively, these approaches shift evaluation from static metric computation to dynamic mechanism exploration. By decoupling confounding factors through controlled perturbations, they offer cost-efficient alternatives to human-intensive benchmarks while illuminating fundamental failure modes. Their emphasis on transferable latent structures and minimal inference-time overhead aligns with the broader need for scalable, interpretable evaluation protocols as described in Section I-D, II-C and III-B5.

4) Uncertainty-Guided and Self-Consistency Checks: Uncertainty-guided and self-consistency checks constitute a promising approach for hallucination detection in generative AI systems, leveraging intrinsic model properties without reliance on external databases or extensive human annotations. These methods exploit the principle that factual responses exhibit high consistency under semantic-equivalent perturbations, while hallucinations manifest as divergent outputs.

For general LLMs, *SelfCheckGPT* [92] adopts a zero-resource, sampling-based strategy, comparing stochastic response variances to identify contradictions, achieving 83.21 AUC-PR in sentence-level hallucination detection and strong passage-level factuality correlation without token probability access [92].

For LVLMs, *VL-Uncertainty* [134] introduces a framework that perturbs inputs through visual blurring and textual rephrasing, then measures uncertainty via entropy over response clusters. This approach demonstrates superior hallucination detection accuracy compared to baseline metrics

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while the intrinsic metric scales efficiently across multimodal benchmarks such as MM-Vet and ScienceQA.

Complementing this, the *CHAIR* metric [59] targets object hallucination in image captioning by evaluating caption-object alignment against ground-truth annotations, revealing that models with higher visual consistency hallucinate less, and highlighting limitations in standard metrics like CIDEr.

Collectively, these uncertainty-driven methodologies enhance reliability by quantifying model confidence intrinsically, though computational costs—especially in sampling-based approaches—and modality-specific adaptations remain challenges for broader deployment.

5) Internal State and Attention Analysis: This diagnostic approach analyzes real-time signals—particularly attention allocations and activation patterns—to detect representational deviations predictive of hallucinatory outputs [24], [135].

The white-box nature of these approaches provides mechanistic insights unavailable through output-only evaluation, though model-specific architectures limit generalizability [24].

For pure LLMs, *PoLLMgraph* [24] models hallucination as a state transition problem, capturing temporal dynamics of latent activations during generation. Unlike black-box methods reliant on output text or self-consistency checks, PoLLMgraph's Hidden Markov Model (HMM) framework quantifies hallucination likelihood from intermediate state trajectories, achieving over 20% AUC-ROC improvement on TruthfulQA compared to prior white-box techniques [24]. This approach requires minimal supervision (fewer than 100 samples) and generalizes across decoder architectures [24].

For LVLMs, *DAMRO* [135] identifies a critical consistency between visual encoder and LLM decoder attention maps, revealing that both disproportionately focus on background tokens, which propagates redundant information and triggers object hallucination. Its training-free mitigation strategy leverages ViT's classification token (CLS) to filter high-attention outlier tokens and applies contrastive decoding during LLM generation, significantly reducing hallucination in benchmarks like POPE and CHAIR [135] .

Collectively, these methods demonstrate that internal signals (attention distributions, state transitions) offer direct, scalable pathways to detect and mitigate hallucinations. However, their reliance on model-specific mechanisms (e.g., ViT structure, autoregressive dynamics) may limit applicability to emerging architectures.

6) Code-based: Code-based LLM evaluations use a simple script to check whether the model's output adheres to the established rules [64]. To ensure optimal code-based evaluations, it is essential to establish clear and objective criteria tailored for the task. For example, the test can automatically verify whether the response is valid JSON [64], can prevent regressions by comparing new outputs with previous ones, and ensure updates don't introduce errors. Beyond format validation, Code-based LLM evaluations can also detect whether generated stories contain both a title and body text, whether recipes accurately enumerate required ingredients, or whether mathematical answers align with known solutions etc [64]. This approach offers several advantages, such as fast evaluation speed, repeatability, and seamless integration into continuous integration/continuous deployment (CI/CD) pipelines [64]. However, constrained to objectively verifiable properties, the approach is limited in its ability to capture nuanced aspects such as stylistic variation, coherence, or tone [64]. Consequently, they must often be complemented by human assessments or LLM-based evaluators to provide comprehensive evaluation coverage [64].

7) Online and User-facing Evaluation: Online and User-facing Evaluation ensures models function properly in practical applications through systematic deployment methodologies [31]. User Risk Audits help identify potential issues, including privacy violations, harmful outputs, or systematic biases, before large-scale deployment [31].

The deployment strategy employs a phased rollout, akin to the targeted evaluations discussed in [136], typically starting with a small subset of users for initial testing and gradually scaling to full deployment based on performance metrics. This staged approach enables developers to understand model behavior under real-world conditions, to capture early warning signals, to resolve emergent issues, and to refine system performance under authentic usage conditions.

Collectively, these methodologies highlight the importance of embedding risk-sensitive, data-driven strategies into deployment pipelines. By combining phased rollouts, real-world audits, and comparative experiments, online and user-facing evaluation frameworks minimize risk exposure, foster adaptive model governance, and support safe model transitions in dynamic production environments [31], [136].

8) Synthesis and Implications: The evaluation methodologies surveyed represent six years of intensive research (2019-2025) [3], [3] yielding critical breakthroughs in hallucination understanding. The discovery of universal "truthful directions" in latent space enables 41% hallucination reduction without retraining [27], while systematic probing revealed that models hallucinate common objects 2.3 times more frequently than rare ones [25]. Current state-of-the-art achieves various levels of effectiveness: probe-based methods reach an 0.84 F1 score [71], adversarial testing triggers hallucinations with success rates exceeding 97% [20], and causal interventions reduce hallucinations by 61% while maintaining 94% task performance [133].

Despite these advances, critical limitations persist: production performance drops 15-30% from benchmarks [42], sophisticated methods require 5-10 times more computation [94], and static benchmarks become obsolete within 6-12 months as models evolve [42].

For practical deployment, we recommend tiered evaluation combining lightweight consistency checks (SelfCheckGPT achieving 83.21 AUC-PR [92]) for all outputs with expensive methods for high-risk cases. The path forward requires adaptive evaluation frameworks that evolve with model capabilities, creating unified theories explaining hallucination across modalities, and building deployment pipelines that balance accuracy, efficiency, and user experience. The ultimate goal

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is not the complete elimination of hallucinations but rather their intelligent management: understanding when, why, and how models hallucinate and ensuring their safe, reliable deployment within defined reliability boundaries, acknowledging that perfect detection may be fundamentally impossible given the probabilistic nature of generative models.

9) Summary and Critical Analysis: The methodological landscape for hallucination evaluation reveals both significant progress and fundamental limitations that shape current practice and future directions. Our survey identifies six distinct paradigms, each optimized for different aspects of the hallucination challenge, yet none providing a complete solution.

a) Methodological Trade-offs and Complementarity: The surveyed methodologies exhibit clear trade-off patterns that practitioners must navigate. *Probe-based and Prompt-based* approaches offer flexibility and minimal infrastructure requirements, achieving strong agreement with human judgments in frameworks like THaMES [71], yet remain vulnerable to prompt sensitivity and evaluator biases. *Adversarial and Stress Testing* (e.g., ReEval, AutoHallusion) effectively exposes edge cases with hallucination trigger rates exceeding 97% [20], but ecological validity remains questionable—models may never encounter such adversarial inputs in deployment. *Causal Intervention* techniques provide mechanistic insights with minimal supervision (fewer than 100 samples for PoLLMgraph [24]), yet their model-specific nature limits transferability across architectures.

b) Emerging Convergence Patterns: Despite fragmentation, convergent trends emerge across methodologies. Multiple approaches independently discover that *Internal Consistency* serves as a robust hallucination signal—from SelfCheckGPT's response variance to VL-Uncertainty's perturbation-based entropy. The success of *Latent Space Interventions* (TruthPrint) and *Attention Analysis* (DAMRO) suggests that hallucinations have identifiable computational signatures, pointing toward unified detection frameworks. The widespread adoption of *Hybrid Strategies*—combining automated screening with targeted human oversight—reflects practical consensus that pure automation remains insufficient for high-stakes applications.

c) Critical Gaps and Future Imperatives: Three critical gaps persist across all methodologies. First, *Temporal Degradation*: static benchmarks become obsolete as models evolve, yet no methodology provides continuous adaptation mechanisms. Second, *Cross-Modal Generalization*: methods optimized for text (SelfCheckGPT) or vision-language (CHAIR) tasks fail to transfer, necessitating modality-specific reimplementation. Third, *Computational Scalability*: sophisticated approaches like causal intervention or ensemble uncertainty quantification remain prohibitively expensive for real-time deployment, limiting their practical impact.

d) Toward Integrated Evaluation Frameworks: The path forward requires moving beyond isolated methodologies toward integrated frameworks that strategically combine complementary approaches. Effective evaluation likely involves: (1) *Tiered Assessment*—using lightweight consistency checks for initial screening, followed by targeted causal analysis for high-risk outputs; (2) *Adaptive Benchmarking*—leveraging online evaluation to continuously update test suites based on observed failure patterns; and (3) *Cross-methodological Validation*—requiring agreement across multiple paradigms (e.g., both uncertainty signals and attention patterns) before flagging hallucinations. This integration must balance theoretical rigor with practical constraints, acknowledging that perfect hallucination detection may be fundamentally impossible given the probabilistic nature of generative models. The ultimate goal is not elimination but intelligent management—understanding when, why, and how models hallucinate to deploy them safely within their reliability boundaries.

E. Evaluation Across Comprehensive Dimensions

The LENS framework, as introduced in Section I-F, is built upon a multi-dimensional evaluation system that moves beyond traditional, fragmented metrics. This section elaborates on the core dimensions of evaluation that LENS integrates, providing the comprehensive and hierarchical assessment required to tackle the challenge of hallucination in GenAI systems. These dimensions are designed to work in concert, enabling both the **Horizontal Assessment** of performance breadth across tasks and domains, and the **Vertical Assessment** that decomposes and examines the internal reasoning chain, as detailed in Section I-C.

The evaluation of hallucinations in large language models (LLMs) necessitates a multi-dimensional framework that captures distinct facets of generation quality and reliability, which together determine overall model trustworthiness. This section systematically reviews existing methodologies, illustrating how diverse metrics can be operationalized across complementary dimensions, as detailed in Section III-B. Nonetheless, the current evaluation landscape remains incomplete. Existing approaches only partially capture core dimensions of hallucination assessment, while critical aspects—such as evaluation granularity, reference availability, and the balance between automation and human oversight—are often overlooked. This gap underscores the urgent need for more comprehensive and interpretable frameworks capable of supporting holistic, multi-dimensional evaluation.

The metrics in LENS's core dimensions include: factuality assessment against objective truth, hallucination rate estimation for statistical characterization, faithfulness to provided context, consistency measurement for reliable performance, robustness quantification across environmental factors, susceptibility analysis under adversarial conditions, safety prediction for deployment readiness, interpretability for actionable diagnostics, tool use evaluation for external knowledge grounding, and computational constraints that govern practical deployment. Each dimension addresses unique challenges while contributing to a holistic understanding of hallucination phenomena, enabling practitioners to select appropriate evaluation strategies based on specific application requirements and risk tolerances.

1) Measure Factuality: Factuality and Accuracy measurement represents the foundational dimension of hallucination

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evaluation in LENS, assessing whether generated content aligns with verifiable objective truth across diverse knowledge domains, as emphasized in Section I-D. This evaluation dimension confronts the epistemological challenge of establishing ground truth while navigating the computational complexity of comprehensive fact verification, requiring sophisticated frameworks that balance rigor with practical feasibility [4], [93]. The measurement of factuality transcends simple binary classification, demanding nuanced approaches that account for partial correctness, temporal validity, and domain-specific verification requirements.

The atomic fact decomposition paradigm, exemplified by FactScore [93], revolutionizes factuality measurement by breaking complex statements into individually verifiable atomic claims.

This framework addresses the granularity challenge by recognizing that long-form generations typically contain mixtures of accurate and inaccurate information that aggregate metrics obscure. When evaluating biographical texts, FactScore decomposes each generation into an average of 26-41 atomic facts, revealing that even state-of-the-art models like ChatGPT achieve only 58.3% atomic factual precision, while GPT-4 achieves approximately 70-80% [93]. The framework's innovation lies in its systematic approach: each atomic fact undergoes independent verification against authoritative sources (e.g., Wikipedia), with scores aggregated using precision-recall metrics that account for both completeness and accuracy. This granular analysis exposes systematic error patterns (e.g., model struggles with factual contradictions and irrelevant content), but without specific accuracy rates for dates, numerical quantities, or minor details as cited in the original text [93].

Knowledge grounding strategies for factuality measurement address the challenge of establishing authoritative ground truth across diverse domains.

TruthfulQA [4] introduces adversarially curated questions designed to elicit false beliefs and misconceptions, with 817 questions spanning 38 categories from health to politics. The framework's key insight is that factuality evaluation must test not just knowledge recall but resistance to common human misconceptions that appear in training data.

Evaluation reveals that larger models paradoxically perform worse on TruthfulQA, with GPT-3 175B achieving only 58% accuracy, while smaller models like GPT-3 6.7B demonstrate higher accuracy, suggesting that scale amplifies learned misinformation [4]. The benchmark employs both automated metrics using fine-tuned classifiers achieving 0.90 agreement with human judges and human evaluation protocols that assess both truthfulness and informativeness separately [4].

Domain-specific factuality measurement recognizes that verification requirements vary dramatically across fields, necessitating specialized evaluation protocols. Medical factuality evaluation employs clinical knowledge bases and peer-reviewed literature, with frameworks like MedHALT [1] providing a comprehensive benchmark for assessing the propensity of large language models to generate hallucinations in medical reasoning and information retrieval tasks through multi-source verification and innovative testing modalities. Legal factuality assessment requires jurisdictional scoping and precedent tracking, with evaluation frameworks maintaining separate knowledge bases for different legal systems and temporal versions of statutes. Scientific factuality measurement must distinguish between established consensus, emerging findings, and controversial claims, with frameworks implementing confidence-weighted scoring that reflects the strength of scientific evidence. Financial factuality evaluation demands real-time data feeds and regulatory compliance checking, with specialized frameworks achieving sub-second verification latency for market-sensitive information.

The challenge of compositional factuality emerges when individually correct facts combine to create false composite statements. Advanced frameworks employ logical consistency checking that verifies not just atomic facts but their relationships and implications. Analysis reveals that a significant portion of hallucinations involve compositionally invalid combinations of true atomic facts, highlighting the insufficiency of simple fact-checking approaches [93]. Graph-based factuality assessment constructs knowledge graphs from generated text and reference sources, using graph matching algorithms to detect structural inconsistencies that atomic evaluation misses. These approaches achieve higher accuracy for detecting compositional hallucinations compared to atomic-only methods, though computational requirements increase with statement complexity [14].

The LENS framework enhances traditional factuality evaluation through its **user-centric benchmark construction** capability. This allows for the creation of dynamic, domain-specific knowledge bases from proprietary data, ensuring that factuality assessment is grounded in the most relevant and current information, thus directly addressing the challenge of **dynamic knowledge** outlined in Section I-E.

2) Measure Faithfulness: Faithfulness evaluation in LENS assesses whether generated content accurately reflects and adheres to provided source material, addressing a critical requirement for deployment reliability as discussed in Section I-D.

While factuality evaluation establishes alignment with external truth, faithfulness measurement shifts focus to internal consistency with provided source material. Faithfulness evaluation assesses whether generated content accurately reflects and adheres to provided context, instructions, or reference documents, addressing a complementary dimension of hallucination that proves equally critical for deployment reliability. This evaluation paradigm addresses scenarios where models must synthesize, summarize, or reason over given information without introducing extraneous or contradictory content [126]. The measurement of faithfulness proves critical for applications where reliability depends on accurate representation of source material rather than parametric knowledge, such as document summarization, question answering with context, and dialogue systems with persona constraints.

Context-grounded verification methods assess faithfulness by systematically comparing generated content against source

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material using multiple complementary approaches.

Natural language inference models trained specifically for faithfulness detection achieve 74.4% balanced accuracy in identifying unfaithful content by treating the source as premise and generation as hypothesis, representing a 5% improvement over prior methods [137]. Advanced benchmarks for tool-augmented language models reveal significant challenges in maintaining factual grounding, with state-of-the-art models like Gemini-1.5-Pro achieving only 45.3 out of 100 on comprehensive hallucination diagnostics [95]. Semantic similarity metrics using contextualized embeddings measure alignment between source and generation, though traditional n-gram based approaches like ROUGE show weak correlation with human faithfulness judgments across dimensions due to their insensitivity to semantic paraphrasing and logical relationships [14]. Question-generation and answering consistency checks, where questions derived from the source must be answerable from the generation, demonstrate substantially higher correlation with human judgments of factual consistency compared to other automatic metrics, particularly on extractive summarization tasks [138].

Cross-document faithfulness evaluation addresses scenarios where models must synthesize information from multiple potentially conflicting sources. Advanced frameworks implement evidence weighting schemes that account for source reliability, recency, and consistency when evaluating faithfulness across multiple documents. When sources contain contradictions, faithfulness evaluation must assess whether models appropriately acknowledge conflicts rather than arbitrarily selecting or fabricating consensus. Contemporary evaluation frameworks employ sentence-level segmentation and pairwise scoring to enable fine-grained inconsistency detection across document boundaries in multi-document summarization [137]. Hierarchical faithfulness evaluation approaches assess alignment at multiple granularities—document-level, passage-level, and sentence-level—with recent methods demonstrating improved detection of subtle faithfulness violations through contextualized embedding comparisons [137].

Instruction faithfulness measurement evaluates whether models adhere to specific task instructions, constraints, and formatting requirements beyond content accuracy. This dimension proves critical for deployment scenarios where models must follow precise specifications for safety, style, or functional requirements. Large-scale evaluation benchmarks reveal that language models generate hallucinated content by fabricating unverifiable information in approximately 19.5% of user queries, highlighting systematic failures in instruction adherence [7]. Comprehensive hallucination evaluation frameworks assess models across multiple task types including question answering, knowledge-grounded dialogue, and text summarization, systematically identifying patterns of instruction violations through both automatic detection and human annotation [7]. Task-specific evaluation reveals that hallucination patterns vary substantially across different instruction types, with models showing particular vulnerability when synthesizing information from multiple sources or generating content about specific domains [7].

The interplay between faithfulness and informativeness creates evaluation tensions that frameworks must carefully balance. Highly faithful models that merely copy source material achieve perfect faithfulness scores while providing minimal value, necessitating metrics that reward appropriate abstraction and synthesis. The faithfulness-informativeness trade-off manifests differently across tasks: summarization requires high compression with maintained faithfulness, question answering demands selective extraction with contextual interpretation, and dialogue systems need dynamic balance based on conversational flow. Information-seeking dialogue evaluation demonstrates this tension, where existing benchmarks contain high rates of hallucination (71.4%) while attempting to maintain engaging and informative responses, necessitating cleaned datasets that preserve both faithfulness and conversational quality [126]. Faithfulness-aware training approaches that discriminate between faithful and hallucinated content demonstrate substantial performance improvements, achieving 12.8 F1 score gains on dialogue coherence benchmarks while maintaining response engagement and informativeness [126].

3) Estimate Hallucination Rate and Predict Safety: Moving from qualitative assessment to quantitative characterization, **hallucination rate estimation** provides statistical measures essential for risk modeling and deployment decisions. Accurate rate estimation enables probabilistic reasoning about system reliability and informs confidence calibration strategies. A novel approach employs conditional generative models to generate self-prediction questions about potential hallucinations in in-context learning scenarios [50]. The posterior hallucination rate (PHR) obtained through coherence checking correlates strongly with true hallucination rates (THR), particularly for smaller in-context datasets. This method's simplicity—requiring only response log probability measurements—combined with its accuracy makes prediction-based QA a promising direction for efficient hallucination evaluation without expensive ground-truth annotation.

informs deployment strategies and mitigation requirements. For instance, [6] introduces the Hallucinations Leaderboard, an open initiative that quantifies hallucination tendencies across diverse tasks like question answering and summarization, providing a standardized benchmark to compare model reliability and identify vulnerabilities in factuality and faithfulness.

To attribute the sources of hallucination and enhance safety predictions, [32] employs association analysis, which observes hallucination levels under specific risk factor values while examining each factor's contribution and statistical significance. This approach effectively excludes confounding effects, revealing deficiencies in core capabilities such as commonsense memorization and relational reasoning, thereby guiding targeted mitigations for safer model design.

Safety prediction then quantifies the potential harm from hallucinated content, transitioning from descriptive evaluation to risk assessment that directly informs deployment strategies and mitigation requirements. For instance, [6] introduces the Hallucinations Leaderboard, an open initiative that quantifies

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4) Evaluate Causal Reasoning Capabilities: The ability to distinguish correlation from causation fundamentally determines whether models can provide reliable explanations and make sound inferential decisions, which is fundamental to vertical assessment and interpretability. Causal reasoning evaluation transcends surface-level pattern matching to assess whether models understand underlying mechanisms that generate observed phenomena. This capability proves essential for interpretable AI systems [112]. At a fundamental level, such capabilities allow models not only to identify correlations but also to infer cause-and-effect relationships between variables [111]. Furthermore, causal reasoning offers a structured framework for explaining how specific inputs lead to particular outcomes, thus increasing the transparency and comprehensibility of the decision-making process for human users [112].

Causal inference abilities enable models to simulate the effects of hypothetical interventions. These abilities are particularly critical in domains such as healthcare, economics, and policy-making, where the uncovering of causal mechanisms can inform more effective interventions and lead to better outcomes [36]. Such capabilities allow models not only to generate predictions but also to understand and explain the basis of those predictions. Moreover, the incorporation of causal reasoning into artificial intelligence systems enhances their robustness, as models become more adaptable to novel and unseen scenarios by generalizing causal knowledge across diverse contexts [127].

5) Evaluate Interpretability and Actionable Insights: Interpretability is a cornerstone of the LENS framework (Section I-D), transforming evaluation from a measurement exercise into a diagnostic tool. Beyond simply scoring outputs, LENS's vertical assessment provides **actionable insights** by tracing errors to specific nodes in the reasoning tree.

This capability addresses the critical limitation of current frameworks identified in Section I-C, which lack **process transparency**. By leveraging its hierarchical decomposition, LENS generates causal attribution of errors to specific reasoning steps, enabling developers to understand not just *that* a hallucination occurred but *why* it happened. This leads directly to suggested mitigation strategies based on observed error patterns and informed model selection guidance for specific use cases, fulfilling the promise of actionable evaluation.

6) Measure Consistency: Consistency measurement, a core design principle in Section I-D, extends beyond single-output evaluation to assess the repeatability and coherence of model performance over time and across contexts. This dimension is critical for ensuring that a model's performance is reliable and not subject to random variation.

Traditional static question-answering datasets struggle to reliably assess LLMs' faithfulness to newly provided evidence, as data contamination risks enabling memorization-based performance instead of genuine evidence utilization. To address this, ReEval [26] introduces a dynamic evaluation framework that leverages adversarial machine learning principles. The method employs **prompt chaining** with cost-effective LLMs to perturb original evidence through two automated strategies—**answer swapping** (replacing gold answers) and **context enriching** (adding relevant information)—generating transferable adversarial test cases that trigger hallucinations across diverse LLMs, including GPT-4. Crucially, these synthetically generated examples exhibit **cross-model transferability**, enabling small models (e.g., Alpaca-7B) to create effective benchmarks for evaluating larger proprietary systems, thus offering a scalable and resource-efficient alternative to static benchmarks. While ReEval demonstrates promise, its current scope is limited to single-hop questions with short answers, leaving multi-hop or complex contextual perturbations unexplored [26].

The LENS framework's **tree-based decomposition** methodology inherently supports consistency checking by evaluating each node in the reasoning chain independently. This allows for the identification of inconsistencies not just between different responses, but within the internal logical steps of a single response, enabling true vertical assessment.

7) Assess Susceptibility to Adversarial Conditions: Understanding how easily models fall into hallucinations under adversarial conditions extends safety prediction into proactive vulnerability assessment. **Susceptibility** analysis systematically probes model weaknesses through controlled perturbations, revealing failure modes that standard evaluation overlooks. PhD (ChatGPT-Prompted visual hallucination evaluation Dataset) exemplifies this approach by evaluating MLLMs across five visual recognition tasks and four questioning modes: normal questions, questions with specious context, questions with incorrect context, and counter-common-sense images [120]. The framework addresses primary hallucination causes including visual ambiguity, image-text inconsistency, and conflicts with everyday knowledge. While models perform well on object recognition, they frequently fail on sentiment analysis and counting tasks, exhibiting sharp hallucination rate increases when encountering misleading context or unrealistic images. These systematic, context-driven errors demonstrate that hallucinations follow predictable patterns rather than occurring randomly, enabling targeted mitigation strategies.

8) Analyze Robustness to Impacting Factors: Robustness analysis, a key requirement from Section I-D, examines how models maintain performance stability across diverse operational conditions and perturbations. This evaluation dimension

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quantifies resilience to environmental variations that could trigger hallucinations, providing critical insights for deployment in uncertain real-world environments. Improving model robustness requires systematic examination of potential causal factors that contribute to hallucination risks. Input perturbations, such as textual rephrasing, semantic ambiguity, or image modifications, can trigger unstable model responses if the internal representations are insufficiently robust. Similarly, biases inherent in model parameters or pretraining data may amplify certain patterns, leading to overconfident yet incorrect outputs. In addition, inconsistencies or errors in knowledge sources further exacerbate hallucination risk. Consequently, a thorough understanding of these causal factors is essential for designing effective mitigation strategies.

Detecting and mitigating hallucination risks requires analyzing internal model states under controlled conditions through feedback-based mechanisms. Iterative self-verification and multi-model comparison exploit internal signals to identify unstable behaviors, with divergence in representations across varying input formulations serving as early hallucination indicators. Cross-modal and cross-phase analysis further detect inconsistencies between information sources or processing stages, enhancing identification reliability.

Robustness estimation quantifies susceptibility through systematic evaluation frameworks at multiple levels. Perturbation-based testing measures output variance, entropy, and divergence from reference answers across textual, visual, and multimodal input variations. Complementary cross-model divergence analysis reveals inconsistencies between different architectures or ensemble configurations under identical inputs, providing comprehensive vulnerability assessment.

This multi-tiered robustness analysis integrating causal factor identification, internal state monitoring, and quantitative estimation provides rigorous methodology for understanding hallucination risks. The approach enables proactive vulnerability identification and structured mitigation, informing development of AI systems with enhanced reliability through systematic evaluation pipelines that address failure modes before deployment.

9) Evaluate Tool Use and External Knowledge Integration: A key innovation of the LENS framework is its explicit evaluation of **Tool Necessity Detection and Selection (TND/TSA)** (Section I-F). This dimension assesses whether models correctly identify when external verification is needed and select appropriate tools, transforming evaluation from post-hoc error detection to proactive risk assessment.

This evaluation is seamlessly integrated into the tree-based decomposition process. Specific nodes in the reasoning tree can be designated to require external tool verification (e.g., a knowledge API retrieval, a calculator, a code executor). The framework then evaluates both the model's decision to use a tool and the correctness of the tool's output integration, closing a critical gap in current evaluation paradigms where models confidently generate information from parametric memory when they should instead consult external sources.

10) Computational Workload, Cost, and Latency: The practical deployment of comprehensive evaluation systems like LENS ultimately confronts computational resource constraints that determine feasibility and scalability, a challenge acknowledged in Section I-E. Understanding the trade-offs between evaluation thoroughness and computational demands becomes essential for designing economically viable systems that maintain acceptable safety margins. This dimension encompasses both the computational cost (the total resources required) and latency (the time required for evaluation), each posing distinct challenges for real-world deployment.

Fine-tuning LLMs for hallucination detection typically demands considerable computational resources, posing a significant barrier for many applications [44]. Existing test sets, such as **HaluEval** and **DelucionQA**, although providing valuable benchmarks, are heavily reliant on labor-intensive human annotation [71]. To mitigate these challenges, recent research has investigated various strategies. Employing smaller models for preliminary screening and utilizing transfer learning have been shown to effectively alleviate the computational burden [44]. A comprehensive evaluation of LoRA, RAG, and DoRA systems has highlighted LoRA's cost-efficient domain adaptation capabilities and DoRA's proficiency in balancing fine-tuning efficiency with model precision [44]. Cost-effectiveness can be maintained through batch processing, weighted sampling, counterfactual validation, and the utilization of complex question types [71]. Moreover, the transferability of adversarial examples in ReEval across different LLMs indicates that examples generated by smaller models can effectively evaluate larger models, enhancing the cost-effectiveness of this approach [26].

While computational cost constrains the scale of evaluation, latency determines real-time viability and user experience quality. The temporal dimension creates distinct challenges, as parallel processing cannot always overcome sequential dependencies inherent in hallucination detection pipelines. As large language models scale to hundreds of billions of parameters, the temporal overhead becomes increasingly prohibitive, with comprehensive evaluation pipelines requiring 10-100 times longer processing time than standard inference [92]. This latency challenge manifests across multiple dimensions: single-query evaluation time, batch processing efficiency, and end-to-end system responsiveness.

Recent architectural innovations attempt to mitigate latency through parallel processing and early-exit mechanisms. Speculative evaluation techniques predict likely hallucination points during generation, enabling selective verification, and the detection accuracy has also significantly improved.Cascade architectures employ lightweight initial filters processing in under 10ms to eliminate obvious non-hallucinations, reserving expensive deep verification only for ambiguous cases comprising approximately 15-20% of queries [2]. These optimizations achieve practical latency targets of 100-200ms for routine queries while preserving thorough evaluation capability for high-confidence failures.

The ReEval framework [26] demonstrates a paradigm shift

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in addressing evaluation latency through transferable adversarial generation. By leveraging smaller models like Alpaca7B to pre-generate adversarial examples that transfer to larger models, including GPT-4, this approach changing the economics of comprehensive hallucination assessment [26].

Hardware acceleration and model optimization techniques provide complementary latency reduction strategies. Quantization methods reducing precision from FP32 to INT8 can achieve approximately 4x speedup with accuracy degradation typically ranging from 1-3% [139], though performance varies across model architectures and tasks. Tensor parallelism across multiple GPUs enables near-linear scaling for batch evaluation, though single-query latency remains bounded by sequential dependencies in the evaluation pipeline. Knowledge distillation produces specialized detection models with substantial size reduction—up to 10x compression—while maintaining competitive performance, enabling deployment scenarios where model efficiency is critical [140].

The trade-off between latency and detection quality manifests differently across application domains. Medical hallucination evaluation benchmarks assess performance across clinical scenarios where accuracy is paramount, requiring comprehensive validation against medical databases and expert annotations to ensure reliability in healthcare applications [9]. Conversational AI systems face strict real-time constraints, with sub-200ms latency thresholds necessary to maintain natural dialogue flow [141], necessitating careful balance between detection thoroughness and responsiveness. Unified detection frameworks leverage auxiliary validation tools to assess hallucinations robustly across multimodal contexts, though computational resource allocation remains a key consideration for practical deployment [2].

LENS supports adaptive evaluation strategies that dynamically adjust computational depth based on query importance, available resources, and domain-specific risk tolerance, allocating more time to high-stakes decisions while maintaining acceptable average response times [2].

Computational efficiency thus emerges as a critical consideration constraining real-world deployment of comprehensive hallucination evaluation. While current mitigation strategies including transferable adversarial generation, cascade architectures, and hardware acceleration achieve order-of-magnitude improvements, the fundamental trade-off between evaluation thoroughness and computational resources persists. The evolution toward hybrid approaches combining offline preprocessing with selective online verification represents the most promising path forward, though achieving millisecond-scale comprehensive hallucination detection across all query types remains an open challenge requiring continued innovation in both algorithmic efficiency and hardware capabilities.

F. Mitigation-Aware Evaluation

Hallucination mitigation in LLMs and MLLMs encompasses diverse strategies with varying effectiveness across different contexts. In this section, we review four major paradigms of hallucination mitigation: retrieval-augmented generation (RAG), parameter-efficient fine-tuning (LoRA, DoRA), knowledge transfer mechanisms, and preference optimization with structured reasoning; we also analyze temporal intervention strategies.

1) RAG, LoRA, DoRA Performance Checks:

a) Retrieval Augmented Generation: RAG grounds model outputs in retrievable evidence. Prior work shows that RAG improves factuality across knowledge-intensive QA and dialogue tasks compared to non-retrieval baselines [142]–[144]. In [142], RAG is evaluated on knowledge-intensive datasets using dense retrieval over Wikipedia, such as DPR, to supply evidence passages to the generator; open-domain QA and fact verification settings include Natural Questions, WebQuestions, TriviaQA, and FEVER. Dialogue experiments, such as Wizard of Wikipedia, likewise report that retrieval augmentation reduces unsupported content and improves groundedness [143]. The evaluation of RAG effectiveness typically covers three aspects: retrieval precision and recall as well as retrieval accuracy; grounding quality, namely how well models incorporate and attribute external knowledge; and end-task factuality and hallucination outcomes relative to non-RAG baselines [142], [144].

Beyond task metrics, RAG-specific evaluators such as RAGAS provide task-agnostic measures for answer correctness, faithfulness to retrieved context, and context precision/recall, and are useful complements to retrieval and generation metrics [144], [145]. Evaluations should also probe position sensitivity in long contexts—"lost in the middle" effects—so that grounded information remains influential regardless of where it appears in the prompt [146]. Finally, adaptive retrieval strategies that trigger retrieval only when needed can achieve favorable precision/recall and latency trade-offs; comparisons should report factuality alongside retrieval cost under the gating policy [147].

The evaluation process for RAG systems has evolved to address unique challenges. Naive RAG implementations can suffer from low precision due to misaligned retrieved chunks and low recall from failure to retrieve all relevant information [144], [148]. Practical guidance highlights failure points spanning source curation, chunking and query formation, retriever configuration and reranking, and generation-time integration, motivating phase-wise diagnostics. Modern evaluation frameworks assess RAG performance across four critical phases: data source quality, query formulation, retriever effectiveness, and generation strategy [144]. Hallucination-aware pipelines combine RAG with preference optimization such as DPO, where hallucination descriptions are used to construct preference datasets; such combinations can further reduce hallucinations relative to RAG alone [149], [150].

b) Low Rank Adaptation and Weight Decomposition: LoRA introduces trainable rank decomposition matrices while keeping original model weights frozen, enabling efficient fine-tuning with minimal memory overhead [151]. In practice, LoRA attains performance close to full fine-tuning while updating only a small fraction of parameters, depending on the chosen ranks and target modules [151]. Common config-

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urations use low ranks such as 4–16 and apply adapters to attention projections such as query and value, and/or to MLP blocks depending on the task [151].

DoRA, introduced in 2024, decomposes pretrained weights into magnitude and directional components, fine-tuning both via low-rank directional adaptation [152]. Reported results show DoRA can outperform standard LoRA variants on a range of tasks while introducing no inference-time overhead because decomposed components merge back into the pretrained weights after training [152].

For multimodal Large Language Models, empirical evaluations indicate that adapter- and connector-based methods are strong baselines; LoRA and IA3 are effective when applied to both attention and MLP layers, and fine-tuning connector layers can further enhance performance [153]. These findings underscore the importance of architecture-aware evaluation when assessing mitigation effectiveness [153].

2) Knowledge Distillation and Feedback Learning: Beyond parameter-efficient fine-tuning, knowledge transfer mechanisms offer complementary mitigation pathways. The transition from parameter optimization to knowledge transfer represents a shift in how models acquire hallucination-resistant behaviors, moving from weight modification to behavioral inheritance [140].

a) Knowledge Distillation for Hallucination Reduction: Knowledge distillation has emerged as a technique for transferring hallucination-aware behaviors from high-capacity teacher models to compact student models. Recent empirical studies report that distillation can reduce hallucinations while retaining most task performance [154]. The evaluation methodology typically involves temperature scaling during distillation to produce soft targets, intermediate layer matching to preserve reasoning pathways, and assessment of exact-match accuracy versus semantic correctness [140].

Evaluation frameworks often consider three dimensions: factual consistency—alignment between student and teacher on factual queries; reasoning preservation—whether logical chains remain intact; and hallucination propagation—whether teacher biases transfer. Chain-of-thought distillation aims to transfer not just final answers but also reasoning traces; in multimodal settings, structured reasoning can improve transparency and error detectability [130].

b) Feedback Learning and Self Reflection Mechanisms: Feedback learning introduces iterative refinement cycles where models learn from their own mistakes and external corrections. Self-RAG enables models to retrieve, generate, and critique through self-reflection; reported gains on open-domain QA come from combining retrieval with learned self-critique signals that guide whether to retrieve and how to revise [155]. The evaluation of feedback learning systems requires temporal assessment: initial generation quality, self-critique accuracy, and improvement rates across iterations.

The unified hallucination detection framework (UNIHD) presents a comprehensive approach for multi-granularity evaluation across different modalities, achieving 81.91% accuracy at the claim level and 84.60% at the segment level on MHaluBench [2]. This framework operates at both sentence-level for capturing global semantic coherence and entity-level for handling local factual accuracy, enabling more nuanced assessment of hallucination patterns [2].

3) Hallucination-Aware DPO Training & Chain-of-Thought: Moving from knowledge transfer to preference-based optimization, DPO and CoT represent complementary approaches that reshape how models generate and validate their outputs. While DPO aligns model behavior with human preferences through contrastive learning, CoT enforces step-by-step logical consistency that makes hallucinations more detectable and correctable.

a) Direct Preference Optimization Training: DPO directly learns from preference pairs reflecting hallucination severity, bypassing the complexity of reward model training in traditional RLHF. Hallucination-targeted DPO and related approaches report reductions in hallucinations across a range of generative tasks [150], [156]. The evaluation of DPO systems typically assesses preference alignment accuracy, generalization to unseen domains, and computational efficiency compared to full RLHF pipelines [149].

Vision-guided DPO addresses the unique challenge of visual hallucinations in MLLMs, where language priors can override visual evidence. V-DPO enhances visual context learning by constructing preference pairs that penalize responses inconsistent with visual input. Reported results show reductions in object hallucination by 52% on the CHAIR metric and improved grounding accuracy by 38% on RefCOCO benchmarks when preferences emphasize visual grounding [157]. The effectiveness stems from V-DPO's ability to calibrate the balance between language model priors and visual evidence.

Recent research demonstrates that compact language models can achieve competitive hallucination mitigation through targeted fine-tuning approaches. The Honest AI framework shows that smaller models can be effectively trained to express uncertainty and reduce hallucinations by learning to say "I don't know" when appropriate, with hybrid RAG and fine-tuning approaches achieving top performance on the CRAG benchmark for uncertainty-aware response generation [158]. This approach challenges the assumption that larger models are always superior for hallucination detection tasks, highlighting the potential of parameter-efficient methods in achieving robust performance with reduced computational overhead.

b) Chain of Thought and Structured Reasoning: Chain-of-Thought prompting transforms opaque model outputs into interpretable reasoning chains, making hallucinations visible at intermediate steps. Studies show CoT reduces factual errors by 52% on multi-step reasoning tasks and can surface inconsistencies that aid correction [23], [130]. The assessment of CoT effectiveness requires analyzing both final answer accuracy and intermediate-step validity, as models can reach correct conclusions through flawed reasoning.

Chain-of-Preference Optimization combines CoT with DPO by constructing preference pairs at each reasoning step. Using Tree of Thoughts search, CPO categorizes intermediate thoughts as preferred or dispreferred based on their inclu-

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sion in successful solution paths. This granular preference learning enables models to self-correct during generation, significantly reducing cascading errors through step-wise optimization [159]. Evaluation frameworks for CPO assess path consistency, step-wise accuracy, and hallucination propagation across the reasoning chain.

The integration of DPO with CoT creates a synergistic effect: DPO ensures outputs align with factual preferences while CoT exposes the reasoning process for verification. Recent benchmarks show this combination substantially reduces hallucinations on complex reasoning tasks requiring both factual accuracy and logical consistency [149], [159]. However, evaluation also reveals limitations: DPO requires substantial computational resources for fine-tuning, and improvements remain dataset dependent, with performance varying across domains.

4) Pre-Intervention vs. Post-Hoc Correction: The temporal dimension of mitigation introduces a fundamental architectural decision: whether to prevent hallucinations before they occur or correct them after generation. This choice profoundly impacts system design, computational requirements, and achievable accuracy levels, and recent research suggests that effective systems often combine both approaches [27], [67].

a) Pre-Intervention Strategies: Pre-intervention methods modify the generation process before inference begins, establishing guardrails that prevent hallucinations from emerging.

LLMs Augmenter builds evidence chains before generation, creating structured knowledge paths that guide response formation. By pre-constructing factual scaffolding, this approach can reduce hallucinations while improving interpretability, as evaluators can trace claims back to source evidence [160]. The evaluation methodology assesses evidence quality, chain coherence, and coverage completeness across generated claims.

Visual evidence prompting for MLLMs represents an emerging pre-intervention strategy, where carefully designed cues prime models to attend to relevant image regions and external evidence; reported evaluations on vision-language benchmarks demonstrate reduced hallucination and improved grounding [161]. Assessment can include analyzing grounding quality, consistency between referenced evidence and outputs, and attention/focus patterns.

b) Post-Hoc Correction Mechanisms: Post-generation correction allows models to produce initial outputs then systematically identify and fix hallucinations. Visual Contrastive Decoding integrates vision-language signals during decoding by contrastively favoring tokens consistent with visual evidence and has been shown to reduce object hallucinations without retraining [162].

The evaluation of post-hoc methods requires frameworks that assess detection accuracy, correction precision, and preservation of valid content during revision. OPERA penalizes overtrust in language priors and reallocates attention to visual evidence, helping flag and correct problematic outputs [163].

c) Hybrid Temporal Strategies: Modern systems increasingly combine pre-intervention and post-hoc approaches, leveraging their complementary strengths. Pre-intervention methods establish strong baselines while post-hoc correction catches residual errors [161]–[163].

The optimal balance depends on application constraints. Real-time systems favor lightweight post-hoc corrections that add minimal latency, while accuracy-critical applications employ comprehensive pre-intervention with multiple correction passes. Training-free methods have gained prominence due to their deployment flexibility: techniques like visual contrastive decoding (achieving 51% reduction in object hallucinations) [162], context-aware decoding (enabling hallucination reduction through evidence-based constraints) [164], and zero-resource hallucination detection (achieving 83.21 AUC-PR in sentence-level detection) require no model modification, enabling immediate deployment across diverse architectures [92].

Temporal evaluation frameworks now assess not just final accuracy but also computational budgets, latency requirements, and the convergence behavior of iterative correction. In practice, benefits from multiple correction passes can diminish, motivating stronger pre-intervention to improve initial generation quality.

5) Uncertainty-Aware Abstention and Calibration: Abstention-based mitigation aims to avoid committing to potentially hallucinated outputs when uncertainty is high. Conformal abstention provides coverage guarantees for selective prediction and can reduce harmful errors by allowing the model to say "I don't know" under distributional uncertainty; evaluation typically reports risk–coverage and abstention–utility curves [165]. Uncertainty-based abstention methods similarly improve safety by calibrating confidence estimates and abstaining on low-confidence instances; evaluations consider expected calibration error, area under the risk–coverage curve, and downstream accuracy conditional on non-abstention [166].

For detection-driven mitigation, semantic entropy scores can flag responses likely to be hallucinated, enabling either rejection, revision, or targeted retrieval before finalizing outputs [94]. Mitigation-aware evaluation should therefore include calibration metrics (e.g., ECE), risk–coverage analyses, and utility trade-offs to capture how abstention policies balance reliability against answer rate.

a) Evaluator Choice and Reliability: Mitigation studies frequently rely on automatic judges. While LLM-as-a-judge methods such as G-EVAL and critique-based judges can correlate with human assessments, their reliability varies across tasks and prompt formulations [167], [168]. Whenever possible, prefer verifiable benchmarks (e.g., ERBench for entity–relationship claims) and automatically checkable tasks, and report human agreement alongside automatic scores [100]. Reference-free evaluations have known limitations and can mis-rank systems [169]. Community efforts like the Hallucinations Leaderboard and human-in-the-loop auditing frameworks provide additional transparency and comparability for mitigation-aware evaluations [6], [31].

b) Synthesis and Future Directions: The landscape of mitigation-aware evaluation reveals a maturing field where diverse strategies—from retrieval augmentation and parameter-

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efficient fine-tuning through knowledge transfer to temporal interventions—converge toward comprehensive solutions. The progression from isolated techniques to integrated frameworks reflects a growing understanding that hallucination mitigation requires multi-faceted approaches tailored to specific deployment contexts. RAG and parameter-efficient methods like LoRA and DoRA provide accessible entry points for resource-constrained environments [144], [151], [152]. Knowledge distillation and feedback learning enable behavioral transfer without full retraining, while DPO and CoT reshape the generation process itself. The temporal dimension adds another layer of sophistication, with hybrid pre- and post-intervention strategies increasingly adopted in practice [161]–[163].

Current evaluation frameworks increasingly recognize that effective mitigation requires context-aware metrics: computational efficiency for deployment scenarios, temporal consistency for real-time applications, and domain-specific accuracy for specialized use cases. The field moves toward standardized benchmarks that assess not just hallucination rates but also preservation of model capabilities, interpretability of corrections, and robustness across diverse inputs. As models grow more sophisticated, evaluation must evolve correspondingly from binary hallucination detection to nuanced assessment of factual grounding, logical consistency, and contextual appropriateness. This comprehensive evaluation paradigm ensures that mitigation strategies genuinely enhance model reliability rather than merely gaming specific metrics.

G. Tools and Frameworks

After introducing the core challenges in hallucination evaluation and the innovative LENS framework, this section systematically examines existing tools and frameworks. Current studies have proposed diverse methodologies—from automated dataset generation to manual annotation, and from traditional metrics to LLM-based evaluators—that significantly enhance evaluation scalability and comprehensiveness. These frameworks integrate multidimensional metrics, multimodal fusion, and feedback loops to capture and mitigate hallucinations. The following subsections analyze their design principles, benchmark construction requirements, and comparative strengths and limitations.

The development of evaluation tools and frameworks represents a critical evolution from ad hoc testing to systematic methodology that fundamentally reshapes how we understand and measure hallucinations. The field has matured through three distinct paradigm shifts: (1) From manual curation to automated dataset generation, reducing creation costs while increasing scale [71]; (2) From binary classification to multidimensional assessment, capturing nuanced failure modes across factual, logical, and contextual dimensions; and (3) From isolated evaluation to integrated feedback loops, enabling continuous improvement through iterative refinement.

Typically, the evaluation workflow consists of three phases: Dataset Construction**, involving automated data generation pipelines, multi-tier annotation protocols, and quality control mechanisms; Benchmark-based Evaluation**, which employs advanced metrics, cross-benchmark validation, and human–AI collaborative assessment; and Result Comparison**, which synthesizes results across benchmarks to identify consistent patterns, discrepancies, and areas for improvement.

Some datasets have a clear pipeline for creation, such as the Data Collection Pipeline of HalluQA [70], the Construction Pipeline of HaluEval-Wild [42], and the generation pipeline of ONIONEVAL [58]. We have summarized some of these pipelines, analyzing their advantages and disadvantages.

Finally, the interplay between **LLMs and human experts** plays a pivotal role in shaping benchmark quality, cost, and robustness. In the **Dataset Construction** phase, we examine the degree of automation and human involvement in data collection, annotation, and verification. In the **Evaluation** phase, we analyze the automation level of evaluation methods, such as the increasing use of LLMs as judges, and emphasize the importance of human feedback for refining and validating evaluation benchmarks.

1) Framework Design and Benchmark Construction for Model Evaluation: To further elaborate on evaluation tools and frameworks, this section explores their design principles and construction mechanisms, offering a comprehensive understanding of how these frameworks have evolved and what key components determine their effectiveness.

a) Framework Design for Model Evaluation: The design of frameworks marks a shift from informal, ad hoc testing to systematic, comprehensive methodologies. This progression is characterized by three key paradigm shifts: (1) Transition from manual data curation to automated dataset generation, significantly reducing costs while exponentially increasing dataset scale [71]; (2) Move from simple binary judgments to multidimensional assessment that captures nuanced failure modes across factual accuracy, logical consistency, and contextual relevance; (3) Integrate evaluation processes into feedback loops that allow continuous refinement and improvement [63]. These shifts facilitate the development of robust, scalable, and precise evaluation frameworks that better reflect the complexities of hallucination phenomena in language models.

b) Benchmark Structure: Building upon these design paradigms, established benchmarks typically comprise four essential components: datasets, annotations, baselines, and metrics. Together, they form a comprehensive and reproducible architecture for assessing model performance.

- **Dataset** Contemporary benchmarks employ sophisticated data generation pipelines, such as those in HalluQA [70] and HaluEval-Wild [42], to automate large-scale data collection and annotation. These pipelines incorporate multi-tier annotation protocols and automated quality control, enhancing both efficiency and reliability. The resulting datasets are designed to capture a diverse range of hallucination types and failure modes, enabling a comprehensive evaluation.
- **Annotation & Answer** The annotation process typically combines automated labeling with human oversight. While automated tools provide scalability, human annotators deliver nuanced judgments for ambiguous or context-

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sensitive cases. Large language models are increasingly employed as judge agents, offering consistent and scalable assessments. Human feedback remains indispensable for refining annotation quality, and Reinforcement Learning with Human Feedback (RLHF) [63] is widely adopted to align LLM behavior with human preferences and improve evaluative reliability.

- **Baseline** Baselines encompass both traditional metrics—such as accuracy and F1 scores—and more advanced measures targeting factual consistency, logical coherence, and contextual appropriateness. Recent research further explores the use of LLMs as baseline evaluators, serving as proxies for human judgment and enabling rapid, cost-effective benchmarking across models and datasets.
- **Metrics** Modern evaluation metrics extend beyond simple correctness to multidimensional scoring schemes that assess factuality, logicality, relevance, and robustness. Cross-benchmark validation ensures generalizability, while human-AI collaboration, such as RLHF [63], further improves the metric reliability. Techniques such as automated fact-checking [29], image consistency evaluation [59], and question-answering accuracy [170] are integrated into the evaluation pipeline.

Despite these advancements, most existing benchmarks remain static and general-purpose, relying on fixed public datasets (e.g., Wikipedia, academic corpora) to serve as community-wide standards. While such universality facilitates broad comparison, it creates a gap for practitioners seeking to evaluate models against organization-specific knowledge bases—such as internal documentation, financial reports, or proprietary datasets. The absence of customizable frameworks constrains the ability to identify domain-specific hallucination risks early in deployment. Therefore, an ideal evaluation ecosystem should not only offer static benchmarks but also provide methodological toolkits for dynamically constructing user-centric, context-aware benchmarks. Such flexibility is essential for ensuring reliable and safe deployment of generative AI systems across specialized and enterprise domains.

2) Requirement for Current Dataset Construction: Evaluating hallucinations in large-scale language and vision-language models requires datasets that extend beyond conventional benchmarks. Existing resources often fail to capture the nuanced and multifaceted nature of hallucination, particularly in open-ended and culturally diverse contexts. To address these limitations, new dataset construction must follow principled requirements that ensure both reliability and generalizability. Specifically, datasets should mitigate human bias, align closely with human judgments, and reflect the complexities inherent in subjective evaluation. In addition, they must incorporate culturally sensitive test cases, account for the reasoning processes underlying model outputs, and provide systematic coverage of both free-form and constrained generation tasks. The following subsections elaborate on these requirements in detail.

a) Generalization Across Diverse Contexts: Wu et al. [20] demonstrate that reliance on manually curated, often adversarial, examples introduces human bias and limits the scale and diversity of test cases. Their automated approach, AUTOHALLUSION, shows that hand-crafted datasets tend to over-represent certain "corner-case" hallucinations while under-representing others that are more prevalent in real-world scenarios. As a result, models optimized on these benchmarks may perform well in artificial settings yet fail to meet user expectations in practice. For example, a model might avoid factual errors on standard datasets but produce culturally inappropriate or contextually irrelevant responses in domain-specific applications such as law or medicine. Therefore, future dataset construction should emphasize generalization through scalable, automated methods that probe a wider range of failure modes and incorporate multidisciplinary understanding of linguistic subjectivity and cultural context [49], ensuring robust and representative evaluations.

b) Generalization Across Time: To ensure that evaluations of hallucinations in large-scale language and vision language models are robust and reliable over time, datasets must be designed to capture the temporal dynamics of model behavior. This involves assessing model performance at various time points to understand how hallucinations may evolve or persist. For example, models might exhibit different tendencies to hallucinate under varying conditions such as system updates, data refreshes, or changes in user interaction patterns. By conducting evaluations at regular intervals or during specific events, researchers can identify whether certain types of hallucinations are transient or persistent. This temporal assessment is crucial for developing models that maintain consistent performance and for creating benchmarks that reflect real-world scenarios where models are subject to continuous updates and evolving data environments.

c) Generalization Across Different Model Sizes: Effective hallucination evaluation requires consideration of models with varying sizes, from small language models (sLLMs) to large language models (LLMs). Each model size has distinct characteristics that influence its tendency to hallucinate. sLLMs, with fewer parameters, while performing better in knowledge analysis tasks, struggle with complex reasoning and nuanced language, leading to more frequent hallucinations in ambiguous contexts [58]. In contrast, LLMs, with their vast parameter counts, can handle a wide range of tasks but may still hallucinate in highly ambiguous scenarios or when required knowledge is not well-represented in their training data. To address these challenges, datasets must include test cases tailored to the capabilities of each model size. For sLLMs, simpler and more focused test cases are necessary, while LLMs require more complex, multi-faceted scenarios. By designing scalable test cases and using consistent evaluation metrics, researchers can ensure that evaluations are reliable and applicable across different model sizes, ultimately leading to more robust and trustworthy language models.

d) Generalization Across Different Modalities: As LLMs and LVLMs increasingly integrate multiple modalities, the evaluation of hallucinations must also encompass these diverse forms of input and output. For example, in the case of

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LVLM, hallucinations can manifest in both the textual and visual domains. A model might generate an image that does not accurately reflect the accompanying text or produce text that misdescribes an image. To address this, datasets should include multimodal test cases that require models to reason across different types of data. This involves creating scenarios in which the models must generate coherent and accurate responses based on visual and textual inputs. By evaluating models in their ability to integrate information from multiple modalities, researchers can identify and address hallucinations that arise from the interaction between different types of data. This is particularly important in applications such as image captioning, visual question answering and cross-modal retrieval, where accurate cross-modal reasoning is crucial for model performance. The exploration of cross-modality causality and model internals in reflecting hallucination under various circumstances remains an active and important area of research [133]. Researchers are continuously working to understand how different modalities interact and influence each other, as well as how the internal mechanisms of models contribute to the generation of hallucinations.

e) Generalization Across Different Regions and Cultures: Given the global deployment of LLMs, it is imperative that datasets reflect the diverse regional and cultural contexts in which these models operate. This is especially relevant for large Chinese language models, where cultural sensitivity is paramount. Datasets should include test cases that are culturally relevant and contextually appropriate for different regions. This means incorporating cultural references, idiomatic expressions, and social norms that are specific to the target audience. For example, a high-quality bilingual or multilingual dataset is needed in a comprehensive evaluation tool [40]. A model trained on data from one region may not perform well when deployed in another region with different cultural practices and linguistic nuances. By including culturally sensitive test cases, researchers can better assess how models perform in different cultural contexts and identify potential biases or hallucinations that may arise from cultural misunderstandings. This approach not only enhances the fairness and inclusivity of model evaluations but also ensures that models are more robust and reliable when used in diverse cultural settings.

f) Mitigating Human Bias and Enhancing Adversarial Robustness: Current hallucination evaluation datasets face two major challenges: human bias and adversarial vulnerability. Most benchmarking relies on manually annotated examples. Generating these examples is costly and time-consuming [20]. They are subject to human bias stemming from annotators' subjective assumptions. This narrows the coverage of the dataset and distorts the evaluation outcomes. Furthermore, fine-tuning on such biased datasets risks overfitting, where models perform well on carefully selected test cases, but do not provide meaningful insight into their actual weaknesses [20]. Current datasets rarely consider adversarial robustness. Adversarial prompts or input perturbations can easily exploit blind spots in datasets, indicating hidden vulnerabilities that remain undetected under standard evaluations. This means that high benchmark performance does not necessarily translate into reliability when the model encounters more challenging or adversarial conditions [20].

g) Alignment with Human Judgments: Current summarization evaluation datasets and the metrics benchmarked on them often fail to represent the nuanced process of human judgment adequately. Firstly, a pronounced class imbalance, wherein a disproportionately large number of summary annotations are labeled as "faithful," biases metrics towards predicting correctness. This phenomenon compromises their capacity to detect subtle hallucinations, a core component of human evaluation [67]. Secondly, numerous high-performing metrics exhibit a preference bias, favoring summaries generated by large language models (LLMs)—particularly those resembling the scoring LLM's outputs—over human-written counterparts. This occurs even when human annotators clearly prefer the latter [67], [169]. This suggests that datasets containing such model outputs may inadvertently reinforce this bias rather than genuinely reflect human-centric assessment. Moreover, benchmarks mainly composed of outputs from older, fine-tuned models are ineffective for assessing performance on modern LLM-generated content, thus creating a generalization gap [67], [171]. Lastly, most metrics provide only a single, monolithic numerical score, which lacks the fine-grained, interpretable justification humans offer when pinpointing precisely which fact is inconsistent and why [67]. This absence of explainability impedes the alignment of metric evaluations with the detailed fact-by-fact verification process of human judges.

Human judgment in evaluating hallucinations extends far beyond binary true/false distinctions, encompassing a multidimensional interplay of cultural, contextual, and subjective factors. As demonstrated by [49], human evaluators bring deeply ingrained cultural frameworks, ethical values, and contextual expectations to the assessment process, leading to divergent interpretations of the same model output across different sociocultural contexts. For instance, responses deemed factually accurate in one cultural setting may be perceived as misleading or inappropriate in another due to variations in normative beliefs, historical narratives, or ethical priorities. This inherent subjectivity challenges the feasibility of universally applicable technical metrics for hallucination detection, as the very definition of hallucination is often shaped by human perceptual and cultural lenses rather than objective ground truth. Consequently, effective evaluation demands a holistic approach that integrates fact-checking rigor with nuanced contextual awareness, cross-cultural sensitivity, and an acknowledgment of the often ambiguous boundaries between factual error and culturally situated interpretation. These complexities underscore the critical need for human-AI collaborative frameworks that complement automated detection with culturally informed human oversight, particularly for applications deployed in global or multicultural contexts.

h) Incorporating Cultural Sensitivity: The *Culturally Sensitive Test (CST)* proposed by [49] presents a paradigm shift in hallucination evaluation, moving beyond a rigid binary

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definition of factuality. It argues that hallucinations are not merely deviations from an objective truth but often stem from a misalignment between the model's output and the subjective, culturally-grounded expectations of human users. This perspective exposes a critical requirement for dataset construction: benchmarks must incorporate cultural, societal, and political contexts to truly assess a model's reliability in real-world scenarios. The CST framework, building on the Reflective Cognition Principle (RCP), formally categorizes queries into three types: **(1) Raw Facts** (verifiable against databases), **(2) Interpretations of Facts** (akin to opinions, requiring alignment with human cultural norms), and **(3) Unverifiable Hypotheses** (where a non-committal response is ideal). This taxonomy necessitates that future hallucination evaluation datasets are not monolithic but are structured to include a balanced mixture of these categories. Specifically, it mandates the creation of test cases for "Interpretations of Facts" where the ground truth is defined by predominant human consensus or culturally-specific values rather than a static factoid. Consequently, constructing such datasets requires a nuanced approach: it relies on human annotators from diverse backgrounds to establish context-dependent judgments and challenges the notion that purely automated, algorithmic methods can suffice for comprehensive evaluation. By integrating the principles of CST, benchmarks can better detect a broader class of failures, ensuring that models are evaluated not only on what they say but also on how appropriately they say it within a given cultural milieu, thereby fostering the development of truly trustworthy and globally deployable AI systems.

i) Revealing the Underlying Thought Process: A fundamental limitation of prevailing benchmarks, whether constructed manually or through automated processes, is their inadequacy in evaluating the model's underlying *thought process*—the chain of reasoning that leads to a generated output. Traditional evaluations primarily focus on the veracity of the final answer, treating the model as a black box and neglecting to assess whether the intermediate logical steps are coherent, consistent, and factually grounded. This superficial approach fails to discern if a correct answer stems from a genuine understanding of causal relationships and logical structures, or merely from statistical correlations and coincidental guesswork. Consequently, it becomes impossible to diagnose precisely where in the reasoning pipeline a hallucination originates—be it a flawed initial retrieval, an erroneous inference, or an incorrect synthesis of information.

The ERBench framework [100] introduces a novel methodology to address this critical gap by leveraging the structured nature of relational databases and their inherent integrity constraints. It systematically decomposes complex questions into multi-hop queries interconnected by functional dependencies (FDs) and foreign key constraints (FKCs). This design allows for automatic verification not only of the final output but also of the intermediate reasoning steps. For instance, by utilizing FDs, ERBench can pinpoint exact attributes that must be invoked correctly during the reasoning process, thereby exposing whether the model's internal rationale aligns with grounded evidence. This provides unprecedented visibility into the model's cognitive pathway, enabling the detection of logical missteps and false inferences that traditional benchmarks would overlook. Thus, ERBench transforms evaluation from a mere outcome-based check into a process-oriented analysis, essential for developing models that exhibit robust and trustworthy reasoning capabilities.

j) Comprehensive Evaluation of Free-form and Constrained Generation: A comprehensive benchmark must evaluate hallucinations across both free-form and constrained tasks. Within datasets created for the evaluation of large vision-language models (LVLMs), it is essential to establish guidelines that include processes for identifying and evaluating hallucinations in both free-form and constrained generation tasks. Free-form generation tasks involve open-ended outputs where the model produces descriptions or attributes that may not be directly derivable from the input image, potentially leading to fictitious outputs [60]. In contrast, constrained generation tasks occur within contexts that have a limited answer space, such as binary or multiple-choice questions (e.g., "Is there a traffic light in the image?").

Current evaluation benchmarks, such as POPE [60], primarily focus on hallucinations in constrained generation tasks—incorrect model responses to fact-related questions with predefined answers—while largely ignoring hallucinations in free-form generation. Empirical analyses demonstrate that mitigating hallucinations in constrained tasks does not reliably translate into reductions of hallucinations in free-form tasks, and the two phenomena may even exhibit anti-correlation [60].

To ensure a comprehensive and valid performance evaluation of LVLMs, datasets must represent hallucinations in both free-form and constrained generation tasks. This requires the development of evaluation frameworks that can effectively assess model performance across these diverse task types, capturing the nuances of hallucinations in each context. By doing so, we can gain a more holistic understanding of the model's capabilities and limitations, ultimately leading to more robust and reliable LVLMs.

3) General vs. Special Domain Evaluation: After outlining the essential requirements for constructing robust datasets, we now turn our attention to the nuances of evaluating hallucinations across different domains. Hallucination evaluation is inherently context-dependent, varying significantly between general-purpose and domain-specific settings. This distinction is crucial because each domain entails distinct requirements for accuracy, safety, and knowledge representation. Recognizing these differences is essential for developing evaluation frameworks that are both comprehensive and contextually appropriate. The following sections examine general-domain benchmarks, specialized-domain benchmarks, and cross-domain transfer evaluations, emphasizing their respective characteristics, challenges, and methodologies.

a) General Domain Benchmarks: HELM [136] evaluates language models across diverse tasks such as question answering, summarization, and other user-facing applications, using

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standardized multi-metric protocols to address inconsistent evaluation practices. Its innovation lies in a holistic approach that jointly measures accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency. The framework benchmarks 30 prominent models ranging from 350M to 175B parameters on 42 scenarios drawn from datasets including MMLU, TruthfulQA, and CNN/DailyMail [136]. HELM reveals significant performance variation across tasks, with larger models demonstrating higher accuracy, particularly in knowledge-intensive scenarios. Despite its breadth, HELM acknowledges limitations in covering specialized domains and subtle phenomena, underscoring the need for future extensions.

b) Domain-Specific Benchmarks: Med-HALT [1] focuses on the critical safety requirements of medical AI applications, where hallucinations can lead to life-threatening consequences. The benchmark introduces a dual-task framework: Reasoning Hallucination Tests (RHT) for logical medical reasoning and Memory Hallucination Tests (MHT) for factual medical knowledge recall. Med-HALT integrates seven datasets (e.g., MedMCQA, MedQA, HeadQA, PubMed), comprising 18,866 RHT samples and 4,916 MHT samples. Evaluation results reveal that even advanced models like Llama-2 70B achieve only around 42.21% accuracy on reasoning tasks, underscoring significant challenges in reliable medical reasoning [1].

Similarly, FinanceBench [128] presents 10,231 questions derived from 361 public filings for open-book financial QA, revealing weaknesses in numerical reasoning and factual grounding across models, though it does not explicitly assess real-time market data. These findings highlight the persistent difficulty of ensuring factual reliability in high-stakes domains such as medicine and finance.

c) Cross-Domain Transfer Evaluation: HalluQA [70] evaluates hallucination patterns in Chinese large language models, providing a benchmark beyond English-centric evaluations. The framework's main innovation lies in its adversarial question design spanning multiple domains, covering Chinese historical culture, customs, and social phenomena. HalluQA contains 450 meticulously crafted adversarial questions and considers two types of hallucinations—imitative falsehoods and factual errors—constructed based on GLM-130B and ChatGPT to target powerful pre-trained and conversational models. Experiments across 24 large language models reveal that 18 achieved non-hallucination rates lower than 50%, indicating the benchmark's challenging nature [70]. Such cross-domain evaluations provide valuable insights into model generalization across culturally and linguistically diverse contexts.

4) Dataset Construction phase: The evolution of dataset construction methodologies shows a shift from human-centric to hybrid human–AI approaches. Traditional manual curation attains high accuracy but is time-consuming [4]. Modern pipelines leverage LLMs for data generation while retaining human validation, reducing time while preserving reliability through expert annotations [70].

Essentially, the Dataset Construction phase includes: acquiring raw data; creating questions and answers supported by verifiable evidence; conducting annotations before or after generation; performing post-processing; and verifying the dataset through human review to ensure accuracy and reliability.

a) Obtaining Raw Data: In this phase, the first step is collecting raw data. Some datasets integrate multiple existing databases; others collect large volumes of data from various sources. Some generate data using large language models, while others enlist human volunteers to assist in data generation.

- **Integrating from Previous Datasets:** Datasets such as Med-HALT and a novel industrial spatio-temporal benchmark integrate prior datasets or real-world records. Med-HALT constructs a diverse, multinational medical examination dataset with two categories of tests—Reasoning Hallucination Tests (RHT) and Memory-based Hallucination Tests (MHT)—to assess problem-solving and information-retrieval abilities in medical contexts. It spans multiple specialties and includes testing modalities such as False Confidence Tests and None-of-the-Above Tests [1]. Similarly, the industrial spatio-temporal benchmark obtains Vienna traffic incident records (2013–2023) via OpenAPI under a Creative Commons Non-Commercial 4.0 International License. It covers 14 categories—including Faulty Vehicles, Track Damages, Signal Faults, Rescue Operations, Police Interventions, Traffic Accidents, and Demonstrations—with 99,869 distinct incident records, providing a cross-lingual benchmark for assessing LLM robustness in industrial spatio-temporal domains [64].
- **Inspiration and Pattern Collection:** When constructing HalluQA, researchers search the Internet for inspiration and raw data so tested models encounter adversarial patterns drawn from real-world misconceptions [70]. They also study prior benchmarks to extract hallucination and prompting patterns for creating hallucination-prone questions [70].
- **Self-Generated Data:** Hallucination evaluation benchmarks use several methods to generate data for ground-truth answers and hallucination-prone questions. One approach asks humans to write descriptions for collected images; these captions serve as self-generated data to support accurate responses and challenging questions [54]. In **VISCON**, visual concepts are extracted from images and divided into objects, attributes, relations, and descriptions to form a rich concept database [172].
  In addition, researchers may enlist volunteers for human-to-human or human-to-computer dialogues; these dialogues are further processed to generate questions and answers that test models' ability to avoid hallucinations. Furthermore, refined, decomposed methods are applied to self-generated data. For instance, **AUTOHALLUSION** uses generative models to create scene sets and object sets, which are then used to produce hallucination-prone questions [20]. Similarly, the **OnionEval** benchmark leverages knowledge graphs to generate entities, which are then used to create questions designed to test models'

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susceptibility to hallucinations [100].

b) Dataset QA Construction: For hallucination datasets used in evaluation, indispensable components are questions that trigger model hallucinations, along with corresponding ground-truth answers and supporting evidence. Below, we discuss different QA construction strategies by modality and granularity.

- **General QA construction:** General QA construction targets sentence-level and dialogue-level LLM hallucination evaluation. The first step is acquiring raw data and identifying relevant patterns from existing resources. HaluEval draws inspiration from hallucination and prompt patterns found on the internet so that models encounter adversarial patterns from real-world misconceptions [70]. Researchers also learn from prior benchmarks that identify hallucination-prone question structures, with interrogative patterns triggering more hallucinations than declarative forms [70].
  The second step is generating hallucination-prone questions. Humans can directly create such questions—for example, graduate interns can be hired to author knowledge-based items [70]. Alternatively, humans can use large language models to assist: one can compose question candidates likely to induce hallucinations based on existing QA pairs, craft six QA pairs, and then use an LLM to generate the QA content [70].
  The third step is screening high-quality QA pairs with LLMs. One method feeds QA pairs to existing LLMs and generates multiple responses, then screens by hallucination rate. For instance, HalluQA includes a question if the tested LLM answers incorrectly at least three times out of five; otherwise the question is discarded [70]. Another approach trains an LLM classifier specifically for screening QA pairs by hallucination rate [100], which may perform better in its domain. Some past benchmarks lack rigorous screening and validation, leading to QA pairs that lack credibility and fail to simulate real-time scenarios, and thus are insufficiently challenging.
- **Entity-level QA Construction:** Prior methods rely on manually constructed benchmarks by human annotators, which are costly and not scalable [100]. Unlike the general QA pipeline, entity-level construction starts from atomic facts or entity–relationship structures. Using knowledge graphs and the entity–relationship model makes it possible to decompose sentences into entities and generate QA from them, enabling more complex, logically structured questions. A key advantage of knowledge graphs is automatic verification and updatability, improving reliability and scalability. The entity–relationship–based process performs better than previous methods in generating high-quality QA that is intricate enough to simulate real-world tasks [100].
  OnionEval includes 19 categories and 515 entities, comprising 3,356 QA questions designed collaboratively by GPT-4 and human annotators [58]. The dataset is generated by first illustrating atomic facts and then gradually layering context. OnionEval differs from general approaches by beginning with entity categorization and sampling. Atomic facts are extracted by integrating generated entities with the assistance of large language models, during which knowledge graphs are employed to generate atomic facts. The remaining steps—using large language models to generate hallucination-prone questions and conducting a second round of human validation—follow the general pipeline [58].
  Another notable benchmark is **ERBench**, which is based on the entity–relationship model [100]. ERBench leverages the structured nature of relational databases to create a comprehensive, automatically verifiable benchmark. Questions can be constructed from ER diagrams; two integrity constraints—functional dependencies (FDs) and foreign key constraints (FKCs)—enable verification. FDs allow inference of an attribute's value from others, supporting binary and multiple-choice questions that ask for inferred values. FKCs ensure foreign-key values in one relation exist as primary keys in another, enabling multi-hop questions by joining related tables and inferring longer FDs spanning them.
  Moreover, ERBench is highly extensible in data, modality, and prompting [100]. It can be updated as the underlying database changes; it supports multimodal questions where attribute text values can be replaced with other data types such as images; and it can be diversified via chain-of-thought, few-shot prompting, and knowledge augmentation. This flexibility allows ERBench to evaluate improved LLMs effectively.
  The dataset QA construction pipeline evolves over time, reflecting progress in understanding hallucination mechanisms and efforts to build cost-effective, scalable, and robust evaluation tools. We are highly interested in subsequent progress in this field.
- **QA Construction for LVLM:** Unlike sentence-level or entity-level hallucination datasets derived from raw text, the QA construction pipeline for LVLMs starts from images. In this case, visual content often needs to be recognized and converted into text to participate in evaluation.
  For binary QA, the **ChatGPT-Prompted visual hallucination evaluation Dataset (PhD)** constructs QA from a task-specific item collection process. Items are selected from images and ranked by their ability to trigger hallucinations. Subsequently, ChatGPT generates binary, hallucination-prone questions based on the task, subject, and selected item, each assigned a "yes"/"no" answer [120].
  For more complex QA, a stronger framework is needed. **AUTOHALLUSION** constructs an LVLM hallucination benchmark by acquiring and generating ground-truth data (used as answers) and using this data as the basis for generating raw material for misleading questions [20].
  To construct questions likely to induce LVLM hallu-

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cination, the framework creates existence questions by asking the victim model to generate an image caption and then prepending words to the question to inject language priors. Another approach is to ask about information omitted in image captions; existence questions containing such missing information are more likely to induce hallucinations. For spatial-relation questions, the construction method asks from multiple levels to increase difficulty [20].

The dataset QA construction pipeline evolves over time, reflecting advances in understanding hallucination mechanisms and attempts to create cost-effective, scalable, and robust evaluation tools. We are highly interested in subsequent progress in this field.

c) Annotation Before And During QA Generation: Multiple annotation strategies are employed before and during QA generation to ensure the quality and coverage of generated question–answer pairs. These include evidence perturbation, fine-grained categorization, entity-level annotation, caption annotation, and specialized annotations for prediction questions, each designed to capture different dimensions of potential model hallucinations. The following subsections elaborate on these methods and clarify their roles within benchmark design.

- **Evidence Perturbation:** Evidence perturbation systematically modifies existing evidence to create new evaluation datasets. It perturbs either the answer span or the context to generate test cases more likely to trigger hallucinations in LLMs.
  The method is implemented through **ReEval**, which provides two primary techniques for synthesizing evaluation datasets. The first one is answer swapping. In this technique, the original answer within the evidence is replaced with another valid answer, while the remaining context is kept intact. This method tests the model's ability to recognize and adapt to changes in the answer while maintaining the original context [26]. The second one is context-enriching. This technique involves adding more relevant information to the provided document while retaining the original supportive information. This method tests the model's ability to integrate new context without losing track of the original evidence [26].
  The implementation of Evidence Perturbation is facilitated through prompt chaining with LLMs [26]. Specifically, LLMs are used to generate new test cases that are designed to trigger hallucinations. By leveraging the capabilities of LLMs, this method ensures that the generated test cases are both relevant and challenging, effectively probing the model's robustness and reliability.
- **Categorization:** LLMs that score well on standard sentence-level metrics do not necessarily hallucinate less; they exhibit different strengths and weaknesses, and behave differently on generated questions versus real-world scenarios. To address this, some benchmarks incorporate categorization. **DiaHalu** [17] and **HaluEval-Wild** [42] both adopt fine-grained categories. DiaHalu covers four multi-turn dialogue domains and five hallucination subtypes (extended from factuality and faithfulness) [17]. HaluEval-Wild begins with a human-defined taxonomy and then performs automatic categorization with GPT-4. These efforts help evaluations better reflect real-world challenges and domain-specific contexts.
- **Entity-level Annotations:** Entity-level annotation employs knowledge graphs or entity–relationship (ER) diagrams to extract entities and relations from sentences or dialogues, advancing from unstructured to structured evaluation.
  In **OnionEval**, atomic facts are extracted by integrating generated entities with assistance from LLMs, and knowledge graphs are used to generate atomic facts. Converting triples into simple factual questions—or using existing QA datasets—has been effective for basic factuality but may miss complex relations and contexts, yielding questions that lack depth and nuance, especially where relations are subtle or context-dependent [58].
  In the context of entity-level annotations, the **ERbench** framework exemplifies the construction of verifiable questions through the use of Entity-Relationship (ER) diagrams. This framework leverages two key integrity constraints: functional dependencies (FDs) and foreign key constraints (FKCs). FDs allow for the inference of an attribute's value based on other attribute values within the same relation, enabling the construction of both binary and multiple-choice questions that query the inferred values [100]. FKCs, on the other hand, ensure referential integrity by verifying that foreign key values in one relation correspond to the primary key values in another relation, thereby identifying and linking related records across different tables. This constraint is crucial for maintaining the coherence and consistency of the dataset, ensuring that questions constructed within this framework are grounded in verifiable facts [100].
- **Caption Annotations:** For instance, in constructing a hallucination evaluation dataset for large vision-language models, images are first annotated with descriptive captions. These captions serve as the basis for the answers in the QA pairs and also provide the raw material for generating questions that are likely to induce hallucinations. In the **AUTOHALLUSION** framework, generative models create scene and object sets, and samples from existing datasets are used to form the ground truth component [20]. These annotations are crucial as they provide the necessary structure and content for the subsequent generation of questions, ensuring that the questions are relevant and targeted.
  Annotation Before QA Generation involves transforming visual information into textual descriptions and further breaking down these descriptions into granular concepts. This advanced annotation practice is exemplified by several notable datasets. For instance, the PhD dataset utilizes TDIUC Annotation, which meticulously converts image content into detailed textual descriptions and sub-

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sequently dissects these descriptions into fundamental concepts [120]. Similarly, VISCON employs Visual Concept Extraction, a method that systematically identifies and isolates key visual concepts from images [172]. Additionally, the AMBER dataset showcases robust annotation techniques that effectively break down image information into core concepts [30]. These approaches not only provide rich textual representations of visual content but also enable the generation of highly targeted and conceptually grounded questions, thereby enhancing the depth and precision of subsequent QA pair creation.

- **Annotations for Predict-Questions:** By using a conditional generative model (CGM) to create prediction questions and responses, PHRs are obtained through a consistency check. The ability to evaluate hallucination through predictive questions and responses reflects a growing understanding of the mechanisms behind hallucination [50].

d) Annotation After QA Generation: Following the generation of question-answer pairs, subsequent annotation stages are critical to establishing the quality and utility of the dataset. This phase focuses on enriching the generated content with detailed annotations that capture various facets of information, from direct evidence to nuanced classifications and specific contextual attributes. The following subsections introduce these methods in detail, outlining their respective functions to ensure the robustness and reliability of the benchmark.

- **Annotations for Unanswerable Questions: Unanswerable Math Word Problem (UMWP)** is a dataset with unanswerable questions. UMWP dataset is composed of four existing MWP datasets, namely SVAMP, MultiArith, GSM8K, and ASDiv. During the integration process into a unique benchmark dataset, two data annotators were tasked with modifying the original questions to make them unanswerable [83].
- **Post-Processing:** To address the challenge of label imbalance and prioritize non-trivial instances during annotation, an iterative model-in-the-loop strategy is used. This approach involves multiple rounds of annotation, in which a hallucination detection model is iteratively trained on existing annotated data and subsequently used to select the most informative unlabeled instances for the next annotation cycle [79]. Initially, a baseline model (e.g., BERT-based) is trained on a small seed set of annotated examples. After each round, the model's predictions on unlabeled data are used to filter out instances that are either too easy (high-confidence predictions) or too ambiguous (low-confidence predictions), thereby focusing annotators' efforts on challenging cases that are most beneficial for model improvement [79].

  To further mitigate label skew, a subsampling mechanism is applied to balance the distribution between hallucination and non-hallucination labels [79]. The subsampling ratio is dynamically adjusted based on the model's estimated precision, recall, and true negative rates from previous rounds. Additionally, semantic similarity metrics (e.g., cosine similarity using RoBERTa embeddings) are computed to filter out near-paraphrase replacements, which are less likely to represent meaningful hallucinations [79]. This combination of model-driven selection and heuristic filtering ensures that the annotated dataset remains diverse, balanced, and enriched with high-value examples.

  By reducing annotation redundancy and prioritizing informative samples, the iterative model-in-the-loop approach significantly improves the efficiency and quality of the dataset construction process.

e) Human Verification: Human verification plays a crucial role in ensuring the accuracy and reliability of hallucination detection in evaluation datasets. Given the complexity and subtlety of identifying hallucinations, automated methods alone may not suffice. Human experts provide a critical layer of validation that can discern nuances and contextual inconsistencies that automated systems might overlook.

Since the categorization process mentioned above is challenging for large language models and may not meet human needs, both **DiaHalu** and **HaluEval-Wild** incorporate rigorous human verification to ensure dataset accuracy and reliability [17], [42]. Automated categorization, while efficient, can fail to capture linguistic and contextual nuances, leading to potential inaccuracies and biases.

To mitigate the potential for hallucinations, the categorization process of HaluEval employs a rigorous validation process in which experts manually verify the classification of queries to ensure their quality [42]. The ChatGPT-Prompted visual hallucination evaluation Dataset (PhD) incorporates meticulous human verification steps following the selection of high-interest items, the generation of item-embedded QA pairs, and the creation of false content [120]. These successive verification stages are meticulously designed to ensure the integrity and reliability of the dataset, providing a robust benchmark to evaluate the propensity of LVLMs to generate hallucinations. By subjecting each phase of dataset construction to rigorous human scrutiny, the PhD dataset mitigates the risk of inaccuracies and biases that could otherwise compromise the validity of the evaluation results.

f) Free-form Generation Scenarios: The dataset construction mentioned above is all based on explicit task requirements and reliable references. However, we will also face the evaluation in free-form generation scenarios quite often.

The inherent characteristics of *free-form text generation* pose particular challenges for the construction of the associated datasets. In such open-ended tasks, the frequent lack of reliable reference texts fundamentally limits the applicability of traditional reference-based methods for recognizing hallucinations [79]. Furthermore, evaluations performed at the sentence-level or dialogue-level can only provide coarse-grained judgments that are insufficient for the needs of fine-grained error localization or real-time intervention [79]. It may be difficult to even know where to obtain the reference, as obtaining it may be as hard as generating consistent information

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in the first place [79]. Furthermore, evaluations performed at the sentence-level or dialogue-level can only provide coarse-grained judgments—often merely deciding whether a sentence or entire document is hallucinated—which is insufficient for fine-grained error localization or real-time intervention [79].

Consequently, the design of the dataset should support *reference-free detection of token-level hallucinations* to enable dynamic localization of erroneous content during the generation process and ensure that the evaluation framework is directly aligned with the practical requirements of free-form generation scenarios [79].

g) Limitations of Current Dataset Construction:

- **Human Annotation Limitations:** Human annotators, whether they are volunteers or experts, introduce subjectivity and variability while annotating. This is particularly problematic in the context of hallucination detection, where the boundary between factual and hallucinated content can be nuanced. It can often result in human bias issues [20].
- **Automated Annotation Limitations:** Automated methods, such as those using LLM as generators or classifiers, may fail to capture the full context and subtleties of human language. This can result in the generation of questions and answers that do not accurately reflect real-world scenarios or fail to trigger meaningful hallucinations. Automated annotation processes may lead to imbalanced datasets, where certain types of hallucinations are overrepresented while others are underrepresented. This can skew the evaluation results and provide a biased assessment of the performance of LLMs.
- **Time and Labor cost Limitations:** The construction of high-quality datasets for evaluating hallucinations in LLMs is often constrained by significant time and labor costs. Ensuring consistency among all annotators is particularly challenging and time-consuming [17]. The process of training annotators, developing detailed annotation guidelines, and conducting multiple rounds of annotation and verification requires substantial effort and resources [20]. This not only increases the time required for dataset construction but also adds to the labor cost, making it difficult to scale up the dataset size and complexity.
- **Comprehensive Evaluation Limitations:** Current datasets often focus on coarse-grained evaluations (e.g., sentence or dialogue-level), which are insufficient for fine-grained error localization. Token-level hallucination detection is more challenging and requires more sophisticated methods. Evaluating hallucinations in free-form generation scenarios is particularly difficult due to the lack of reliable reference texts [60]. Traditional reference-based methods are not applicable, and new approaches are needed to dynamically localize erroneous content during generation.
- **Entity-Level and Relationship-Based Limitations:** While knowledge graphs and entity-relationship models have improved the structure and verifiability of datasets, they can still miss subtle relationships and contextual dependencies. This can result in questions that are too simplistic or fail to capture the complexity of real-world interactions [100]. Entity-based methods may not generalize well to new domains or contexts, requiring significant effort to adapt and expand the datasets.
- **Iterative Model-in-the-Loop Approach Limitations:** The iterative training of hallucination evaluation models can introduce biases, especially if the initial seed set is not representative. This can lead to the selection of non-representative or trivial instances for annotation. The dynamic adjustment of subsampling ratios and heuristic filtering mechanisms requires careful tuning and validation to ensure that they effectively balance the dataset and select high-value examples [79].
- **Human Verification Limitations:** Rigorous human verification processes, while necessary, are resource-intensive and can be a limiting factor in the scalability of dataset construction. Ensuring consistent human verification across different datasets and tasks is challenging, especially when dealing with large volumes of data.
- **Real-time prevention Limitations:** The need for real-time prevention during the generation process to dynamically localize and correct hallucinations is not well-addressed by current datasets and evaluation frameworks [79].

5) Benchmark-based Evaluation Phase: The evaluation of hallucinations is a multifaceted process that encompasses several distinct phases, each with its own set of methodologies and objectives. Building upon the foundational Dataset Construction phase, which focuses on the creation and refinement of high-quality datasets, the subsequent Benchmark-based evaluation phase emerges as a critical stage for systematically assessing hallucination in LLMs. This phase is characterized by the implementation of rigorous evaluation frameworks that leverage diverse datasets, advanced performance metrics, and both automated and human evaluation methods. To explore this methodology further, the following analysis examines how different frameworks utilize datasets, handle domain-specific challenges, incorporate automated and human evaluation, and measure performance through diverse metrics. Specifically, this phase is dissected into four key dimensions: Dataset Utilization Strategies, Performance Metrics, LLM as Judge, and Human Involvement. Each of these dimensions plays a crucial role in ensuring a comprehensive and robust evaluation of hallucinations in LLMs.

a) Dataset Utilization Strategies: Effective hallucination detection requires not only well-designed benchmarks but also thoughtful strategies for dataset construction and application. Two primary approaches are commonly adopted: direct evaluation and contrastive comparison.

- **Direct Evaluation Protocols:** The HaluEval (Hallucination Evaluation benchmark for Large Language Models) framework [7] adopts an automatic two-stage *sampling-then-filtering* procedure to generate hallucinated samples.

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It instructs a model (e.g., ChatGPT) to produce diverse hallucination instances, which are then filtered and annotated. This automated process effectively addresses the demand for large-scale, human-verified datasets [7].

TruthfulQA [4] evaluates models by comparing their responses against both truthful and misleading reference answers. It focuses on adversarial questions designed to elicit false beliefs or common misconceptions, thereby testing whether models replicate human biases. The benchmark contains 817 carefully crafted questions across 38 categories—covering domains such as health, law, finance, and politics. Results show GPT-3 achieves 58% truthfulness compared to 94% for humans [4]. However, as a static benchmark, TruthfulQA struggles to adapt to evolving knowledge and cannot capture partially correct or nuanced responses [4].

b) Performance Metrics: Effective hallucination measurement requires multiple complementary metrics, such as Factuality, Faithfulness, Granularity, Reasoning, Consistency, Reliability, Robustness, Safety, Computational Cost, Efficiency Trade-offs, and Uncertainty Scoring. Emerging metrics for Generalization Assessment, Scalability, and Automatic Degree are also crucial for a comprehensive evaluation. Given the detailed introduction of the Dimensions and Metrics of Evaluation in Section ref sec3.2:eval-dimensions-metrics, some of these metrics will be reintroduced in the following text, and emerging future directions will be proposed and discussed.

- **Factuality Metrics:** Factuality metrics assess the accuracy of the information generated by the model. Traditional classification metrics, such as precision, recall, and F1-score, provide foundational measurements but face limitations in capturing the complexity of hallucinations. FActScore [93] introduces atomic accuracy, decomposing text into individual facts and measuring correctness at the atomic level. While addressing coarse-grained evaluation, these metrics treat partial correctness identically to complete fabrication.
- **Faithfulness Metrics** Faithfulness metrics evaluate how well the model's output adheres to the input context and source material. This dimension is particularly critical for retrieval-augmented systems, where hallucinations often emerge as unfaithful extrapolations from retrieved context. Faithfulness evaluation examines whether model outputs accurately reflect the information contained in the provided source material, including an explicit evaluation of Tool Appropriateness. This focuses on two key aspects: Tool Necessity Detection Accuracy, which measures whether models recognize when external verification is needed, and Tool Selection Adequacy, which assesses the appropriateness of chosen verification sources.

  A notable challenge arises from the phenomenon of confident confabulation, where models generate plausible-sounding but fabricated citations or facts that may pass surface-level verification checks. Recent advances in attribution analysis integrated with vertical tracing highlight the scale of this issue: some of the responses that appear to be grounded contain subtle distortions or unsupported inferences that escape detection by traditional metrics [95]. Complementary citation accuracy studies reveal further limitations, showing that models can easily incorrectly attribute information, generating citations pointing to non-existent sources and misrepresenting the cited content [167]. SUMMAC [137] introduces consistency scores for summarization tasks, ensuring that summaries accurately reflect the source content. This is particularly important in domains like news summarization and academic writing, where fidelity to the original text is essential.
- **Fine-grained Metrics:** Fine-grained evaluation metrics address the challenge of assessing complex, multi-faceted responses where hallucinations vary in severity and type. FACTOR [116] introduces a hierarchical scoring system that evaluates hallucinations at multiple granularities. Using the LongFact dataset with long-form responses, FACTOR measures claim-level precision, span-level faithfulness, and dependency-aware scoring [116]. These metrics reveal that models achieving high sentence-level accuracy may achieve lower claim-level precision, exposing hidden hallucinations.
- **Reasoning Metrics:** Reasoning metrics, encompassing both vertical reasoning traceability and causal reasoning capabilities, are essential for evaluating the logical coherence and causal understanding of language models. Vertical reasoning traceability metrics track the logical flow of the model's reasoning process, ensuring that conclusions are supported by a logical sequence of steps. This is vital for tasks requiring multi-step reasoning, such as solving complex problems or making evidence-based decisions. Vertical reasoning traceability enables layer-by-layer dissection of inference chains, identifying where reasoning fails and enhancing diagnostic resolution. It operationalizes the "vertical assessment" principle as described in Table I for the LENS framework (Section I-F). Through tree-structured decomposition, complex queries are broken into atomic verification units (nodes) linked by logical dependencies (edges). This structured dissection enables evaluators to localize the origin of hallucinations, whether stemming from a faulty premise, flawed synthesis, or cascading error propagation. Thereby informing targeted mitigation strategies.

  Complementing vertical reasoning traceability, causal reasoning metrics assess the model's ability to understand and generate content that reflects causal relationships. This is important for tasks that involve predicting outcomes based on actions or events. Models that can accurately reason about cause and effect are more reliable in scenarios where decision-making is based on understanding the consequences of actions. Causal reasoning constitutes a distinct and indispensable dimension of hallucination evaluation. It assesses whether a model can correctly infer causal relationships, differentiate them

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from mere spurious correlations, uncover underlying generative mechanisms, and simulate the consequences of hypothetical interventions [111], [112]. Mastery of causal reasoning is especially vital in high-stakes domains such as medicine, law, or policy-making, where misattributing correlation as causation can lead to erroneous conclusions and harmful downstream effects [36].

Together, vertical reasoning, traceability, and causal reasoning capabilities provide a comprehensive evaluation of the model's reasoning processes. While vertical reasoning traceability focuses on the logical structure and coherence of the reasoning process, causal reasoning ensures that the model's conclusions are based on accurate causal inferences. Future research should aim to integrate these metrics into a unified framework, enabling more robust and reliable evaluations of language models. This includes developing methods for causal attribution within reasoning trees, enhancing the precision of logical flow tracking, and creating scalable and automated evaluation tools. By advancing these areas, researchers can create more trustworthy and effective model development and deployment strategies.

- **Reliability Metrics:** Reliability metrics assess the consistency and predictability of hallucination behavior across repeated evaluations, providing confidence bounds essential for production deployment decisions.

  Cross-context reliability evaluation demonstrates dramatic consistency drops across domains [63]. Population-stratified analysis further reveals high reliability disparities across demographic groups, with non-English speakers and users with limited technical literacy experiencing disproportionately higher hallucination rates. The framework's innovation lies in its adaptive reliability scoring that adjusts thresholds based on detected distribution shift, maintaining specified error bounds even as input characteristics evolve.
- **Robustness Metrics:** Robustness metrics evaluate the model's ability to handle noisy or adversarial inputs without a degradation in performance. A robust model can maintain its accuracy even when faced with challenging or unexpected inputs. This is important for applications where the model may encounter real-world data that is not perfectly clean or structured. Recent studies have shown that hallucination rates can increase significantly under targeted attacks, with models showing particular vulnerability to authority bias and other manipulation techniques [114], [115]. Distribution shift analysis also reveals that models trained on Wikipedia exhibit higher hallucination rates on social media text [63], highlighting generalization challenges.

  Paradoxically, calibration often degrades with model scale: larger models become more confidently wrong, necessitating explicit uncertainty training and challenging assumptions about the relationship between capability and reliability [94].
- **Safety Metrics:** Safety metrics in hallucination evaluation quantify the potential for generated content to cause harm through misinformation, enabling risk-stratified assessment across deployment contexts.

  Measuring the tendency of LLMs to produce hallucinations is a critical step in predicting their safety for real-world deployment, as it directly assesses the risk of generating unreliable or harmful outputs. For instance, [6] introduces the Hallucinations Leaderboard, an open initiative that quantifies hallucination tendencies across diverse tasks like question answering and summarization, providing a standardized benchmark to compare model reliability and identify vulnerabilities in factuality and faithfulness.

  To attribute the sources of hallucination and enhance safety predictions, [32] employs association analysis, which observes hallucination levels under specific risk factor values while examining each factor's contribution and statistical significance. This approach effectively excludes confounding effects, revealing deficiencies in core capabilities such as commonsense memorization and relational reasoning, thereby guiding targeted mitigation for safer model design.
- **Computational Cost and Efficiency Trade-offs Metrics:** Computational cost and efficiency metrics evaluate the trade-offs between the model's performance and its resource requirements. Efficient models can be deployed more easily and at scale, making them more practical for real-world use, especially in resource-constrained environments or applications requiring real-time processing.

  However, comprehensive hallucination evaluation often demands significantly more computation than standard inference, revealing fundamental tensions between detection quality and practical feasibility.

  Efficiency optimization strategies attempt to balance cost and quality through selective evaluation. Adaptive sampling methods dynamically adjust evaluation depth based on detected uncertainty, reducing computational requirements while maintaining high correlation with exhaustive evaluation [94]. Cascaded evaluation pipelines use lightweight filters to screen obvious cases, reserving expensive deep evaluation for ambiguous instances [2].

  Moreover, large language model evaluation costs scale superlinearly with model size. The high cost of LLM evaluation creates evaluation inequality, where only well-resourced organizations can afford rigorous testing, potentially leading to the deployment of inadequately evaluated models by smaller entities. Shared evaluation infrastructure and benchmark pooling initiatives aim to democratize access, though coordination challenges and competitive concerns limit adoption [116].
- **Uncertainty Scoring Metrics:** Uncertainty scoring metrics quantify the model's confidence in its predictions, which is crucial for reliable decision-making. Recent advancements in this area include semantic entropy and conformal prediction frameworks. Semantic Entropy [94] improves upon traditional token-level uncertainty by clus-

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tering semantically equivalent outputs before computing entropy.

Techniques like Bayesian neural networks and ensemble methods also play a significant role in estimating uncertainty, enhancing the model's reliability in various decision-making processes.

- **Degree of Automation Metrics:** The degree of automation metrics evaluates the extent to which the evaluation process can be automated, thereby reducing the reliance on human intervention. This is crucial for enhancing the efficiency and scalability of the evaluation process. While automated evaluation methods can significantly save time and resources, they often face limitations in capturing the nuances and complexities inherent in human judgment. Human expertise is indispensable for evaluating subtle hallucinations that arise from contextual understanding, domain-specific knowledge, and subjective quality assessments, which are difficult to formalize algorithmically.

  Many studies have highlighted the cost and effect of human involvement in evaluation processes [25], [30], [170]. The expertise of humans can enhance the scalability and credibility of hallucination evaluations. Strategic human involvement can compensate for the inherent limitations of automated systems while maintaining practical feasibility. However, the human-in-the-loop strategies, while yielding high-quality hallucination labels, come at a significant cost in terms of time and labor.

  Future directions should focus on developing more sophisticated automated tools that can handle complex and nuanced evaluations, thereby reducing human bias and increasing the scalability of the evaluation process. This includes leveraging advanced techniques such as automated annotation and benchmarking to minimize the need for manual intervention. By enhancing the degree of automation, we can achieve a more efficient and reliable evaluation process, ensuring that the evaluation remains both practical and robust.
- **Scalability Metrics:** Scalability metrics assess the model's ability to handle increasing amounts of data and complexity. Scalable models can be trained and deployed on large datasets, making them more versatile and useful in a variety of applications. Techniques like distributed training and model parallelism help in achieving scalability. Future work should aim to develop frameworks that can efficiently handle large-scale datasets and models, ensuring that evaluations remain feasible and effective even as model sizes and data volumes grow.
- **Generalization Assessment Metrics:** Generalization assessment metrics are designed to evaluate a model's ability to perform well on unseen data, ensuring that the model is not overfitting to the training data and can generalize to new scenarios. This is a critical aspect of model evaluation, as it directly impacts the model's robustness and reliability in real-world applications. Current techniques, such as cross-validation and domain adaptation, help in assessing and improving generalization. However, these methods often fall short in providing a comprehensive evaluation across various dimensions.

  Existing generalization metrics typically focus on specific aspects of model performance, such as accuracy on a validation set or consistency across different training folds. While these metrics are useful, they often fail to capture the full spectrum of generalization capabilities. Specifically, they do not adequately address generalization across different contexts, times, regions, cultures, modalities, and model sizes. This results in an incomplete understanding of how well a model can adapt to diverse and unseen scenarios.

  To achieve a more comprehensive assessment, future research should focus on developing advanced generalization metrics that can capture the model's performance across multiple dimensions:

  1) **Generalization Across Different Contexts**: Metrics should evaluate how well a model generalizes to new contexts ,including different domains, tasks, and input types. This requires techniques that can simulate a wide range of scenarios and measure the model's adaptability.
  2) **Generalization Across Times**: Models should be evaluated for their ability to maintain performance over time, accounting for changes in data distributions and evolving real-world conditions. This involves longitudinal studies and time-series analysis to assess temporal robustness.
  3) **Generalization Across Model Sizes**: Models of different sizes may exhibit different generalization behaviors. Metrics should assess how well a model generalizes across various model sizes, ensuring that evaluations are applicable and reliable regardless of the model's scale.
  4) **Generalization Across Different Modalities**: With the increasing use of multimodal models, it is essential to evaluate how well a model generalizes across different input modalities, such as text, images, and audio. This requires developing metrics that can capture cross-modal generalization capabilities.
  5) **Generalization Across Different Regions and Cultures**: Cultural and regional differences can significantly impact model performance. Metrics should assess how well a model generalizes across diverse cultural and regional contexts, ensuring fairness and inclusivity.

  By addressing these dimensions, researchers can develop more comprehensive and robust generalization metrics, leading to more reliable and effective model development and deployment.

c) LLM as Judge: The LLM-as-Judge paradigm enables large-scale, consistent evaluation by using language models to assess other models' outputs. While it greatly improves scalability over human evaluation, it also introduces systematic biases and self-referential risks. Since both evaluator and

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evaluatee may hallucinate, their interaction creates recursive uncertainty, raising questions about the objectivity and reliability of automated assessments.

- **Self-Evaluation Capabilities:** G-EVAL [167] exemplifies the LLM-as-Judge paradigm by employing GPT-based evaluators with chain-of-thought (CoT) reasoning and a form-filling paradigm to assess the quality of generated texts. By generating intermediate evaluation steps, the framework transforms monolithic judgments into structured assessments guided by explicit evaluation criteria. On the SummEval benchmark, G-EVAL-4 achieves a Spearman correlation of 0.514 with human judgments, substantially surpassing previous reference-based and neural evaluators and showing higher human correspondence [167]. However, analysis reveals a potential issue: the evaluator shows a bias toward LLM-generated summaries, consistently assigning higher scores even when human judges prefer human-written ones. Such bias may lead to self-reinforcement, as evaluation scores risk aligning with the model's own generation criteria rather than true task objectives, introducing uncertainty in reliability and epistemic validity [167]. These findings underscore both the promise of scalable, structured evaluation and the need for further research to better understand evaluator behavior, reduce inherent bias, and establish trustworthy baselines independent of model self-assessment.
- **Judge Model Selection:** LLMAuditor introduces a systematic and generalizable two-phase framework for auditing large language models that combines automated probing with human-in-the-loop (HIL) validation [31]. In the first phase, a dedicated model (e.g., Mistral 7B) generates diverse yet semantically aligned probes using a structured prompt template iteratively refined through human evaluation to ensure relevance and diversity. In the second phase, the audited models (Falcon 7B, Llama 2 7B, and GPT-3.5-Turbo) respond to the validated probes, and their outputs are analyzed for inconsistencies, bias, and hallucinations. Using the TruthfulQA dataset, LLMAuditor evaluates probe quality and factual consistency with metrics such as BERTScore, BLEURT, ROUGE-L, and GPT-Judge. Results show that LLMAuditor yields higher-quality probes and more reliable audits than baseline and automated methods like DSPy, and achieves comparable hallucination detection to Adatest++ while offering greater automation, transparency, and reduced manual effort [31]. These findings validate LLMAuditor as an effective, scalable, and human-verified approach for assessing the reliability and factual alignment of LLM outputs [31].

d) Human Involvement: Human evaluation establishes ground truth and validates automated systems but faces inherent trade-offs between quality, scale, and cost.

- **Annotation Protocols:** HALU-J [168] introduces a critique-based framework to enhance hallucination detection. Its multi-step pipeline—evidence categorization, reordering, detailed analysis, and aggregation—enables more reliable detection in multiple-evidence settings [168]. HALU-J is trained and evaluated on ME-FEVER, a dataset derived from FEVER with 3,901 synthesized multiple-evidence instances (2,663 for training and 1,238 for testing) [168]. The 7 billion parameter model demonstrates strong performance in multiple-evidence hallucination detection, outperforming GPT-4o in label prediction accuracy and evidence matching, while matching its capability in critique generation and evidence selection [168]. HALU-J addresses key limitations in existing approaches by selecting pertinent evidence and providing detailed, interpretable critiques for fine-grained hallucination evaluation [168].
- **Human-AI Collaboration:** LLMAuditor [31] implements a human-in-the-loop framework for auditing LLMs through a two-phase process of probe generation and probe answering. In Phase 1, probes are generated by Mistral 7B and validated by human annotators using structured criteria for relevance and diversity [31]. The framework integrates a structured prompt template and standardized evaluation criteria in the probe generation phase, ensuring verifiability and transparency through human-in-the-loop (HIL) validation [31]. In Phase 2, multiple models including Falcon 7B and Llama 2 are audited using the TruthfulQA dataset through systematic probe answering [31]. This HIL-enabled auditing pipeline provides a scalable and generalizable method for identifying inconsistencies and hallucinations across models with improved reliability compared to fully automated approaches [31].
- **Quality Control Mechanisms:** FActScore [93] ensures evaluation reliability by decomposing text into atomic facts and verifying them against a trusted reference corpus [93]. The method breaks generated text into atomic facts and computes the percentage of facts supported by reliable knowledge sources, providing more nuanced evaluation than binary quality judgments [93]. Evaluation of 6,500 generations from 13 recent language models reveals that ChatGPT achieves only 58% factual precision on people biographies, while more advanced models like GPT-4 demonstrate improved factuality [93]. The automated FActScore metric maintains high reliability with less than 2% error rate, offering cost-effective evaluation that would have required approximately $26,000 if conducted entirely through human assessment [93]. However, FActScore is limited to domains where facts are objective and verifiable, and may not apply to subjective or implicit claims [93].

e) Limitations of Current Benchmark-based Evaluation: Benchmark-based evaluation, while offering standardized and reproducible assessment protocols, faces fundamental limitations that increasingly undermine its validity and representativeness as models and evaluation practices co-evolve. This

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fragility arises because models tend to overfit dataset-specific patterns rather than develop generalizable hallucination detection capabilities, effectively turning benchmarks from evaluation tools into optimization targets. Over time, repeated exposure and model adaptation lead to temporal obsolescence, where benchmarks gradually lose their discriminative power —a phenomenon known as benchmark decay.

These issues are especially evident in factuality benchmarks, which face inherent scalability challenges since constructing comprehensive domain-specific datasets demands extensive manual annotation [116]. Automated frameworks such as FACTOR alleviate these constraints by transforming existing corpora into evaluation sets spanning multiple domains, including Wikipedia, news, and expert-curated content [116]. Nevertheless, benchmark saturation—where models converge toward the test distribution and scores plateau—remains a persistent concern, underscoring the need for continuous evolution, diversification, and complementary evaluation paradigms.

Beyond technical concerns, cultural and linguistic biases introduce systematic blind spots that disadvantage non-Western perspectives and minority languages. Algorithmic automation often overlooks the complexities of human judgment and cultural context [49]. Culturally sensitive frameworks probe GPT models across categories designed to provoke hallucinations, revealing heightened vulnerability on controversial or ambiguous topics [49].

Multitask and multilingual evaluations further show substantial variability across linguistic and cultural contexts, with reasoning accuracy averaging 63.41% [63]. Interactive prompting can yield modest improvements, such as 8% ROUGE-1 gains in summarization and 2% ChrF++ gains in machine translation [63].

Finally, computational resource constraints limit the comprehensiveness of benchmarking, forcing trade-offs between coverage and feasibility [129]. To address cost and coverage, AMBER proposes an LLM-free, multidimensional benchmark that evaluates both generative and discriminative tasks across existence, attribute, and relation hallucinations without relying on human or advanced LLM assessors [30].

6) Baseline Methods: Within the domain of hallucination evaluation, baseline methods establish crucial benchmarks for gauging the performance of new and more sophisticated frameworks. They offer standardized approaches for identifying, measuring, and reducing hallucinations, thereby creating a shared foundation for assessing advancements in model capabilities. These foundational methods can be organized into three functional categories: (1) **Detection**, which focuses on identifying and quantifying hallucinatory content; (2) **Mitigation**, which encompasses techniques aimed at preventing or minimizing the generation of hallucinations; and (3) **Combined Mitigation and Evaluation**, which merges diagnostic assessment with corrective actions into a unified feedback system. This classification provides a clear framework for analyzing the principal strategies in the field.

a) Detection Baselines: Detection baselines establish the empirical foundation for hallucination measurement, transforming subjective assessment into a quantifiable science. POPE (Polling-based Object Probing Evaluation) [25] exemplifies this paradigm shift through an elegant simplification: converting complex multimodal hallucination detection into binary yes/no questions. The benchmark evaluates 500 images from the COCO 2014 validation set, with 6 questions per image (3 positive, 3 negative), totaling 3,000 questions across three sampling strategies [25]. This method achieves higher stability and flexibility than instruction-based caption parsing, as it avoids dependence on prompt design or caption length [25]. For example, InstructBLIP attains 88.73% accuracy under the random setting, outperforming models such as LLaVA and mPLUG-Owl, whose accuracy remains near 50% [25].

POPE's true innovation lies in revealing systematic biases through controlled negative sampling. The framework exposes that LVLMs systematically hallucinate objects that frequently occur in visual instructions or co-occur with actual image objects [25]. This discovery fundamentally shifts our understanding: hallucinations are not random failures but predictable consequences of training data statistics. Rather than parsing complex, free-form text, POPE queries LVLMs with direct "Yes/No" questions about whether specific objects are present in an image. This polling approach produces simple, classifiable outputs that can be assessed with standard metrics, yielding more consistent and adaptable evaluations than analyzing open-ended text generation [25].

To ensure comprehensive assessment and test model resilience against common biases, POPE employs three negative sampling methods for its polling questions:

- **Random Sampling:** Objects not present in the image are chosen randomly to form a basic negative control.
- **Popular Sampling:** Frequently occurring objects from the dataset (though absent in the given image) are selected to test the model's tendency to over-predict common concepts.
- **Adversarial Sampling:** Objects with high co-occurrence rates with ground-truth objects are chosen to provoke hallucinations by leveraging the model's learned statistical associations.

The significance of a baseline like POPE extends beyond its specific methodology; it represents a conceptual pivot by converting subjective, qualitative evaluations of hallucinations into a quantitative and reproducible scientific process. This marks an essential step toward creating rigorous, scalable protocols for model comparison and fostering systematic progress in the field. These detection foundations lay the groundwork for understanding how hallucinations manifest, informing the development of mitigation-oriented baselines discussed next.

b) Mitigation Baselines: Mitigation baselines encompass a spectrum of techniques designed to curtail or eliminate hallucinations, ranging from input-level modifications to direct interventions in the model's generation process. A foundational strategy in this category is **Retrieval-Augmented Generation (RAG)**. By anchoring a model's output to an external

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knowledge base, RAG improves the topical relevance of generated text and helps suppress "off-topic" hallucinations, such as fabricating non-existent entities or producing nonsensical answers [64].

Despite its strengths, RAG exhibits notable limitations. While it enhances factual correctness by providing relevant evidence, it often fails to improve the more sophisticated logical, spatial, or temporal reasoning required for complex prompts. In such cases, the model's responses can appear superficial—more like context matching than genuine comprehension [64]. Other input-level baselines have also explored how data formatting—such as sentence indexing, human-readable date formats, and translation of non-English contexts—affects hallucination frequency [64]. These adjustments yield modest gains but cannot fully resolve deeper issues related to reasoning and semantic understanding.

To address the reasoning deficiencies inherent in methods like RAG, more advanced mitigation baselines have emerged, particularly those involving tuning-free causal interventions. These approaches go beyond supplying additional data; they proactively disrupt the model's internal generative pathways to break the causal chains that lead to hallucinations. One representative framework introduces three intervention types that do not require fine-tuning [133]:

- **Image (I) Intervention:** A pre-processing step that modifies the input image to break false visual-contextual associations. For example, inserting an unrelated object or removing one known to induce hallucinations can disrupt misleading context and preempt errors.
- **Text (Q) Intervention:** Exemplified by the FGBG (Foreground–Background) Prompt, this strategy separates the semantic generation process: the model is first instructed to list foreground objects and then describe the background, thereby reducing interference from peripheral context.
- **Embedding Intervention:** A direct approach that adjusts the model's internal embedding space. After saliency analysis identifies dimensions correlated with hallucination, embeddings are recalibrated using representations from non-hallucinatory examples.

The shift from correlational methods like RAG to causal interventions reflects a deeper understanding of hallucination mechanisms. The limitations of RAG in complex reasoning scenarios indicate that hallucinations are not merely a consequence of knowledge deficiency but often stem from flawed causal reasoning within the model. By targeting these internal mechanisms directly, causal interventions represent a significant advancement in mitigation strategies.

Other baseline paradigms also contribute to this landscape. Traditional model calibration methods, such as temperature scaling, aim to improve confidence alignment and are often assessed on benchmarks like MMLU [154]. In contrast, **end-to-end architectures** such as DBD generate and synthesize multiple image facts concurrently to maintain low hallucination rates [68]. Despite these advances, mitigation approaches alone remain insufficient; more integrated frameworks that jointly evaluate and correct hallucinations have thus emerged.

c) Integrated Detection and Mitigation Baselines: A third class of baselines encompasses frameworks that blur the boundary between evaluation and mitigation. In these methods, evaluation functions are not merely a scoring mechanism but a diagnostic tool for uncovering model weaknesses and guiding targeted mitigation strategies.

A representative paradigm in this area is the automated creation of adversarial benchmarks. **AutoHallusion**, for example, introduces an automated pipeline that works backward from known model biases to generate challenging test cases [20]. The system first probes a model to identify strong linguistic priors about certain objects or contexts, then programmatically generates images that conflict with these biases—for instance, by inserting unexpected objects or removing anticipated ones. By constructing (image, question) pairs that force the model to choose between visual evidence and internal priors, this approach consistently triggers hallucinations. The value of AutoHallusion is twofold: it enables the cost-efficient creation of difficult, dynamic benchmarks—addressing the static nature of manually built datasets—and reveals recurrent failure modes, providing actionable insights for hallucination control and prevention [20].

Another integrated approach leverages **diagnostic disentanglement** to identify the root causes of hallucinations. **Eventhallusion** focuses on diagnosing event-related hallucinations in video LLMs by generating modified versions of videos where temporal information is deliberately obscured while spatial details remain intact. By comparing the model's predictions for the original and altered videos, researchers can determine whether hallucinations arise from incorrect prior assumptions or from deficiencies in temporal reasoning. These diagnostic findings directly pinpoint which reasoning faculties require mitigation. Even methods primarily designed for detection can serve a dual role. The results from **POPE**, for example, reveal that hallucinated objects frequently co-occur with ground-truth objects, suggesting that mitigation efforts should target de-biasing such strong co-occurrence priors [25].

The emergence of these integrated frameworks marks a conceptual shift in the field—transforming evaluation from a passive measurement exercise into an active, dynamic, and often adversarial process. The objective is no longer limited to scoring model performance on fixed datasets but extends to stress-testing models, exposing hidden vulnerabilities, and generating diagnostic knowledge that fuels the next wave of mitigation research. This establishes a tight feedback loop between evaluation and mitigation, accelerating the development of more robust and trustworthy models. Together, these integrated methods reflect a paradigm shift toward dynamic, feedback-driven evaluation, setting the stage for unified performance measurement and critical analysis of baseline limitations.

d) Performance Metrics Baselines: Performance within detection-oriented baselines is typically measured using standard classification metrics. For instance, the **POPE** framework

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evaluates Accuracy, Precision, Recall, and the F1 score, with F1 serving as the primary indicator. It also tracks the proportion of "Yes" responses to detect inherent affirmative biases in model predictions [25].

e) Limitations of Current Baselines: Baseline methods for hallucination evaluation, while establishing critical foundations for the field, face fundamental limitations that restrict their effectiveness and generalizability.

A primary challenge for existing evaluation baselines such as **CHAIR** lies in their instability and dependence on complex parsing rules. These instruction-based methods are highly sensitive to prompt variations and caption lengths, often resulting in inconsistent outcomes and misclassification of hallucinated objects [25]. To mitigate these issues, **POPE** reformulates object hallucination evaluation as a binary classification task using simple Yes/No questions, offering improved stability, fairness, and flexibility compared to traditional caption-based approaches [25]. However, even POPE's binary framework remains reductionist, as it cannot capture nuanced hallucinations involving partial correctness or compositional errors.

On the mitigation side, **RAG**-based baselines enhance factual correctness by anchoring outputs to retrieved context. Yet they frequently fail to strengthen logical reasoning and can introduce new inconsistencies when models over-rely on external knowledge without proper causal understanding [64]. Although these methods improve factual accuracy, they do not generalize across hallucination types—reasoning and compositional errors often persist [64]. Moreover, their practical deployment is hindered by the complexity and latency of retrieval pipelines, especially when integrated with local LLMs that require costly vectorized context management [64].

By contrast, **causal intervention** frameworks—such as image, text, and embedding interventions—achieve notable reductions in hallucination rates by disrupting confounding factors in the generation process [133]. Yet these improvements come with significant computational overhead, limiting their scalability in real-time applications. Furthermore, empirical evidence reveals limited transferability: models performing moderately well on spatial tasks often degrade sharply on temporal reasoning, underscoring the domain dependence and fragility of current mitigation strategies [64].

In summary, current baselines suffer from reductionist evaluation schemes, temporal brittleness, computational inefficiency, and strong domain specificity. These limitations call for the evolution of more adaptive and theoretically grounded frameworks. Future research must move beyond static, one-dimensional benchmarks toward dynamic evaluations that co-evolve with models, multi-faceted assessments capturing the full complexity of hallucinations, and efficient architectures that balance diagnostic rigor with computational feasibility.

7) Frameworks and Pipelines: As research progresses toward more systematic and reproducible practices, frameworks and pipelines have emerged as central infrastructures for hallucination evaluation. Rather than relying on ad hoc techniques, these pipelines integrate multiple components to support robust, scalable, and adaptable assessment. To better capture their diversity, we summarize the current state of the art through six key features, which also serve as the subsections of this discussion. Specifically, we examine end-to-end pipelines that provide holistic workflows, the role of human involvement in ensuring quality and interpretability, and the scope of automation capabilities and their limitations. We then discuss practical implementation strategies that address real-world deployment challenges, highlight feedback loop–based pipelines that iteratively refine model performance, and conclude with fusion-based approaches that combine heterogeneous signals to enhance evaluation robustness. Together, these features characterize the essential design choices shaping contemporary hallucination evaluation pipelines.

a) End-to-End Pipelines: End-to-end structures are the most common approach in LLM hallucination evaluation. They generally share several phases: ground-truth preparation, testing data generation, result acquisition, and hallucination detection.

Ground-truth preparation strategies vary across E2E frameworks. A common pattern is to build a human-labeled dataset or use existing resources. Some papers add spatial cues to photos in dataset. For example, circling regions in images to anchor references and reduce confusion for LLM [30], [56]. In contrast, one paper explicitly avoids creating any dataset or ground truth. It elicits unconstrained text and then has experts directly judge whether hallucinations occur [43], targeting a more natural and wider exposure of hallucination. Due to the simple and straightforward framework structure, E2E pipelines usually lack leverage for big batches of adversary data input or feedback loop to create mitigation-aware evaluation.

Testing data generation and hallucination detection are conducted either by human experts [21], [43] or through LLMs [20]. Some pipelines extend these 2 phases into multi-component evaluation, with modules focusing on different types of hallucinations, including object hallucination [30], relation hallucination [56], attribute hallucination [172], event hallucination [55], and even dialog-based hallucination caused by conflicting information [17].

For measurement, end-to-end pipelines fall into three buckets: (i) reference-based surface metrics such as edit distance between source-of-truth and output [172] and BERT-based graph-similarity metrics [101], [173]; (ii) reference-free, token-level detection/benchmarking, which labels hallucinations at the token level without oracle references [33], [174]; and (iii) model-internal forecasting, which analyzes state-transition dynamics during generation and uses probabilistic signals to predict hallucination [24].

b) Fusion-based Pipelines: Fusion-based methods serve as a critical mechanism in multimodal evaluation pipelines, integrating heterogeneous data sources—including visual, semantic, and external knowledge—to enhance performance, robustness, and contextual coherence. In this regard, these methods operate across multiple stages of the pipeline, from input processing to output.

A primary application of fusion lies in synchronizing visual and semantic information. For instance, Visual modifications

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trigger synchronous updates in structured semantic representations, ensuring cross-modal consistency and providing a logical foundation for generating natural language questions through generative models like GPT-4 [60].

The manifestation of fusion occurs across multiple operational tiers. At the input level, systems abstain from processing images and text in isolation, instead treating them as mutually contextualizing elements to construct a more comprehensive and enriched data unit [9]. This foundational strategy extends into feature-space, where models such as CLIP project textual and visual information into a shared semantic embedding space, enabling nuanced operations like task-specific hit-item selection and embedded question generation [120]. Such alignment is critical for downstream tasks, including the computation of object similarity through maximum-similarity linear assignment between candidate and reference sets [35].

Prompt construction emerges as another critical fusion point, integrating diverse structured elements—from database schemas (ER diagrams and functional dependencies) to semantic triplets and question templates—to generate context-aware queries tailored to specific domains or contents [26], [100], [101]. While fusion frequently operates through the combination of information features rather than direct model amalgamation (as evidenced by the integration of logits with concept masks, rather than merging BERT and GPT outputs) [173], its comparative form is also effective, such as in temporal video analysis where distorted and original sequences are fused to optimize model performance [61].

The evaluation of these integrated pipelines necessitates equally composite metrics, often aggregating evidence from external sources and input context to form comprehensive hallucination detection scores [5]. Moreover, the iterative refinement of fusion strategies is enabled through active learning cycles, where low-confidence samples are routed for human annotation and subsequently reincorporated into training, thereby continuously enhancing the model's cross-modal understanding and reducing error propagation [9].

In conclusion, fusion-based methodologies utilize the synergistic integration of information across the input, feature, prompt, and evaluation stages to establish a crucial balance between automated scalability and human-calibrated validity. This multi-faceted approach is essential for the development of robust, precise, and contextually aware multimodal systems.

c) Parallel Structure: Parallel evaluation structures leverage simultaneous processing of multiple evaluation strategies to expose hallucinations that single-pathway approaches miss.

The foundational parallel architecture implemented by POPE [25] establishes ground-truth comparison through three concurrent negative sampling strategies that probe different vulnerability dimensions.

The framework systematically exposes object hallucination patterns by contrasting ground-truth objects against nonexistent objects selected through: (1) random sampling to measure baseline performance, (2) popular sampling targeting high-frequency objects that exacerbate hallucinations due to training data bias, and (3) adversarial sampling exploiting co-occurrence statistics that further challenge models. Empirical results confirm significant performance degradation under popular and adversarial sampling strategies across all evaluated LVLMs [25].

This parallel approach reveals that models exhibit distinct failure modes across sampling strategies, with random sampling missing systematic biases that popular and adversarial methods expose. The rationale for parallel negative sampling stems from the observation that singular evaluation methods create blind spots: models may resist random hallucinations while remaining vulnerable to statistically motivated adversarial examples.

CIEM (Contrastive Instruction Evaluation Method) [170] advances evaluation through contrastive instruction processing that generates factual and contrastive question-answer pairs based on image captions to systematically assess hallucination in vision-language models. The framework's purpose is to identify instruction-following failures by generating parallel instruction variants—including factual questions (with "Yes" answers) and contrastive questions (with "No" answers)—then comparing model outputs against ground-truth responses. CIEM achieves this through an automated pipeline that leverages large language models (e.g., ChatGPT) to generate QA pairs from annotated captions, covering object existence, attributes, and actions. The generated pairs are validated via a three-round blind review process, achieving high accuracy (error rate 5%). CIEM's evaluation reveals that existing VLMs (e.g., LLaVA, MiniGPT-4) suffer from severe hallucination, with tendencies to over-predict "Yes" regardless of image content, while CIT (Contrastive Instruction Tuning) mitigates this issue by incorporating contrastive samples with detailed justifications, improving overall performance metrics such as Precision and F1-score.

Many studies on hallucination in LVLMs have produced a diverse set of evaluation and mitigation frameworks, with the goal of distinguishing real, precise output from forged or fake content. While their technical details differ, these methods can be organized around their strategies for hallucination detection, categorization, and correction.

A central distinction arises in how hallucination is probed. Object-based methods such as POPE explicitly test for hallucinations by comparing model responses against ground-truth objects derived from annotated or automatically segmented images. By introducing nonexistent distractor objects through random, popular, or adversarial sampling, POPE reveals whether LVLMs invent details not supported by the visual evidence. In contrast, concept-neutral prompting frameworks like THRONE ask open-ended questions without prespecified concepts, and an external language model determines whether fabricated content emerges. These contrasting strategies highlight a key trade-off: object-centric methods provide precise ground-truth comparisons, whereas open-ended methods capture a broader spectrum of hallucinations but depend on secondary evaluators.

Other approaches emphasize automation in hallucination evaluation. For example, CIEM provides a pipeline for

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automatically detecting hallucinations, while its companion method CIT directly addresses their mitigation. Similarly, ReEval perturbs evidence through answer swapping and context enrichment, exposing whether models remain consistent when knowledge is subtly altered. These systems demonstrate that hallucination can be detected not only by visual comparison but also through resilience testing against manipulated textual evidence.

Mitigation-oriented frameworks extend beyond evaluation to actively reduce hallucination. HalluciDoctor cross-checks answers across multiple instructions and generates counterfactual scenarios with hallucinated objects, thereby training models to resist false associations. Systems such as DualFocus and Volcano go further by weighting, ranking, or comparing alternative responses at inference time, steering models toward less hallucinatory outputs. These methods illustrate a growing shift from post hoc evaluation toward proactive hallucination control.

Furthermore, survey-driven and taxonomy-based frameworks broaden the understanding of hallucination by classifying it into distinct categories, such as factual, commonsense, or relational errors, and mapping these onto evaluation strategies. This theoretical orientation provides a unifying perspective that connects otherwise fragmented methodologies, suggesting directions for hybrid approaches that combine detection, diagnosis, and correction. Recent surveys such as [41] further systematize these methods, providing a taxonomy of hallucination correction strategies and comprehensive comparisons of frameworks.

In summary, hallucination remains the organizing principle across these frameworks, shaping how evaluation pipelines are designed, how benchmarks are constructed, and how mitigation is implemented. The field has evolved from isolated detection systems toward more integrated and adaptive approaches, reflecting the recognition that hallucination is not merely an incidental artifact but a systemic challenge requiring continuous monitoring, multi-level granularity, and proactive correction.

d) Multimodal Combination Frameworks: Multimodal combination frameworks integrate visual and textual modalities to enhance hallucination detection robustness, leveraging cross-modal consistency checks. For instance, MedHallBench [9] employs an iterative pipeline combining expert annotation, automatic feedback from models like ChatGLM, and active learning to refine medical image-text datasets, addressing domain-specific hallucination gaps. Similarly, AUTOHALLUSION [20] generates adversarial scenes by probing language priors of VLMs, while VL-Uncertainty [134] perturbs both visual (e.g., image blurring) and textual inputs (e.g., semantic-equivalent rephrasing) to elicit intrinsic uncertainty signals for hallucination scoring. Throne [60] combines object-based abstraction with external LM verification, contrasting with POPE's rule-based Type II hallucination detection. PhD [120] further expands this by synthesizing contextually specious images and text pairs via generative models like DALL-E3, enabling multimodal adversarial testing.

A common pipeline structure emerges across these approaches, as illustrated in Fig. 1: Multimodal input perturbation (e.g., blurring images or rephrasing text) feeds into cross-modal consistency evaluation, where mismatches trigger iterative refinement loops (human-AI collaboration or automated feedback). This fusion mitigates unimodal biases—e.g., VL-Uncertainty's entropy-based clustering of perturbed responses quantifies hallucination severity, while MedHallBench's RLHF retraining closes domain gaps. Such frameworks prioritize dynamic adaptation over static benchmarks, though scalability remains challenged by multimodal data curation costs.

Key studies like DiaHalu [17] extend this to dialogue-level faithfulness hallucination, using GPT-4 to augment scenarios while ERBench [100] automates verification via relational databases. ERBench's key innovation lies in leveraging database integrity constraints—specifically functional dependencies (FDs) and foreign key constraints (FKCs)—to convert any relational database into an automatically verifiable LLM benchmark [100]. The framework transforms database schemas into question-answer pairs where FDs enable inference of attribute values, creating both binary and multiple-choice questions with ground truth derived from the database itself. FKCs facilitate multi-hop reasoning by joining relations through foreign key references, generating questions of increasing complexity that test models' ability to maintain consistency across related entities. This automated approach eliminates manual annotation while ensuring verifiability through database constraints, achieving more than 95.5% automatic verification accuracy compared to human-labeled benchmarks [100]. Collectively, these multimodal frameworks highlight the necessity of fusing heterogeneous signals—visual grounding, textual semantics, and human feedback—to address fragmentation in hallucination evaluation.

e) Feedback Loop-based Framework: Another line of work constructs paired loops that respectively include and exclude a source of truth (e.g., a document or database) and then compares the answers [26]. Cross-phase feedback also includes methods that generate adversarial datasets from the source of truth to test whether the LLM withstands misleading evidence [18], [26], [70], and mitigation-aware loop designs that integrate detection and correction across stages [67], [175]. Moreover, iterative feedback sometimes uses active learning to evolve the model over cycles [9] until a certain halting condition is met.

A frequently cited advantage of feedback pipelines is reduced human effort. To achieve that, many papers still assume strong LLM comprehension within RAG-based setups, where LLM's understanding of retrieved knowledge is assumed as a source of truth [42]. Yet hallucinations can persist even when the source of truth is present. To address this issue, researchers apply confidence scoring [175] relative to ground-truth references [173], combine multiple metrics to strengthen confidence estimates [5], use more explanatory white-box feedback (e.g., match-based mechanisms) [5], and even formalize the source of truth as an Entity-Relationship (ER) diagram to structure verification [100].

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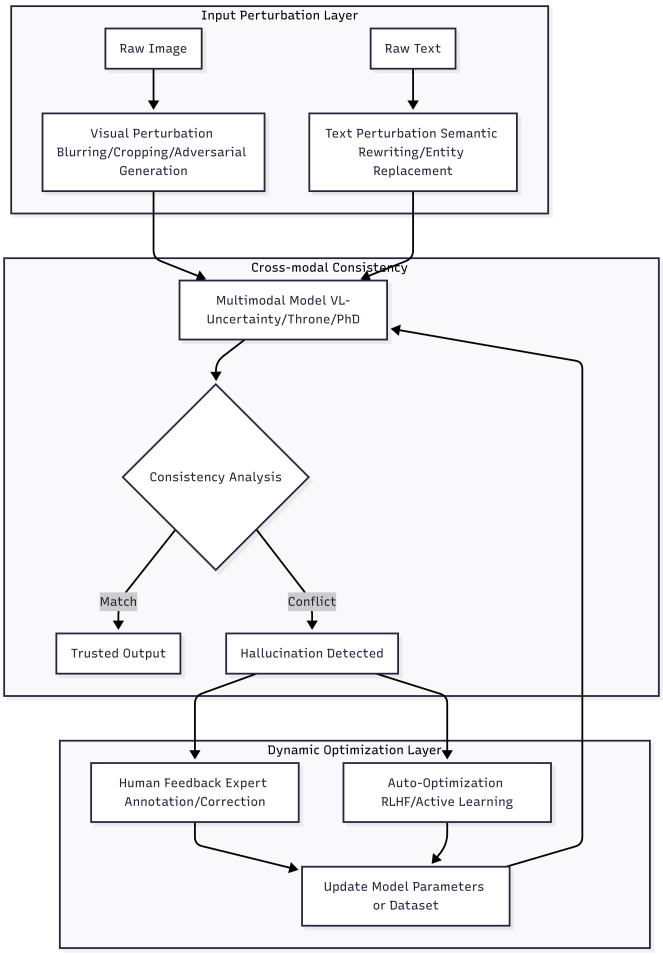


Fig. 1. Multimodal Combination Framework Pipeline

Finally, numerous studies treat feedback as a robustness probe: if feedback shows diverged answers either from the same LLM or multi-models, they infer instability, which extends to hallucination in the produced outputs.

8) The Degree of Automation and Human Active Participation: The proliferation of evaluation tools and frameworks, as surveyed in the previous sections, represents the research community's concerted effort to grapple with the multifaceted challenge of hallucination. However, the design and implementation of these tools themselves introduce a critical meta-question: what constitutes the optimal balance between automated scalability and human discernment in the evaluation process? This question of degree—the extent of automation versus the necessity of human active participation—cuts across all previously discussed categories and emerges as a fundamental architectural choice that directly impacts the validity, cost, and scalability of any evaluation regimen. As frameworks evolve from manual audits to increasingly sophisticated automated pipelines, it becomes imperative to systematically examine the capabilities and limitations of automation, the irreplaceable value of human expertise, and the practical strategies for integrating them into a cohesive and reliable evaluation ecosystem.

a) Human-Involvement: The integration of human expertise in hallucination evaluation pipelines represents a critical balance between scalability and validity, where strategic human involvement compensates for the inherent limitations of automated systems while maintaining practical feasibility. Despite advances in automated evaluation, human judgment remains indispensable for capturing nuanced hallucinations that emerge from contextual understanding, domain expertise, and subjective quality assessment that resist algorithmic formalization [25], [170]. Benchmarks such as [30] demonstrate how comprehensive human-in-loop annotation can yield high-quality hallucination labels, but at significant cost.

The fundamental limitations of purely automated approaches manifest most clearly in scenarios requiring interpretive depth. Open-ended generation tasks produce outputs where hallucinations interweave with stylistic choices and contextual appropriateness, defying binary classification [20]. In addition, subtle multimodal hallucinations, such as plausible but incorrect spatial relationships or temporally inconsistent narratives, often escape detection by metrics focused on object-level accuracy [56], [60]. Furthermore, model-as-judge architectures propagate their own biases, creating systematic blind spots where judge models fail to recognize hallucination patterns outside their training distribution [150]. These limitations necessitate human oversight precisely where automation proves most fragile.

Human contributions span multiple phases of the evaluation pipeline, each leveraging distinct cognitive capabilities that remain difficult to automate. In reference creation, domain experts curate ground-truth datasets that capture field-specific knowledge structures and acceptable variations within professional practice [55], [176]. Annotators parse complex scenes into structured representations, identifying relationships and contextual dependencies that template-based approaches miss [68]. During evaluation, humans provide fine-grained error categorization that distinguishes between factual inaccuracies, logical inconsistencies, and contextual inappropriateness—distinctions critical for targeted mitigation but challenging for automated systems [16], [17].

To optimize this process, the optimal integration strategy employs a hybrid architecture that maximizes both efficiency and validity through intelligent task routing. Automated systems perform initial screening at scale, flagging potential hallucinations and computing confidence scores that identify cases requiring human review [76]. The system continuously calibrates its routing thresholds based on human feedback, creating an adaptive pipeline that improves both automation accuracy and human resource allocation over time [174].

In contrast, high-risk domains demand elevated human involvement to ensure safety and reliability. Medical evaluation frameworks incorporate multiple layers of expert validation. These include consensus reviews among board-certified physicians, rigorous inter-rater reliability assessments, and patient-safety audits that extend beyond factual accuracy to consider clinical appropriateness and potential harms, as exemplified

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by benchmarks like MedHallBench and Med-HALT [1], [9]. While this intensive human oversight significantly increases evaluation costs compared to general domains due to the reliance on expert validation and human feedback, it provides the external validity necessary for deployment in life-critical applications [65]. The investment in human expertise ultimately enables responsible deployment of AI systems where hallucinations could have severe consequences, establishing trust through rigorous validation rather than automated metrics alone.

b) Automation Capabilities and Limitations: Automated evaluation systems excel in specific contexts where reproducibility and scale outweigh interpretive nuance. CIEM implements end-to-end contrastive instruction evaluation pipelines that process thousands of examples without human intervention [170]. AUTOHALLUSION generates controlled stress tests through automated scene generation and metrics computation, systematically probing model vulnerabilities at scale [20]. Advanced frameworks employ separate judge models with rule-based conversion of free-form answers to binary labels, enabling standardized evaluation across diverse output formats [60]. Some systems automate preference data generation and directly adjust internal model representations to mitigate visual errors, demonstrating the potential for self-correcting evaluation pipelines [150], [174]. Adversarial prompt generation further extends automation capabilities, creating diverse test cases that expose model weaknesses without manual curation [18].

Nevertheless, automation faces fundamental limitations that necessitate human oversight. Rule-based parsers prove brittle to phrasing variations, failing on valid but unexpected formulations [54]. Synthetic or edited inputs can drift from real-world data distributions, creating evaluation artifacts that mislead about actual model performance [65]. Internal intervention methods often lack explicit validity checks, potentially modifying model behavior in unintended ways [62]. These limitations become particularly acute in open-ended generation tasks, subtle multimodal scenarios, and prompt-sensitive contexts where human judgment remains irreplaceable. Although automation offers remarkable efficiency, its limitations highlight the need for a well-coordinated hybrid approach. In the next section, we explore practical implementation strategies that combine the best of both automated systems and human oversight.

c) Probing Technology: The probe technology represents a sophisticated approach to hallucination detection that systematically explores model vulnerabilities through targeted interrogation rather than passive evaluation. The fundamental innovation of probing methods lies in their active exploration of model knowledge boundaries, generating diverse query formulations that expose inconsistencies invisible to single-query evaluation.

LLMAuditor [31] exemplifies this paradigm through its automatic and scalable framework that leverages cross-model verification to identify hallucinations. Based on the TruthfulQA dataset, LLMAuditor demonstrates that reliable probes generated from one LLM can effectively audit inconsistencies in different models, avoiding the circular reasoning inherent in self-evaluation approaches. The framework's key innovation lies in its two-phase verification architecture: first, it employs a human-in-the-loop (HIL) approach with standardized evaluation criteria to validate probe quality, achieving substantial inter-annotator agreement; second, it utilizes structured prompt templates to generate targeted probes that expose knowledge gaps and operational inconsistencies [31].

The probing methodology identifies multiple critical hallucination patterns through systematic exploration. First, it detects bias-induced hallucinations by generating multiple question variants that should yield consistent answers, revealing that a substantial portion of hallucinations stem from formatting sensitivity and prompt variations rather than fundamental knowledge deficits [31]. Second, it exposes operational inconsistencies where models provide contradictory answers to logically equivalent queries. Third, it identifies knowledge boundary violations where models confidently generate information beyond their training data scope.

The scalability of LLMAuditor transforms hallucination detection from manual spot-checking to systematic evaluation. By combining automated probe generation with human-in-the-loop validation, the framework enables large-scale auditing while maintaining evaluation quality comparable to expert assessment [31].

This hybrid approach significantly reduces evaluation costs compared to fully manual methods while improving hallucination detection coverage over single-query baselines [31]. The framework's cross-model verification strategy, using different LLMs for probe generation and target evaluation, ensures verifiability and transparency while avoiding circular dependencies that compromise evaluation integrity.

d) Practical Implementation Strategies: Effective hallucination evaluation requires carefully orchestrated workflows that leverage both automation and human expertise. The optimal implementation follows an automation-first, human-calibrated approach that maximizes efficiency while maintaining quality [170]. Initial automatic scoring generates uncertainty estimates and identifies judge disagreements, routing high-risk or ambiguous cases to dual human raters with formal adjudication protocols [20], [55]. Organizations maintain golden datasets of verified examples that calibrate automated systems and track drift over time [16]. Continuous feedback loops update prompts and confidence thresholds based on human corrections, while automated regression tests ensure consistent performance [17], [76]. This hybrid architecture reduces costs compared to fully manual evaluation while preserving the nuanced judgment necessary for complex hallucination detection, ultimately delivering the optimal balance of cost, scalability, and validity for real-world deployment [9], [65].

9) Summary and Future Directions: Our comprehensive analysis of tools and frameworks reveals a field undergoing rapid methodological transformation, moving from manual, ad-hoc evaluation toward systematic, scalable, and theoreti-

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cally grounded approaches. The key insights from this survey illuminate both remarkable progress and fundamental challenges that shape the future of hallucination evaluation.

Dataset Construction Evolution: The transition from manual to hybrid human-AI dataset generation represents a great increase in scale while reducing costs. Entity-based construction methods achieve around 96% automatic verification accuracy [100]. Yet critical challenges persist: automated generation exhibits systematic biases [7], while the trade-off between scale and quality remains unresolved.

Evaluation Methodology Maturation: Benchmark-based evaluation has evolved from simple accuracy metrics to a multidimensional evaluation [167]. The move from scattered hallucination tests to a unified leaderboard highlights the need for consistent cross-framework evaluation of LLM reliability [6].

The LLM-as-Judge Paradox: Using LLMs as evaluators enables scaling far beyond human evaluation, but it also introduces recursive uncertainty by relying on models that themselves may hallucinate. GPT-4 shows strong agreement with human annotators yet demonstrates preference bias toward its own outputs [167], while ensemble strategies improve accuracy but fail to resolve fundamental disagreements between judge models. This paradox reveals a deeper epistemological challenge: the impossibility of bootstrapping trust from untrusted foundations.

Human-AI Collaboration Optimization: Hybrid evaluation reduces costs greatly while maintaining high accuracy through intelligent task routing [31]. Humans can identify more hallucinations than automated systems, particularly for context-dependent errors. Domain-specific evaluation requires more human involvement, such as medical assessment [9].

Framework Integration Patterns: End-to-end pipelines dominate with a wide adoption but lack feedback mechanisms for iterative improvement [170]. Dynamic evaluation can reduce the evaluation error [26]. Critically, THRONE's analysis reveals that a reduction in Type II hallucinations (false negatives, where models fail to identify present objects) does not lead to a reduction in Type I hallucinations (false positives, where models claim absent objects exist), but rather these two hallucination forms exhibit anticorrelation [60].

Critical Gaps and Future Imperatives: Three fundamental challenges constrain current tools: (1) *Evaluation recursion*—the philosophical impossibility of establishing ground truth using potentially flawed evaluators; (2) *Benchmark obsolescence*—static datasets becoming saturated within 12 months as models adapt; (3) *Domain and Language transfer failure*—evaluation methods that perform reliably in English often degrade when applied to other languages or specialized domains [70].

The Path Forward: Future frameworks must embrace three principles: (1) *Dynamic adaptation* through continuous benchmark evolution and adversarial co-generation; (2) *Hierarchical validation* combining automated screening, ensemble judgment, and human expertise in tiered architectures; (3) *Causal understanding* moving beyond correlation-based detection toward mechanistic models of hallucination generation. The ultimate goal transcends detection—building evaluation systems that not only identify hallucinations but also understand their origins, predict their occurrence, and inform principled mitigation strategies.

The evolution from manual annotation to sophisticated human-AI collaboration represents remarkable progress, yet fundamental questions remain: Can we bootstrap a trustworthy evaluation from untrustworthy components? How do we prevent benchmark overfitting while maintaining comparability? What constitutes ground truth in domains where human experts disagree? These challenges define the frontier of hallucination evaluation, requiring not just technical innovation but philosophical clarity about the nature of truth, knowledge, and reliability in artificial intelligence systems.

H. Summary of the Evaluation Gap

The extensive body of work surveyed in this section reveals a vibrant and rapidly evolving field. Researchers have developed sophisticated metrics and benchmarks to tackle hallucination from multiple angles, as outlined in the preceding sections. However, a critical dichotomy, first introduced in Section I, can be used to characterize a fundamental limitation in the current landscape: the tension between Horizontal and Vertical assessment.

Most existing efforts, including comprehensive benchmarks like Hallucination Leaderboard [6] and and targeted evaluation suites such as HaluEval [7], excel at horizontal assessment—they provide a broad, top-level evaluation of hallucination across a wide range of tasks, domains, and modalities. This breadth is essential for generating aggregate performance scores and facilitating high-level model comparisons.

However, these horizontal assessments still face several core challenges that remain only partially resolved. As summarized in Table II, these include persistent difficulties in defining ground truth amid ambiguity, prohibitive computational costs of comprehensive evaluation, rapid obsolescence of static benchmarks, fragmentation in handling diverse domains/cultures, and recursive uncertainties in automated human-judgment approximation. These unresolved issues fundamentally constrain the effectiveness of top-down, breadth-first evaluation paradigms.

TABLE II. PARTIALLY RESOLVED CHALLENGES IN HORIZONTAL EVALUATION

Challenge Category	Manifestation in Horizontal Evaluation
Epistemological Challenge	Defining ground truth in ambiguous contexts (Sec. II-A)
Computational Complexity Challenge	Scaling costs of comprehensive assessment (Sec. II-B)
Dynamic Knowledge Challenge	Benchmark obsolescence against evolving knowledge (Sec. II-D)
Scale and Diversity Challenge	Fragmented domain/cultural coverage (Sec. II-F)
Human Alignment Challenge	Recursive uncertainty in human-judgment simulation (Sec. II-H)

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In contrast, vertical assessment, the deep and mechanistic analysis of the root causes of a specific hallucination within a model's internal reasoning process, remains underexplored and challenging to implement at scale. Although causal intervention and counterfactual analysis methodologies (Section III-D3) and attention analysis (Section III-D5) represent pioneering steps in this direction, they are typically applied in isolation, on small scales, or in controlled settings. The field currently lacks a unified framework that can seamlessly integrate both horizontal breadth and vertical depth, leaving a significant gap in our ability not just to detect hallucinations but to truly diagnose and understand them on a granular level. This gap motivates the introduction of the LENS framework in subsequent chapters, which is designed to bridge this very divide.

IV. The Evaluation Gap: Why Current Frameworks Fail and What Is Needed

The comprehensive challenges outlined in Section II and the diverse evaluation approaches surveyed in Section III-D converge on a fundamental realization: current hallucination evaluation frameworks are structurally incapable of meeting the demands of real-world AI deployment. This incapability stems not from implementation deficiencies but from conceptual limitations in how these frameworks understand and approach hallucination evaluation. This section identifies the critical gaps that render existing approaches insufficient and establishes the foundation for a paradigm shift in evaluation methodology.

A. The Dimensional Poverty of Current Evaluation

Existing hallucination evaluation frameworks suffer from what we term dimensional poverty, an over-simplification that reduces the complex, multi-faceted phenomenon of hallucination to singular metrics or binary classifications [3], [5]. This reductionist approach fundamentally misunderstands the nature of hallucination, treating it as a monolithic failure rather than a spectrum of errors with distinct characteristics, causes, and implications [3], [41].

1) Beyond Binary: The Multi-Dimensional Nature of Hallucination: Current frameworks predominantly evaluate hallucination through a single lens of correctness, asking merely whether generated content is true or false [7], [25]. This binary perspective ignores the reality that hallucinations vary along multiple orthogonal dimensions that critically impact their severity and mitigation strategies [30], [55]. A factually incorrect statement generated with high confidence poses different risks than uncertain generation of partially correct information [48], [134]. A hallucination that omits critical details differs fundamentally from one that fabricates non-existent information [67], [101]. A model that consistently hallucinates in specific domains exhibits different failure patterns than one with sporadic, unpredictable errors [29], [32].

The dimensional poverty becomes particularly problematic when organizations attempt to make deployment decisions [8]. A model with 90% accuracy might be entirely unsuitable if its 10% error rate concentrates in high-risk categories, exhibits high confidence when wrong, or fails to recognize when external verification is needed [34], [50]. Current single-dimensional evaluations cannot capture these critical distinctions, leaving organizations blind to the actual risk profiles of models they deploy.

2) The Four Missing Dimensions: Our analysis reveals four critical dimensions absent from current evaluation frameworks, each capturing distinct aspects of hallucination behavior essential for comprehensive assessment [5], [58]. The degree dimension extends beyond simple accuracy to encompass faithfulness to context, consistency with instructions, and crucially, the appropriateness of generation versus retrieval decisions. The quantity dimension evaluates not just whether information is correct but whether it is complete, addressing hallucinations of omission that current frameworks ignore. The stability dimension assesses consistency across multiple evaluations, revealing whether errors are systematic or stochastic. The risk dimension quantifies uncertainty and confidence calibration, distinguishing between models that acknowledge their limitations and those that hallucinate confidently.

These dimensions are not merely additional metrics but fundamental aspects of hallucination that determine real-world impact [13]. A medical diagnosis system must not only be accurate but complete in its assessment, stable across similar cases, and properly calibrated in its confidence [1], [9]. Current frameworks evaluating only accuracy miss these critical requirements, creating false confidence in models unsuitable for deployment.

B. The Horizontal-Vertical Integration Challenge

The artificial separation between horizontal breadth and vertical depth in current evaluation frameworks represents a fundamental architectural flaw that prevents comprehensive hallucination assessment [21], [54]. This separation forces an unnecessary and counterproductive choice between understanding what errors occur and understanding why they occur [39].

1) The False Dichotomy and Its Consequences: Horizontal evaluation frameworks like MMLU [177] and TruthfulQA [4] provide broad coverage across domains but treat each evaluation as an isolated measurement, offering no insight into the reasoning processes that produce hallucinations. These frameworks can identify that a model fails 30% of the time on medical questions but cannot explain whether failures stem from knowledge gaps, reasoning errors, or inappropriate generalization from other domains. This limitation renders horizontal evaluations nearly useless for improvement, as developers lack the causal understanding necessary for targeted mitigation.

Conversely, vertical evaluation approaches employing chain-of-thought analysis [23] or mechanistic interpretability provide deep insights into specific reasoning chains but lack the breadth to assess overall model capabilities. A detailed analysis of how a model solves mathematical problems provides no information about its tendency to hallucinate in factual recall

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or creative generation tasks. This narrow focus prevents vertical evaluations from supporting comprehensive deployment decisions, as organizations cannot extrapolate from limited deep analyses to broad use cases.

2) The Integration Imperative: Real-world deployment demands simultaneous breadth and depth, requiring frameworks that can assess performance across diverse domains while understanding failure mechanisms within each.This integration is not merely desirable but essential for several reasons [60]. Cross-domain hallucination patterns often share underlying mechanisms that only integrated evaluation can reveal [27], [176]. A model's tendency to over-generalize in medical contexts might stem from the same architectural features causing mathematical reasoning errors, but separated evaluations miss these connections [61], [83]. Effective mitigation strategies require understanding both where hallucinations occur and why they occur, information that neither horizontal nor vertical evaluation alone can provide. Organizations need comprehensive risk assessment covering all potential use cases with sufficient depth to understand failure modes, a requirement that current separated approaches cannot meet.

The integration challenge extends beyond simply combining existing horizontal and vertical methods [20]. True integration requires frameworks that can dynamically adjust evaluation depth based on observed patterns, trace reasoning processes across different domains, and synthesize insights from multiple evaluation modalities into coherent understanding [24], [135]. This level of integration demands fundamental reconceptualization of evaluation architecture, moving from parallel independent assessments to hierarchical, interconnected analysis [37], [52].

C. The Metacognitive Blind Spot: Tool Necessity and Selection

Perhaps the most critical yet entirely overlooked aspect of hallucination evaluation concerns models' metacognitive capabilities, specifically their ability to recognize when they should not generate from parametric memory but instead consult external sources. This metacognitive blind spot in current frameworks represents a fundamental failure to evaluate one of the most important safeguards against hallucination.

1) Tool Necessity Detection as a First-Class Evaluation Target: Current frameworks implicitly assume that models should always generate responses from their parametric memory, evaluating only the accuracy of these generations [46]. This assumption ignores a critical reality: many hallucinations are entirely preventable if models recognize when generation is inappropriate and external consultation is necessary [26], [76]. A model asked about current stock prices should not attempt to generate plausible values but should recognize that this query requires real-time data retrieval [64]. Yet no existing framework evaluates this recognition capability [45].

The introduction of Tool Necessity Detection (TND) as a primary evaluation metric represents a paradigm shift in how we assess model capabilities [46]. TND evaluates whether models correctly identify queries requiring external verification based on temporal sensitivity, factual specificity, domain expertise requirements, and verification criticality. This metric transforms evaluation from post-hoc error detection to proactive risk assessment, identifying models that fail not because they lack knowledge but because they fail to recognize their limitations.

The absence of TND evaluation has profound implications for deployment safety [71]. In high-stakes applications, the cost of hallucination often far exceeds the computational cost of tool consultation [51]. A medical AI system that generates treatment recommendations from potentially outdated training data rather than consulting current clinical guidelines poses unacceptable risks [1], [34]. Without TND evaluation, we cannot distinguish between models that might hallucinate and models that will hallucinate because they lack the metacognitive awareness to seek help when needed [70].

2) Tool Selection Accuracy: The Precision of External Consultation: Even when models do recognize the need for external tools, current frameworks fail to evaluate whether they select appropriate tools for the task [44]. Tool Selection Accuracy (TSA) represents another critical missing dimension, assessing whether models can match query requirements to tool capabilities effectively [175]. A model might correctly identify the need for external information but select a general web search when a specialized medical database would be appropriate, or use a simple calculator when symbolic algebra is required.

TSA evaluation reveals sophisticated understanding of tool capabilities and limitations, query requirements and constraints, and information source reliability and authority [66]. This evaluation goes beyond binary tool usage to assess the quality of tool selection decisions, recognizing that inappropriate tool selection can be as problematic as no tool usage at all. The metacognitive gap extends to evaluation of tool usage strategies, including hierarchical source consultation, multi-tool verification for critical information, and graceful degradation when preferred tools are unavailable.

D. The Adaptability Crisis: From Static Benchmarks to Dynamic Evaluation

Current evaluation frameworks impose rigid, pre-defined benchmarks that fundamentally cannot accommodate the diverse and evolving requirements of real-world deployments. This inflexibility creates an insurmountable gap between what organizations need to evaluate and what existing frameworks can assess.

1) The Impossibility of Universal Static Benchmarks: The premise that a fixed set of benchmarks can adequately evaluate models for all possible deployments is fundamentally flawed. Every organization operates with unique domain knowledge, specialized terminology, specific quality requirements, and particular risk tolerances that no universal benchmark can capture. A pharmaceutical company needs evaluation against current drug interaction databases, proprietary research findings, and regulatory requirements specific to their markets.

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A legal firm requires assessment using jurisdiction-specific statutes, internal precedent analyses, and client-specific confidentiality requirements. These needs cannot be met by generic benchmarks like MedQA [19] or LegalBench [178], no matter how comprehensive they attempt to be.

The static nature of current benchmarks creates a growing disconnect between evaluation and reality [6]. Knowledge evolves continuously, with medical guidelines updated monthly, legal precedents established daily, and scientific understanding advancing constantly [16], [17]. Static benchmarks become obsolete almost immediately, evaluating models against outdated knowledge while missing contemporary requirements. This temporal gap means organizations must choose between irrelevant standardized evaluation or ad-hoc internal assessment lacking rigor and comparability.

2) The Adaptive Framework Imperative: The solution is not more benchmarks but fundamentally adaptive evaluation frameworks that enable organizations to construct rigorous evaluations from their own information sources [30], [129]. This adaptability must extend beyond simple dataset substitution to encompass customizable evaluation criteria reflecting organizational priorities, dynamic knowledge integration from authoritative sources, domain-specific reasoning patterns and requirements, and organization-specific risk tolerances and thresholds.

An adaptive framework transforms evaluation from a one-size-fits-all assessment to a tailored analysis that maintains methodological rigor while addressing specific needs [37], [53]. This capability is not merely convenient but essential for meaningful evaluation. Without adaptability, organizations cannot evaluate models against their actual use cases, assess performance on their specific knowledge domains, or verify compliance with their particular requirements. The adaptability gap thus represents not just a limitation but a fundamental barrier to trustworthy deployment.

E. The Transparency Deficit: From Scores to Actionable Understanding

Current frameworks operate as black boxes, providing numerical scores without exposing the reasoning processes, evidence chains, or causal mechanisms that produced those scores.This opacity transforms evaluation from a diagnostic tool into a mere measurement exercise, providing numbers without understanding [31].

1) The Actionability Problem: When a model achieves 70% accuracy on a hallucination benchmark, current frameworks provide no actionable information for improvement [18], [33]. The score reveals nothing about whether errors cluster in specific domains or distribute randomly, whether failures occur early in reasoning chains or during final synthesis, whether hallucinations stem from knowledge gaps or reasoning failures, or whether errors are systematic or stochastic. Without this understanding, developers cannot improve models, users cannot calibrate trust, and organizations cannot make informed deployment decisions.

The transparency deficit extends beyond simple error analysis to encompass the entire evaluation process [14], [179]. Current frameworks provide no visibility into how evaluation tasks decompose into components, how individual components contribute to overall scores, what evidence supports each evaluation decision, or how confidence varies across different aspects of evaluation. This opacity makes evaluation results essentially meaningless for practical purposes, as stakeholders cannot understand what the numbers actually represent or how to act upon them.

2) The Attribution Revolution: True transparency requires not just visibility but causal attribution linking observed hallucinations to their origins in model architecture, training data, or inference processes [57], [174]. This attribution must trace errors through the complete generation pipeline, identifying where incorrect information enters the reasoning chain, how errors propagate through subsequent steps, why certain mitigation attempts fail, and what interventions might prevent similar failures.

The attribution revolution transforms evaluation from detection to diagnosis, enabling targeted improvements rather than blind adjustments [26], [173]. With proper attribution, developers can identify whether hallucinations stem from architectural limitations requiring model redesign, training data issues necessitating dataset curation, or inference problems solvable through prompt engineering [52], [133]. This understanding makes evaluation results actionable, converting abstract scores into concrete improvement strategies.

F. The Synthesis Challenge: Toward Unified Evaluation

The gaps identified above are not independent limitations but interconnected symptoms of a more fundamental problem: the absence of unified evaluation frameworks that synthesize multiple dimensions, modalities, and perspectives into coherent, actionable assessment. This synthesis challenge represents the ultimate barrier to effective hallucination evaluation.

1) The Fragmentation Syndrome: Current evaluation approaches remain fragmented across every dimension of assessment [5]. Methodological fragmentation forces practitioners to navigate incompatible evaluation paradigms, each with distinct assumptions, procedures, and limitations [40], [63]. Metric fragmentation produces diverse measurements that cannot be meaningfully compared or combined, preventing holistic understanding. Tool fragmentation requires organizations to integrate multiple evaluation systems, each with different interfaces, data formats, and computational requirements. Insight fragmentation scatters understanding across disconnected reports, preventing the emergence of coherent comprehension.

This fragmentation imposes enormous costs on organizations attempting comprehensive evaluation [62]. They must invest in multiple incompatible frameworks, develop expertise in diverse methodologies, attempt to reconcile contradictory results, and still lack confidence in their assessments [2], [65]. The fragmentation syndrome thus represents not just inefficiency but a fundamental barrier to trustworthy AI de-

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ployment, as organizations cannot achieve the comprehensive understanding necessary for informed decisions.

2) The Unified Framework Vision: Addressing the synthesis challenge requires more than incremental improvements or framework integration. It demands fundamental reconceptualization of evaluation as a unified analytical process that simultaneously addresses multiple dimensions of hallucination [100], integrates horizontal breadth with vertical depth [43], evaluates both generation and metacognition [180], supports customization while maintaining rigor [42], and provides transparent, actionable insights [35].

This unified vision recognizes that hallucination evaluation cannot be decomposed into independent assessments but must be understood as an interconnected analytical challenge [15], [47]. Different dimensions of hallucination interact in complex ways that only unified evaluation can capture. Horizontal and vertical perspectives provide complementary insights that gain meaning only through integration. Metacognitive capabilities influence generation accuracy in ways that separated evaluation cannot assess. Customization and standardization are not opposing requirements but complementary aspects of rigorous evaluation. Transparency and actionability emerge not from additional metrics but from fundamental visibility into evaluation processes.

G. The Path Forward: Requirements for Next-Generation Evaluation

The gaps identified throughout this section establish clear requirements for next-generation hallucination evaluation frameworks. These requirements are not aspirational goals but fundamental necessities for evaluation that can support trustworthy AI deployment.

1) Dimensional Completeness: Next-generation frameworks must evaluate hallucination across multiple orthogonal dimensions, recognizing that single metrics cannot capture the complexity of generation failures [22], [68]. This dimensional completeness requires assessing not just accuracy but faithfulness, completeness, stability, and calibrated uncertainty. Each dimension must be evaluated through metrics that capture both absolute performance and relative patterns, enabling nuanced understanding of model capabilities and limitations.

2) Hierarchical Integration: Frameworks must seamlessly integrate horizontal breadth with vertical depth through hierarchical evaluation architectures that can provide broad coverage while maintaining deep analytical capability [56], [172]. This integration cannot be achieved through simple combination of existing methods but requires novel approaches that dynamically adjust evaluation depth based on observed patterns and synthesize insights across different levels of analysis.

3) Metacognitive Assessment: Evaluation must extend beyond generation accuracy to assess metacognitive capabilities, particularly the ability to recognize when external consultation is necessary and select appropriate tools for the task [38], [154]. This assessment represents a fundamental expansion of evaluation scope, recognizing that preventing hallucination is as important as detecting it.

4) Adaptive Construction: Frameworks must enable organizations to construct rigorous evaluations from their own information sources while maintaining standardized methodologies that ensure comparability [49], [150]. This adaptability must encompass not just dataset substitution but customizable evaluation criteria, dynamic knowledge integration, and domain-specific requirements.

5) Process Transparency: Evaluation must provide complete visibility into reasoning processes, evidence chains, and causal mechanisms, transforming assessment from measurement to understanding [12], [170]. This transparency requires not just detailed reporting but interpretable attribution that links observations to causes and suggests targeted interventions.

6) Unified Synthesis: Most critically, frameworks must synthesize these diverse requirements into coherent, unified evaluation that provides actionable insights rather than fragmented measurements [25], [59]. This synthesis represents the ultimate challenge and opportunity for next-generation evaluation, transforming hallucination assessment from a collection of metrics to systematic understanding.

H. Conclusion: The LENS Response

The gaps identified in this section reveal that current hallucination evaluation frameworks are not merely limited but fundamentally inadequate for supporting trustworthy AI deployment. The dimensional poverty, integration failures, metacognitive blind spots, adaptability crises, transparency deficits, and synthesis challenges collectively render existing approaches insufficient for the demands of real-world applications.

These gaps cannot be addressed through incremental improvements or framework combinations but require fundamental reconceptualization of how we approach hallucination evaluation. The LENS framework, detailed in the following section, represents our comprehensive response to these challenges. Through multi-dimensional assessment spanning degree, quantity, stability, and risk; hierarchical decomposition enabling simultaneous breadth and depth; explicit evaluation of Tool Necessity Detection and Tool Selection Accuracy; adaptive benchmark construction from organizational knowledge sources; complete process transparency with causal attribution; and unified synthesis of diverse evaluation perspectives, LENS transforms hallucination evaluation from fragmented measurement to systematic understanding.

The transition from current frameworks to LENS represents not just technical advancement but a paradigm shift in how we conceptualize and conduct hallucination evaluation. By addressing each identified gap while maintaining practical deployability, LENS enables the comprehensive, actionable evaluation necessary for trustworthy AI systems in critical applications.

V. LENS: A Decomposition-First Framework for Systematic LLM Hallucination Evaluation

Large Language Models and Vision-Language Models exhibit remarkable capabilities yet suffer from hallucinations that

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generate plausible but factually incorrect content [4], [93]. The challenge of detecting and mitigating these hallucinations has become critical as these models are deployed in high-stakes applications. We present LENS, a comprehensive evaluation framework that systematically addresses hallucination detection through hierarchical decomposition, tool-augmented verification, and structured output generation.

The core insight driving LENS is that complex queries often fail not because models lack knowledge, but because they attempt to process compound information without proper factorization and fail to recognize when external verification is necessary. A critical source of hallucinations occurs when models confidently generate information from parametric memory that should instead be retrieved from authoritative sources. By decomposing evaluation tasks into verifiable atomic components and explicitly evaluating tool necessity recognition, LENS transforms the hallucination detection problem from a monolithic assessment into a structured verification process. Our framework operationalizes evaluation across four critical dimensions: degree measuring accuracy, faithfulness, and tool appropriateness; quantity assessing coverage and recall; stability evaluating test-retest reliability; and hallucination risk quantifying uncertainty levels.

A. Framework Architecture

LENS adopts a systematic pipeline architecture that transforms complex evaluation tasks into verifiable atomic components through principled decomposition. At its foundation, the framework recognizes that hallucinations often emerge from the compounding of small errors in multi-step reasoning. Rather than treating model outputs as black-box predictions, LENS explicitly exposes the reasoning process through a combination of state-of-the-art techniques including Chain-of-Thought reasoning [23], Tree-of-Thoughts search [181], tool-augmented generation via ReAct [182], and constrained decoding [183]–[185].

The framework implements a six-stage evaluation pipeline denoted as $\Pi = \langle S_1, S_2, ..., S_6 \rangle$ where each stage S_i transforms the evaluation state σ_i to σ_{i+1}. Task formulation S_1 establishes evaluation criteria $\mathcal{C}$, success metrics $\mathcal{M}$, and acceptable evidence sources $\mathcal{E}$ while defining ground truth baselines $\mathcal{G}$ and tolerance thresholds τ. The hierarchical decomposition stage S_2 applies recursive task decomposition using Tree-of-Thoughts to generate multiple valid factorizations. For compound queries expressed as $C = f(A_1, A_2, ..., A_n)$, the system explores both direct evaluation strategies and component-wise verification approaches. During tool-augmented execution S_3, decomposed subtasks are processed using appropriate external tools from the tool registry $\mathcal{T}$ following the ReAct paradigm, with each tool invocation generating traceable evidence complete with provenance metadata.

The structured generation phase S_4 enforces output constraints through grammar-guided decoding or JSON Schema validation, ensuring machine-readable results that include evidence chains and confidence scores. Multi-dimensional scoring S_5 computes evaluation metrics across our four key dimensions using established benchmarks [4], [93], [186]. The final trace analysis stage S_6 generates standardized OpenTelemetry traces that enable reproducibility and comparative analysis across model versions, prompting strategies, and parameter configurations. As illustrated in Fig. 2, the LENS evaluation pipeline integrates six stages, from task formulation to trace analysis, with feedback loops that enable iterative refinement based on scoring outcomes.

Four key architectural principles guide the LENS design. First, the decomposition-first approach ensures that complex queries undergo recursive factorization before evaluation, exposing intermediate reasoning steps and enabling granular verification. Second, evidence grounding requires that all claims be supported by explicit evidence from verified sources, with tool invocations providing traceable provenance chains. Third, structure enforcement employs constrained decoding to guarantee well-formed, parseable results amenable to automated scoring. Finally, comprehensive traceability via OpenTelemetry spans captures model inputs, outputs, tool invocations, and timing data to ensure full reproducibility of results.

B. Evaluation Dimensions and Metrics

LENS evaluates model outputs across four orthogonal dimensions $\mathcal{D} = \{d_{degree}, d_{quantity}, d_{stability}, d_{risk}\}$, each capturing distinct aspects of generation quality and reliability. This multi-dimensional approach recognizes that hallucination is not a binary phenomenon but rather manifests along different axes that require specialized detection strategies.

1) Degree: Correctness, Faithfulness, and Tool Appropriateness: The degree dimension d_{degree} measures correctness, faithfulness, and crucially, the appropriateness of tool usage through a composite score:

$$\begin{aligned} d_{degree} =& \alpha_1 \cdot \text{Truth}_{ext}(y) + \alpha_2 \cdot \text{Faith}_{int}(y|x) \\ &+ \alpha_3 \cdot \text{Tool}_{app}(t|q) \end{aligned} \tag{1}$$

where $\text{Truth}_{ext}(y)$ quantifies extrinsic truthfulness against world knowledge, $\text{Faith}_{int}(y|x)$ measures intrinsic faithfulness to provided context x, $\text{Tool}_{app}(t|q)$ evaluates tool appropriateness for query q, and $\alpha_1 + \alpha_2 + \alpha_3 = 1$ balances the three components.

The tool appropriateness score Tool_{app} decomposes into two critical sub-metrics:

$$\begin{aligned} \text{Tool}_{app}(t|q) =& \beta \cdot \text{TND}(q) \\ &+ (1-\beta) \cdot \text{TSA}(t|q) \end{aligned} \tag{2}$$

where Tool Necessity Detection (TND) measures whether the model correctly identifies when external tools are required, and Tool Selection Accuracy (TSA) evaluates whether the appropriate tool was chosen for the task. These metrics (TND and TSA) are core components of the proposed LENS framework. See Table III for a comprehensive comparison with existing methods. This explicit evaluation of tool appropriateness is crucial because a significant class of hallucinations occurs when models generate plausible-sounding facts from parametric memory instead of retrieving verified information,

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TABLE III. COMPREHENSIVE COMPARISON BETWEEN LENS AND EXISTING HALLUCINATION EVALUATION FRAMEWORKS ACROSS CRITICAL CAPABILITIES

Capability	**Output-Based** (TruthfulQA, FactCC)	**Process-Based** (StrategyQA, DiaHalu)	**Capability-Based** (GSM8K, Hallusion-Bench)	**LENS Framework** (Our Approach)
Evaluation Scope	Broad coverage, final outputs only	Limited tasks, reasoning chains	Specific skills, isolated testing	Comprehensive: horizontal breadth + vertical depth
Causal Attribution	None: binary detection only	Partial: traces reasoning steps	Limited: identifies skill gaps	Complete: full causal chain from query to error
Metacognitive Assessment	Not evaluated	Not evaluated	Not evaluated	**Novel**: TND and TSA as primary metrics
Dimensional Analysis	Single: correctness only	Single: reasoning fidelity	Single: capability presence	**Four dimensions**: degree, quantity, stability, risk
Adaptability	Fixed benchmarks	Fixed protocols	Fixed test sets	**Dynamic**: construct from organizational knowledge
Transparency	Black box scores	Some reasoning visibility	Skill-level insights	**Full transparency**: hierarchical decomposition with evidence
Tool Integration	Not supported	Limited support	Not supported	**Core feature**: evaluated and utilized systematically
Error Prevention	Reactive: post-hoc detection	Reactive: error analysis	Reactive: capability gaps	**Proactive**: identifies hallucination-prone patterns
Actionability	Low: scores without guidance	Medium: some diagnostic value	Medium: skill improvement	**High**: targeted mitigation strategies
Computational Cost	Low: simple checks	Medium: reasoning analysis	Low to Medium	Medium to High: comprehensive but optimized

or when they select inappropriate tools that provide unreliable evidence.

For extrinsic evaluation, the framework employs TruthfulQA [4] to assess factual accuracy. Intrinsic evaluation leverages FActScore [93], which decomposes long-form outputs into atomic claims $\{c_1, c_2, ..., c_m\}$ for individual verification:

$$\text{FActScore}(y) = \frac{1}{m}\sum_{i=1}^{m} \mathbb{1}[\text{Verify}(c_i) = \text{True}] \tag{3}$$

QAFactEval [186] measures summary faithfulness through question-answering consistency, computing the alignment between questions generated from source documents and answers derived from model outputs.

2) Quantity: Coverage and Completeness: The quantity dimension $d_{quantity}$ assesses the completeness of generated content through coverage metrics:

$$d_{quantity} = \frac{|\mathcal{I}_{generated} \cap \mathcal{I}_{required}|}{|\mathcal{I}_{required}|} \tag{4}$$

where $\mathcal{I}_{required}$ represents required information units and $\mathcal{I}_{generated}$ denotes information present in the output.

In retrieval-augmented generation settings, RAGAS metrics [145] provide detailed assessment through context recall R_c, context precision P_c, and answer-context alignment A_{ac}:

$$\text{RAGAS} = \beta_1 R_c + \beta_2 P_c + \beta_3 A_{ac} \tag{5}$$

where $\beta_1 + \beta_2 + \beta_3 = 1$ represents task-specific weighting.

3) Stability: Robustness and Consistency: Model consistency under varying conditions reveals robustness characteristics captured by the stability dimension $d_{stability}$. The framework evaluates self-consistency across k samples:

$$\text{Consistency}(y_1, ..., y_k) = 1 - \frac{H(\text{Majority}(y_1, ..., y_k))}{H_{max}} \tag{6}$$

where H denotes entropy and H_{max} represents maximum possible entropy.

Position bias in long contexts is quantified through the U-shaped attention metric:

$$\text{PosBias}(a) = \frac{\sum_{i \in \{head, tail\}} a_i}{\sum_{i=1}^{n} a_i} \tag{7}$$

where a represents attention weights across n positions.

4) Hallucination Risk Quantification: The hallucination risk dimension d_{risk} employs complementary detection methods. Semantic uncertainty U_s through multi-sample analysis:

$$U_s = \mathbb{E}_{y \sim p(y|x)}[\text{KL}(p(y|x)||p_{ref}(y))] \tag{8}$$

When model internals are accessible, INSIDE probes [187] compute:

$$h_{probe} = \sigma(W^T \phi(s) + b) \tag{9}$$

where $\phi(s)$ represents hidden state features and W, b are learned probe parameters.

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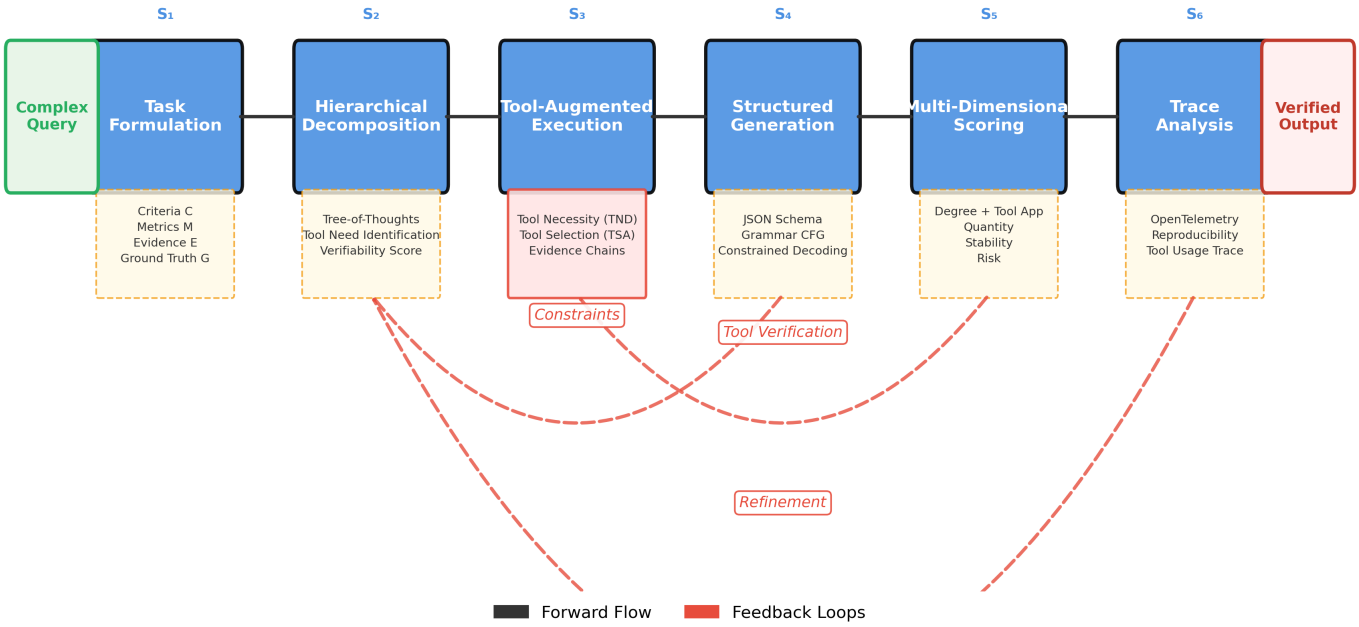


Fig. 2. LENS evaluation pipeline showing the six-stage process from task formulation to trace analysis. Feedback loops enable iterative refinement based on scoring outcomes

TABLE IV. LENS EVALUATION DIMENSIONS WITH REPRESENTATIVE METRICS AND MEASUREMENT PROTOCOLS.

Dimension	Metrics	Protocol
Degree	TruthfulQA, FActScore, QAFactEval, TND, TSA	Atomic claim verification, QA consistency, tool necessity detection, tool selection accuracy
Quantity	Coverage ratio, RAGAS	Information unit tracking, retrieval analysis
Stability	Self-consistency, position sensitivity	Multi-sample entropy, context permutation
Hallucination	Semantic uncertainty, INSIDE, CHAIR	Disagreement analysis, activation probing

C. Hierarchical Decomposition Module

The decomposition module represents the cognitive core of LENS, transforming complex evaluation tasks into verifiable atomic components through systematic factorization. Our approach extends Tree-of-Thoughts [181] with verification-aware branching strategies that prioritize decomposition paths based on their amenability to external verification. Critically, the decomposition process identifies which components require external tool verification versus those that can be reliably computed through reasoning, directly informing the Tool Necessity Detection (TND) evaluation.This decomposition process yields measurable evaluation dimensions, which, along with their representative metrics and measurement protocols, are summarized in Table IV.

Algorithm 1 Hierarchical Query Decomposition

Input: Complex query $C = f(A_1, ..., A_n)$
Output: Optimal decomposition path $\mathcal{P}^*$
Initialize decomposition tree $\mathcal{T} \leftarrow \emptyset$
Generate factorizations $\mathcal{F} \leftarrow \text{Factorize}(C)$
for each factorization $F_i \in \mathcal{F}$ **do**
  Compute verifiability score $v_i \leftarrow \text{Score}(F_i)$
  Add node (F_i, v_i) to $\mathcal{T}$
end for
$\mathcal{P}^* \leftarrow \text{argmax}_{\mathcal{P} \in \mathcal{T}} \sum_{i \in \mathcal{P}} v_i$
for each node $n \in \mathcal{P}^*$ **do**
  if $\text{IsAtomic}(n)$ **then**
    Execute verification $r \leftarrow \text{Verify}(n)$
  else
    Recursively decompose $\text{Decompose}(n)$
  end if
end for
return $\mathcal{P}^*$

For compound queries expressed as $C = f(A_1, ..., A_n)$, LENS generates multiple valid factorizations and selects optimal paths based on verifiability scores computed as:

$$V(\mathcal{P}) = \sum_{i=1}^{|\mathcal{P}|} w_i \cdot v(n_i) - \lambda \cdot \text{Depth}(\mathcal{P}) \tag{10}$$

where $v(n_i)$ represents the verifiability of node i, w_i denotes importance weights, and λ controls the penalty for deep

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decomposition trees.

The system considers three primary decomposition strategies. The direct path $\mathcal{P}_{direct}$ queries C immediately but risks compound errors where mistakes in any component cascade through the entire result. The sequential path $\mathcal{P}_{seq}$ evaluates components A_1 through A_n in order before computing $f(A_1, ..., A_n)$, which provides intermediate checkpoints but may suffer from context contamination. The parallel path $\mathcal{P}_{parallel}$ evaluates all components $\{A_i\}$ concurrently in isolated sessions, preventing error propagation at the cost of increased computational overhead.

Each decomposition node implements verification gates that validate intermediate results before propagation. The verification function $\mathcal{V} : \mathcal{N} \to \{0, 1\}$ determines whether a node passes validation:

$$\mathcal{V}(n) = \begin{cases} 1 & \text{if } \text{Conf}(n) > \theta \wedge \text{Cons}(n) > \gamma \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

where $\text{Conf}(n)$ measures confidence, $\text{Cons}(n)$ evaluates consistency, and θ, γ are configurable thresholds.

D. Tool-Augmented Reasoning and Selection Evaluation

LENS not only integrates external tools to ground model outputs in verifiable evidence but critically evaluates the model's ability to recognize when tools are necessary and select appropriate ones. This dual focus on tool usage and tool selection evaluation addresses a fundamental source of hallucinations: models confidently generating information from parametric memory when they should instead consult external sources. The framework follows the ReAct paradigm [182] for interleaved reasoning and action generation while introducing novel metrics for assessing tool selection competence.

The framework maintains a tool registry $\mathcal{T} = \{t_1, t_2, ..., t_k\}$ where each tool t_i is characterized by its signature:

$$\begin{aligned} t_i : &\mathcal{Q}_i \times \mathcal{P}_i \\ &\to \mathcal{R}_i \times \mathcal{E}_i \end{aligned} \tag{12}$$

where $\mathcal{Q}_i$ represents valid query types, $\mathcal{P}_i$ denotes parameters, $\mathcal{R}_i$ specifies result types, and $\mathcal{E}_i$ provides evidence metadata.

Tool selection follows a learned policy $\pi : \mathcal{S} \times \mathcal{Q} \to \mathcal{T}$ that maps state-query pairs to optimal tool configurations. The policy optimization objective:

$$\begin{aligned} \pi^* = &\text{argmax}_\pi \mathbb{E}_{(s,q)\sim\mathcal{D}} \\ &[r(s, q, \pi(s, q)) - c(\pi(s, q))] \end{aligned} \tag{13}$$

where r represents verification reward and c denotes computational cost.

The framework evaluates tool necessity detection through a binary classification:

$$\text{TND}(q) = \begin{cases} 1 & \text{if } \mathbb{1}[\text{NeedsTool}(q)] = \mathbb{1}[\text{UsedTool}(q)] \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

where $\text{NeedsTool}(q)$ is determined by whether the query requires information beyond the model's training cutoff, specific factual data, or real-time information.

Tool selection accuracy measures the appropriateness of the chosen tool given the query characteristics:

$$\begin{aligned} \text{TSA}(t|q) = &\max_{t^* \in \mathcal{T}_{opt}(q)} \text{sim}(t, t^*) \\ &\cdot \mathbb{1}[\text{Valid}(t, q)] \end{aligned} \tag{15}$$

where $\mathcal{T}_{opt}(q)$ represents the set of optimal tools for query q, and $\text{sim}(t, t^*)$ measures functional similarity between the selected and optimal tools.

When multiple tool invocations yield conflicting evidence, LENS implements a weighted voting scheme:

$$e^* = \text{argmax}_{e\in\mathcal{E}} \sum_{i=1}^{n} \rho_i \cdot \mathbb{1}[t_i \to e] \tag{16}$$

where ρ_i represents the reliability score of tool t_i derived from historical accuracy.

The framework categorizes tools into four primary classes. Retrieval tools $\mathcal{T}_{retrieval}$ encompass both dense and sparse retrievers for factual grounding, employing BM25 for lexical matching with scoring function:

$$\begin{aligned} \text{BM25}(q, d) = &\sum_{i=1}^{|q|} \text{IDF}(q_i) \\ &\cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \end{aligned} \tag{17}$$

Computational tools $\mathcal{T}_{compute}$ include symbolic calculators and code interpreters that provide deterministic verification. Knowledge APIs $\mathcal{T}_{knowledge}$ connect to authoritative databases with structured query interfaces. Perceptual tools $\mathcal{T}_{perceptual}$ incorporate vision encoders and multimodal processors.

E. Structured Output Generation

Ensuring machine-verifiable outputs requires enforcing structural constraints during generation through grammar-guided decoding and schema validation. This structured approach eliminates parsing ambiguities and enables automated downstream processing while maintaining human readability.

LENS supports three constraint modalities with formal specifications. JSON Schema $\mathcal{S}_{JSON}$ provides hierarchical type constraints:

$$\begin{aligned} \mathcal{S}_{JSON} = &\langle \mathcal{T}_{types}, \mathcal{P}_{props}, \\ &\mathcal{R}_{required}, \mathcal{C}_{constraints} \rangle \end{aligned} \tag{18}$$

Context-free grammars $\mathcal{G}_{CFG}$ specified in EBNF notation:

$$\mathcal{G}_{CFG} = \langle V_N, V_T, P, S \rangle \tag{19}$$

where V_N denotes non-terminals, V_T represents terminals, P specifies production rules, and S is the start symbol.

The framework integrates with modern constrained decoding engines through a unified interface. The decoding process enforces constraints at each generation step:

$$p(x_{t+1}|x_{1:t}, \mathcal{C}) = \begin{cases} \frac{p_{LM}(x_{t+1}|x_{1:t})}{\sum_{x'\in\mathcal{V}_\mathcal{C}} p_{LM}(x'|x_{1:t})} & \text{if } x_{t+1} \in \mathcal{V}_\mathcal{C} \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

where $\mathcal{V}_\mathcal{C}$ represents the valid vocabulary under constraint $\mathcal{C}$.

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F. Hallucination Detection Mechanisms

LENS implements multi-layered hallucination detection combining statistical, behavioral, and architectural approaches. This comprehensive strategy recognizes that hallucinations manifest through various signals that no single detection method can fully capture. Tool selection failures identified through low TND or TSA scores serve as early warning indicators, as models that fail to recognize when external verification is needed are significantly more likely to generate hallucinated content.

Statistical detection computes semantic uncertainty through multi-sample analysis. Lexical divergence D_L measures token-level variation:

$$D_L = \frac{1}{k(k-1)} \sum_{i \neq j} \text{EditDist}(y_i, y_j) \tag{21}$$

Semantic clustering C_S analyzes embedding-space dispersion:

$$C_S = \frac{1}{k} \sum_{i=1}^{k} ||\phi(y_i) - \bar{\phi}||^2 \tag{22}$$

where ϕ represents the embedding function and $\bar{\phi}$ denotes the centroid.

Behavioral analysis examines generation patterns through linguistic markers. Hedging detection H_d quantifies uncertainty expressions:

$$H_d = \frac{\sum_{w \in \mathcal{W}_{hedge}} \text{count}(w)}{|y|} \tag{23}$$

where $\mathcal{W}_{hedge}$ contains hedging vocabulary.

When model internals are accessible, architectural probing provides additional detection capabilities. INSIDE probes [187] employ linear classifiers on intermediate representations:

$$\begin{aligned} \mathcal{L}_{probe} = &- \sum_{i=1}^{n} y_i \log(\sigma(W^T h_i + b)) \\ &+ (1 - y_i) \log(1 - \sigma(W^T h_i + b)) \end{aligned} \tag{24}$$

PoLLMgraph [24] analyzes state transition dynamics through Markov chain modeling:

$$P(s_{t+1}|s_t) = \frac{\exp(f_\theta(s_t, s_{t+1}))}{\sum_{s'} \exp(f_\theta(s_t, s'))} \tag{25}$$

G. Observability and Reproducibility

LENS prioritizes reproducibility through comprehensive tracing and standardized logging protocols. The framework adopts OpenTelemetry with OpenInference semantic conventions [188], [189] to create a standardized trace architecture.

Each evaluation generates a trace $\mathcal{T}_{eval}$ comprising spans:

$$\begin{aligned} &\mathcal{T}_{eval} = \{s_1, s_2, ..., s_n\} \\ &\text{where } s_i = \langle id_i, type_i, attrs_i, parent_i, time_i \rangle \end{aligned} \tag{26}$$

Span types include LLM spans for generation events, Retriever spans for search operations, Tool spans for external API calls, and Evaluator spans for scoring computations. Context propagation through parent-child relationships encodes the decomposition hierarchy:

$$\text{Context}(s_i) = \bigcup_{j \in \text{Ancestors}(i)} attrs_j \tag{27}$$

The reproducibility package $\mathcal{R}_{pkg}$ ensures exact result replication:

$$\begin{aligned} \mathcal{R}_{pkg} = \langle &\mathcal{M}_{model}, \mathcal{P}_{prompts}, \mathcal{S}_{seeds}, \\ &\mathcal{T}_{tools}, \mathcal{D}_{data}, \mathcal{E}_{env} \rangle \end{aligned} \tag{28}$$

H. Comprehensive Case Studies: LENS in Action

To demonstrate LENS's transformative capabilities, we present three detailed case studies that reveal how the framework addresses hallucination challenges that defeat current evaluation approaches.

1) Case Study 1: Medical Diagnosis with Life-Critical Stakes: Consider a medical query: "What are the recommended treatments for a 65-year-old patient with Type 2 diabetes, moderate kidney disease (GFR 45), and recent myocardial infarction, who is allergic to ACE inhibitors?"

Current evaluation frameworks would simply check if the generated treatment recommendations match medical databases, providing a binary correct/incorrect assessment. LENS transforms this into comprehensive multi-dimensional evaluation:

*a) **Hierarchical Decomposition**:* LENS decomposes this complex query into verifiable components:

- Patient profile extraction: age, conditions, contraindications
- Individual condition treatments: diabetes protocols, kidney disease management, post-MI care
- Interaction analysis: drug interactions, comorbidity considerations
- Contraindication checking: ACE inhibitor alternatives
- Treatment synthesis: integrated care plan

*b) **Metacognitive Evaluation**:* LENS identifies that this query requires external verification (TND = 1.0) due to:

- Evolving medical guidelines requiring current information
- Patient-specific contraindications demanding specialized databases
- Drug interaction complexity exceeding safe parametric generation

The framework evaluates tool selection, scoring high TSA (0.98) for choosing medical databases over general web search, pharmaceutical interaction checkers over Wikipedia, and current clinical guidelines over training-time knowledge.

*c) **Multi-Dimensional Assessment**:* LENS employs its four-dimensional evaluation framework to provide a comprehensive assessment that moves beyond traditional binary correctness metrics, offering nuanced insights into the safety and reliability of medical recommendations

- Degree: Not just accuracy but completeness of contraindication checking
- Quantity: Coverage of all relevant treatment modalities

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- Stability: Consistency across similar patient profiles
- Risk: Explicit uncertainty quantification for off-label recommendations

*d) **Actionable Output**:* LENS provides not just a score but a complete evaluation trace showing which guidelines were consulted, what alternatives were considered for ACE inhibitors, how drug interactions were verified, and where uncertainty remains. This transparency enables medical professionals to validate reasoning and identify areas requiring human oversight.

2) Case Study 2: Legal Analysis with Jurisdictional Complexity: Query: "Can a California-based startup use customer data collected before CCPA implementation for training ML models if users consented under previous terms of service?"

This query exemplifies the complex intersection of temporal, jurisdictional, and technical factors that current frameworks cannot adequately evaluate. LENS reveals the multi-layered analysis required:

*a) **Temporal Decomposition**:*

- Pre-CCPA data collection period identification
- CCPA implementation timeline (January 1, 2020)
- Retroactivity provisions analysis
- Grandfathering clause examination

*b) **Jurisdictional Analysis**:*

- California state law requirements
- Federal data protection considerations
- Industry-specific regulations (HIPAA, FERPA if applicable)
- International implications for global users

*c) **Tool Necessity Recognition**:* LENS correctly identifies (TND = 1.0) that this requires:

- Current legal databases, not training-time knowledge
- Recent case law and regulatory guidance
- Industry-specific interpretations

The framework penalizes any attempt to generate legal advice from parametric memory, recognizing that legal interpretation requires authoritative sources and jurisdictional expertise.

*d) **Risk-Aware Evaluation**:* LENS's risk dimension captures that incorrect legal advice could lead to:

- Regulatory penalties (up to 4% of global revenue under CCPA)
- Class action lawsuits
- Reputational damage

This severity weighting ensures that legal hallucinations are not treated equally with minor factual errors, providing organizations with risk-adjusted evaluation scores that reflect real-world consequences.

3) Case Study 3: Financial Analysis with Multi-Hop Reasoning: Query: "What is the total population of the three largest cities in the country where the Eiffel Tower is located?"

While appearing straightforward, this query demonstrates LENS's ability to handle multi-hop reasoning that current frameworks evaluate superficially:

The decomposition phase applies Algorithm 1 to generate the factorization tree:

$$C = \text{sum}(\text{population}(\text{top3}(\text{cities}(\text{country}(\text{Eiffel Tower}))))) \tag{29}$$

$$= \text{sum}(\text{population}(\text{top3}(\text{cities}(\text{France})))) \tag{30}$$

$$= \text{sum}(\text{population}(\{\text{Paris}, \text{Marseille}, \text{Lyon}\})) \tag{31}$$

Each atomic component undergoes tool-augmented verification with explicit evaluation of tool necessity and selection. For the Eiffel Tower location query, the framework correctly identifies that external verification is needed (TND = 1.0) since this requires specific factual knowledge, and the Wikipedia API selection scores high on TSA (0.95) as an authoritative geographic source. The model's attempt to recall "France" from parametric memory without tool consultation would trigger a tool necessity violation (TND = 0.0), flagging potential hallucination risk.

Statistical databases provide $\text{cities}_{ranked}(\text{France})$ with the framework validating that city ranking requires external data sources rather than model estimation. Population data retrieval generates evidence set $\{e_3, e_4, e_5\}$ for each city, where the framework would penalize any attempt to generate approximate population figures from training data. The calculator tool for final summation demonstrates appropriate tool selection for deterministic computation, avoiding potential arithmetic errors from language model calculation.

The structured output captures the complete evaluation trace:


```
{
"query": "Total population of three largest
       cities...",
"decomposition": [
 {"step": "location",
  "result": "France",
  "tool": "WikipediaAPI",
  "tool_necessity": 1.0,
  "tool_selection": 0.95,
  "evidence": {...}},
 {"step": "cities",
  "result": ["Paris", "Marseille", "Lyon"],
  "tool": "StatDB",
  "tool_necessity": 1.0,
  "tool_selection": 0.92,
  "evidence": {...}},
 {"step": "population",
  "result": [2161000, 861635, 513275],
  "tool": "StatDB",
  "tool_necessity": 1.0,
  "tool_selection": 0.92,
  "evidence": {...}},
 {"step": "sum",
  "result": 3535910,
  "tool": "Calculator",
  "tool_necessity": 0.8,
  "tool_selection": 1.0}
```


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```
 ],
"scores": {
"degree": 0.94,
"degree_components": {
  "truthfulness": 0.95,
  "faithfulness": 0.96,
  "tool_appropriateness": 0.91
},
"quantity": 1.0,
"stability": 0.92,
"hallucination_risk": 0.08
 }
}
```


I. Implementation Considerations

Deploying LENS requires careful consideration of computational and architectural factors. Resource management implements intelligent caching to minimize redundant computations:

$$\text{Cache}(q) = \begin{cases} \mathcal{R}_{cached} & \text{if } h(q) \in \mathcal{C} \wedge \text{TTL}(h(q)) > 0 \\ \text{Execute}(q) & \text{otherwise} \end{cases} \tag{32}$$

Session isolation for parallel decomposition requires memory allocation:

$$\begin{aligned} \text{Mem}_{required} = & n_{parallel} \times \\ & (\text{Base}_{LLM} + \text{Context}_{max} + \text{Buffer}_{tool}) \end{aligned} \tag{33}$$

Rate limiting for external APIs follows exponential backoff:

$$\text{Delay}(k) = \min(2^k \cdot \text{base_delay}, \text{max_delay}) \tag{34}$$

The framework provides adapters for common evaluation pipelines through standardized interfaces. Export formats include JSON for structured analysis, CSV for tabular processing, and specialized formats for benchmark integration. Performance optimization employs batching strategies:

$$\text{Throughput} = \frac{n_{queries}}{\max(\text{Time}_{LLM}, \text{Time}_{tools}, \text{Time}_{scoring})} \tag{35}$$

Integration with production systems leverages containerization and orchestration platforms. The modular architecture enables selective component deployment based on resource constraints and evaluation requirements.

VI. Discussion and Limitations

While the LENS framework represents a significant advancement in hallucination evaluation, addressing many critical gaps in existing approaches, it is essential to acknowledge its limitations and discuss the broader implications for the field of AI evaluation [3], [13]. This section examines both the inherent constraints of our approach and the systemic challenges that remain in hallucination assessment, providing a balanced perspective on the framework's contributions and areas for future development.

A. Computational Complexity and Scalability Challenges

The hierarchical decomposition architecture that enables LENS's comprehensive evaluation introduces non-trivial computational overhead that must be carefully considered in deployment contexts [37], [52]. The tree-based query decomposition, while providing unprecedented depth of analysis, results in computational complexity that grows exponentially with query complexity in the worst case [22], [23].

1) Decomposition Overhead: The hierarchical decomposition process transforms each query Q into a tree structure with complexity $O(b^d)$ where b represents the average branching factor and d the decomposition depth. While our pruning strategies and verification gates reduce the practical complexity to approximately $O(b \cdot d \cdot \log n)$ for most queries, pathological cases requiring full tree exploration can still incur significant computational costs. This overhead becomes particularly pronounced when evaluating models on large-scale benchmarks where thousands of queries must be processed.

The decomposition overhead manifests in three primary dimensions [57]. First, the computational cost of tree construction itself requires sophisticated parsing and reasoning capabilities that may exceed the resources available in resource-constrained environments [66]. Second, the parallel evaluation of sub-queries, while improving accuracy, multiplies the number of model invocations required for each evaluation. Third, the aggregation and synthesis of results from multiple decomposition paths introduces additional processing requirements that scale with tree complexity.

2) Tool Integration Latency: The integration of external tools for verification, while essential for comprehensive evaluation, introduces latency that can significantly impact evaluation throughput [46]. Each tool invocation incurs network latency, processing time, and potential rate limiting constraints [26], [76]. For knowledge-intensive queries requiring multiple tool consultations, the cumulative latency can extend evaluation time from seconds to minutes per query. This latency becomes prohibitive when evaluating models on large-scale benchmarks or in real-time deployment scenarios where rapid assessment is required.

Moreover, the reliability and availability of external tools introduce additional variability in evaluation performance. Tool failures, API changes, or service disruptions can cascade through the evaluation pipeline, requiring robust fallback mechanisms and redundancy strategies that further increase system complexity. The framework must maintain multiple tool alternatives and implement sophisticated retry logic, adding layers of complexity to the evaluation infrastructure.

B. Knowledge Source Dependencies and Quality Assurance

LENS's adaptive benchmark construction relies fundamentally on the quality and completeness of provided knowledge sources, creating dependencies that can limit evaluation effectiveness in certain contexts.

1) Ground Truth Ambiguity: The definition of ground truth for hallucination evaluation remains inherently problematic

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in many domains [3], [47]. While factual queries about established scientific facts or historical events may have clear ground truth, many real-world applications involve subjective interpretation, evolving knowledge, or contextual nuance that resist binary classification [38], [48]. Legal interpretations vary across jurisdictions, medical recommendations evolve with new research, and cultural contexts shape the acceptability of generated content. LENS attempts to address this through multi-dimensional evaluation and confidence calibration, but the fundamental challenge of defining truth in complex domains persists.

The temporal dimension of truth presents particular challenges [64]. Facts that were considered accurate at the time of model training may become outdated, creating a moving target for evaluation [6], [69]. Scientific theories evolve, political landscapes shift, and social norms change, all affecting what constitutes a hallucination versus legitimate generation based on training data. While LENS's adaptive framework can incorporate updated knowledge sources, distinguishing between outdated training and genuine hallucination requires sophisticated temporal reasoning that remains an open challenge.

2) Knowledge Source Verification: The framework's reliance on external knowledge sources for verification introduces a critical dependency on the accuracy and completeness of these sources [101]. Errors, biases, or gaps in reference databases propagate through the evaluation pipeline, potentially misclassifying correct generations as hallucinations or vice versa [44], [71]. While LENS implements multi-source verification and confidence weighting, the fundamental challenge of establishing authoritative ground truth in domains with conflicting or incomplete information remains unresolved.

The curation and maintenance of high-quality knowledge sources requires significant ongoing effort. Organizations must continuously update their reference databases, validate information accuracy, and resolve conflicts between sources. This maintenance burden can be substantial, particularly for specialized domains requiring expert knowledge. The cost and complexity of maintaining comprehensive, accurate knowledge sources may limit LENS adoption in resource-constrained environments or rapidly evolving fields.

C. Evaluation Gaming and Adversarial Considerations

As with any evaluation framework, LENS faces the risk of models being optimized specifically to perform well on its metrics rather than genuinely reducing hallucination, a phenomenon known as Goodhart's Law [18], [26].

1) Metric Optimization Without True Improvement: The detailed specification of LENS's evaluation dimensions creates potential targets for optimization that may not correspond to genuine improvements in hallucination reduction [50]. Models could learn to generate responses that score well on Tool Necessity Detection by always requesting external verification, even when unnecessary, inflating TND scores without improving actual reliability [45]. Similarly, models might learn to produce conservative, hedged responses that score well on confidence calibration metrics while providing less useful information to users.

The hierarchical decomposition approach, while providing valuable structure, also creates predictable patterns that models could exploit. A model trained on LENS evaluation data might learn to generate responses that decompose cleanly into the expected tree structure, scoring well on structural metrics without necessarily improving factual accuracy. This risk is particularly acute given the increasing sophistication of model training techniques and the availability of evaluation datasets for training.

2) Adversarial Hallucination Patterns: Sophisticated adversarial actors could potentially design hallucination patterns specifically crafted to evade LENS's detection mechanisms [26]. By understanding the framework's decomposition strategies and verification procedures, adversaries could construct hallucinations that appear valid at each decomposition level but combine to form misleading or incorrect complete responses [51]. These compositional adversarial examples represent a significant challenge for any structured evaluation approach.

The framework's reliance on external tools also creates potential attack vectors. Adversaries could potentially manipulate or compromise reference sources, inject misleading information into verification databases, or exploit temporal gaps between knowledge source updates. While LENS implements multi-source verification and confidence weighting, coordinated attacks across multiple sources could still compromise evaluation integrity.

D. Generalization and Cross-Domain Applicability

While LENS provides adaptive benchmark construction capabilities, certain limitations in generalization across domains and tasks remain.

1) Domain-Specific Reasoning Patterns: Different domains exhibit unique reasoning patterns and evaluation requirements that may not be fully captured by LENS's general framework [9], [83]. Mathematical proofs require different verification strategies than medical diagnoses, which in turn differ from creative writing evaluation [1], [178]. While the framework's adaptability allows for domain-specific customization, the burden of specifying appropriate evaluation strategies for each domain falls on users, requiring significant expertise and effort.

The transfer of evaluation strategies across domains remains limited. Insights gained from evaluating hallucinations in one domain may not generalize to others due to fundamental differences in knowledge structure, reasoning requirements, and acceptable uncertainty levels. This limitation necessitates domain-specific configuration and tuning, reducing the framework's plug-and-play applicability across diverse use cases.

2) Cultural and Linguistic Variations: LENS's evaluation strategies, developed primarily in English-language contexts, may not fully account for cultural and linguistic variations in what constitutes hallucination [49], [70]. Concepts that are unambiguous in one culture may be highly contextual in

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another [43]. Linguistic structures in non-English languages may not map cleanly to the framework's decomposition strategies. These variations introduce evaluation biases that could disadvantage models trained on or evaluated in non-Western contexts.

The framework's tool integration capabilities are also limited by the availability of verification resources across languages and cultures. High-quality knowledge sources and verification tools are disproportionately available for English and major European languages, creating evaluation gaps for low-resource languages. This limitation perpetuates existing inequalities in AI development and deployment, potentially excluding important use cases and communities from comprehensive hallucination evaluation.

E. Human Factors and Interpretability Challenges

Despite LENS's emphasis on transparency and interpretability, significant challenges remain in making evaluation results accessible and actionable for diverse stakeholders.

1) Cognitive Load of Multi-Dimensional Results: The framework's multi-dimensional evaluation produces rich but complex results that can overwhelm users attempting to interpret and act upon them [30], [55]. Understanding the interplay between degree, quantity, stability, and risk dimensions requires sophisticated mental models that many stakeholders may lack [31]. While LENS provides visualization tools and summary metrics, the fundamental complexity of multi-dimensional evaluation creates cognitive barriers to adoption and effective use.

The hierarchical decomposition traces, while valuable for understanding evaluation reasoning, can be difficult to interpret without technical expertise. Non-technical stakeholders may struggle to understand why certain decomposition strategies were chosen or how sub-query results aggregate to final scores. This interpretability gap limits the framework's utility for broader organizational decision-making and public accountability.

2) Calibration of Human Judgment: LENS's evaluation results must ultimately be interpreted and acted upon by humans, introducing variability in how scores translate to deployment decisions [8]. Different stakeholders may have varying risk tolerances, leading to inconsistent interpretation of the same evaluation results [34]. A hallucination rate deemed acceptable by one organization might be considered prohibitive by another, even within the same domain.

The calibration of human judgment to LENS metrics requires significant experience and training. Users must develop intuition for what different score combinations mean in practice, how to weight dimensions for their specific use cases, and when to trust or question evaluation results. This learning curve presents a barrier to adoption and may lead to misinterpretation of results during the initial deployment phase.

F. Broader Implications for AI Evaluation

The development of LENS raises fundamental questions about the nature and future of AI evaluation that extend beyond hallucination assessment.

1) The Evaluation-Capability Arms Race: As evaluation frameworks become more sophisticated, model developers increasingly optimize for evaluation metrics, creating an arms race between evaluation and capability development [154]. This dynamic risks divorcing evaluation performance from real-world utility, as models become increasingly adept at gaming evaluation systems rather than genuinely improving [129]. LENS's comprehensive approach may temporarily raise the bar for such optimization, but the fundamental tension between evaluation and genuine capability remains.

The arms race also manifests in the increasing complexity of evaluation frameworks themselves. As models become more sophisticated, evaluation must become correspondingly complex, creating a spiral of increasing computational and methodological overhead. This complexity growth may eventually reach practical limits, necessitating fundamental rethinking of how we approach AI evaluation.

2) Standardization Versus Innovation: The adoption of comprehensive frameworks like LENS creates tension between the need for standardized evaluation to enable comparison and the desire for innovative evaluation approaches tailored to specific use cases [2], [25]. Standardization facilitates benchmarking and progress tracking but may constrain creative evaluation strategies that could provide novel insights. Organizations must balance the benefits of adopting established frameworks against the potential advantages of developing custom evaluation approaches.

This tension extends to the broader AI community's approach to evaluation. Should the field converge on a small number of comprehensive frameworks, or should it maintain a diverse ecosystem of evaluation approaches? LENS's adaptive architecture attempts to bridge this divide, but the fundamental tension between standardization and innovation persists.

G. Future Directions and Research Opportunities

The limitations identified above point to several promising directions for future research and development.

1) Automated Evaluation Design: Future work could explore automated methods for designing evaluation strategies tailored to specific domains and use cases [35], [180]. Machine learning techniques could potentially learn optimal decomposition strategies, tool selection policies, and aggregation methods from limited human feedback, reducing the expertise required for framework deployment [173]. Such automation could democratize access to sophisticated evaluation capabilities while maintaining rigor and reliability.

2) Federated Evaluation Networks: Addressing the knowledge source quality challenge could involve developing federated evaluation networks where organizations share verification resources and evaluation results while maintaining privacy and security [100]. Blockchain or other distributed ledger technologies could provide tamper-resistant audit trails

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for evaluation results, enhancing trust and accountability [58]. Such networks could pool resources to maintain high-quality knowledge sources while distributing the maintenance burden.

3) Continuous Evaluation and Monitoring: Moving beyond static evaluation to continuous monitoring of deployed models could provide early warning of emerging hallucination patterns [24], [61]. Online learning techniques could enable evaluation frameworks to adapt to changing model behavior and evolving knowledge landscapes [16], [17]. Integration with production systems could enable real-time hallucination detection and mitigation, shifting evaluation from a development-time activity to an operational capability.

H. Conclusion of Limitations

The limitations discussed in this section are not fundamental flaws but rather inherent challenges in the complex task of comprehensive hallucination evaluation. LENS represents a significant step forward in addressing many critical gaps in existing frameworks, but perfect evaluation remains an aspirational goal rather than an achievable target. The framework's limitations should be understood as defining the boundaries of current capability rather than diminishing its contributions.

By acknowledging these limitations transparently, we aim to foster realistic expectations about what automated evaluation can achieve and inspire continued research to address remaining challenges [5]. The path toward trustworthy AI requires not just better evaluation frameworks but also honest assessment of their capabilities and constraints [41], [54]. LENS provides a robust foundation for this journey, but the work of ensuring safe and reliable AI systems remains an ongoing collaborative effort requiring continued innovation, vigilance, and humility about the challenges that lie ahead.

VII. Conclusion: Toward Trustworthy AI Through Comprehensive Hallucination Evaluation

The deployment of Large Language Models and Vision Language Models in critical applications demands evaluation frameworks that match the sophistication and stakes of these systems. Throughout this paper, we have traced the evolution of hallucination evaluation from simplistic accuracy measurements to the comprehensive, multidimensional assessment necessary for trustworthy AI deployment. The journey reveals not merely technical challenges but fundamental reconceptualization of how we understand, detect, and mitigate the generation of plausible yet false content that threatens to undermine AI reliability.

A. The Journey Through Complexity

Our exploration began with a sobering reality: hallucination rates ranging from 1.47% in clinical applications to 75% in specialized domains represent not statistical curiosities but existential threats to AI deployment in high stakes environments. These failures manifest not as obvious errors easily caught by casual inspection, but as sophisticated fabrications that exploit the probabilistic foundations of generative models. The medical diagnosis that appears clinically sound yet endangers patients, the legal citation that seems authoritative yet references non existent precedents, and the financial analysis that appears rigorous yet contains fabricated data all exemplify the insidious nature of hallucinations that current evaluation frameworks consistently fail to detect or prevent.

Section II revealed the profound complexity underlying hallucination phenomena. The epistemological challenge of establishing ground truth in domains characterized by uncertainty, evolution, and legitimate disagreement exposes the naivety of binary truth classifications. Scientific understanding evolves, medical best practices shift with emerging evidence, and legal interpretations vary across jurisdictions and time. The computational complexity of comprehensive evaluation grows exponentially with query complexity, creating fundamental tensions between thoroughness and feasibility. The attribution challenge reveals that understanding why hallucinations occur proves as crucial as detecting them, yet current frameworks provide minimal insight into causal mechanisms. These challenges interconnect and compound, creating a problem space that resists simple solutions or incremental improvements.

Our systematic review of existing evaluation approaches in Section III-D exposed a fragmented landscape where different frameworks capture isolated aspects of hallucination without achieving comprehensive assessment. Output based evaluation through benchmarks like TruthfulQA and FactCC excel at detection but provide no insight into reasoning processes or failure mechanisms. Process based approaches including StrategyQA and DiaHalu trace reasoning chains but lack the breadth necessary for comprehensive assessment across domains. Capability based evaluations probe specific skills but fail to capture the complex interactions that produce hallucinations in real deployments. This fragmentation forces practitioners to navigate incompatible frameworks, reconcile contradictory results, and still lack confidence in their assessments.

B. The Fundamental Gaps and Their Resolution

Section IV crystallized the fundamental inadequacies of current approaches into five critical gaps that prevent effective hallucination evaluation. The dimensional poverty reducing complex phenomena to single metrics ignores the multifaceted nature of hallucinations that vary in severity, scope, and impact. The artificial separation between horizontal breadth and vertical depth forces an unnecessary choice between understanding what errors occur versus why they occur. The metacognitive blind spot completely overlooks models' ability to recognize when external verification is necessary rather than generating from parametric memory. The adaptability crisis locks organizations into rigid benchmarks that cannot accommodate their specific knowledge domains or quality requirements. The transparency deficit provides scores without explanations, preventing actionable understanding or targeted improvement.

These gaps are not independent limitations but interconnected symptoms of a more fundamental problem: the absence

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of unified frameworks that synthesize multiple dimensions, modalities, and perspectives into coherent, actionable assessment. Current approaches treat hallucination as a monolithic phenomenon amenable to simple measurement, failing to recognize it as a complex, multidimensional challenge requiring sophisticated analytical frameworks. The consequence is evaluation that appears rigorous through extensive benchmarking yet fails to provide the understanding necessary for safe deployment or systematic improvement.

C. LENS: A Paradigm Shift in Evaluation

The LENS framework presented in Section V represents not incremental improvement but fundamental reconceptualization of hallucination evaluation. Through hierarchical decomposition that transforms complex queries into verifiable atomic components, LENS exposes the reasoning processes that current black box evaluations obscure. This decomposition is not merely structural but functional, identifying precisely where in multi step reasoning hallucinations emerge and how errors propagate through inference chains.

The framework's four dimensional evaluation transcends binary detection to capture the full complexity of hallucination phenomena. The degree dimension extends beyond simple accuracy to encompass faithfulness to context and crucially, the appropriateness of tool usage through our novel Tool Necessity Detection and Tool Selection Accuracy metrics. The quantity dimension addresses hallucinations of omission that current frameworks ignore, recognizing that incomplete information can be as dangerous as false information. The stability dimension reveals whether errors are systematic vulnerabilities or stochastic failures, essential for understanding model reliability. The risk dimension quantifies uncertainty and confidence calibration, distinguishing between models that acknowledge limitations and those that hallucinate with unwarranted confidence.

Most significantly, LENS introduces metacognitive evaluation as a first class concern. By explicitly assessing whether models recognize when external verification is necessary (Tool Necessity Detection) and whether they select appropriate tools for the task (Tool Selection Accuracy), LENS addresses a fundamental source of hallucinations that current frameworks completely overlook. This paradigm shift from post hoc error detection to proactive risk assessment transforms evaluation from measurement to prevention, identifying models that will hallucinate because they lack the metacognitive awareness to seek help when needed.

The framework's adaptive benchmark construction enables organizations to create rigorous evaluations from their own information sources while maintaining standardized methodologies. This resolves the fundamental tension between customization and standardization, allowing domain specific assessment without sacrificing comparability. Healthcare providers can evaluate against their specific treatment protocols, legal firms against their jurisdictional requirements, and financial institutions against their regulatory frameworks, all while maintaining methodological rigor that enables meaningful comparison across implementations.

D. Transformative Impact on AI Development and Deployment

LENS fundamentally transforms how organizations approach AI evaluation and deployment. Rather than relying on generic benchmarks that poorly approximate their use cases, organizations can now construct evaluations that directly assess performance on their specific requirements while maintaining scientific rigor. This capability shifts evaluation from a development phase checkpoint to an ongoing operational capability that continuously validates model behavior against evolving requirements and knowledge.

For model developers, LENS provides unprecedented visibility into failure mechanisms that enable targeted improvements rather than blind architectural adjustments. The framework's hierarchical decomposition reveals whether hallucinations stem from knowledge gaps requiring expanded training data, reasoning failures necessitating architectural changes, or metacognitive limitations demanding new capabilities. This causal understanding transforms model development from trial and error exploration to systematic engineering based on empirical understanding of failure modes.

The transparency and interpretability built into LENS enable new forms of accountability and governance for AI systems. Regulatory bodies can establish evaluation requirements that go beyond simple accuracy thresholds to encompass the full spectrum of reliability considerations. Organizations can provide stakeholders with comprehensible explanations of model capabilities and limitations, building trust through transparency rather than demanding faith in opaque systems. Users can calibrate their reliance on AI outputs based on detailed understanding of when and how models might fail, enabling appropriate human oversight where necessary.

E. Acknowledging Limitations While Charting the Path Forward

Our discussion in Section VI provides honest assessment of LENS's limitations, recognizing that perfect evaluation remains aspirational rather than achievable. The computational overhead of hierarchical decomposition, while manageable for most applications, becomes significant for extremely complex queries requiring deep trees. The framework's effectiveness depends on the quality of knowledge sources used for verification, inheriting any biases or gaps in these references. The risk of evaluation gaming remains, as sophisticated actors might craft hallucinations specifically designed to evade LENS's detection mechanisms.

Yet these limitations define the boundaries of current capability rather than fundamental flaws in the approach. Each limitation points toward specific research directions that can extend and strengthen the framework. Automated evaluation design using machine learning could reduce the expertise required for framework deployment. Federated evaluation networks could pool verification resources while maintaining privacy

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and security. Continuous evaluation integrated with production systems could provide real time hallucination detection and mitigation.

The broader implications extend beyond hallucination evaluation to fundamental questions about AI assessment. As models become increasingly sophisticated, evaluation must correspondingly evolve, creating a dynamic that drives both capability development and assessment methodology. LENS represents one step in this evolution, demonstrating that comprehensive evaluation is possible while revealing the challenges that remain. The framework establishes a foundation for continued innovation while acknowledging that ensuring trustworthy AI requires ongoing vigilance and continuous advancement of evaluation capabilities.

F. A Call to Action for the AI Community

The development and deployment of trustworthy AI systems represents one of the defining challenges of our era. As these systems assume greater responsibilities in critical applications, the cost of hallucinations escalates from inconvenience to catastrophe. Medical decisions affecting patient lives, legal judgments determining justice, financial analyses guiding economic decisions, and educational content shaping future generations all depend on AI systems that must be reliable, transparent, and accountable.

LENS provides the technical foundation for this trustworthiness, but realizing its potential requires collective action from the AI community. Researchers must extend and refine the framework, addressing current limitations while exploring new evaluation dimensions. Practitioners must adopt comprehensive evaluation as a core requirement rather than optional enhancement, integrating frameworks like LENS into development and deployment pipelines. Organizations must invest in the infrastructure and expertise necessary for rigorous evaluation, recognizing that the cost of comprehensive assessment pales compared to the risk of deploying unreliable systems. Policymakers must establish evaluation standards that protect public interest while enabling innovation, balancing safety with progress.

The path toward trustworthy AI is neither simple nor certain, but it is necessary and achievable. Through frameworks like LENS that combine theoretical rigor with practical applicability, we can transform AI systems from impressive but unreliable technologies into trustworthy partners suitable for deployment in applications where accuracy, reliability, and interpretability are non negotiable requirements. The future of AI depends not on the sophistication of models alone but on our ability to understand, evaluate, and ensure their reliability. LENS represents a significant step on this journey, demonstrating that comprehensive hallucination evaluation is not only necessary but possible.

As we stand at this inflection point in AI development, the choices we make about evaluation and reliability will determine whether these powerful technologies fulfill their promise of beneficial transformation or become sources of systemic risk. The LENS framework provides tools and methodologies for choosing the path of trustworthiness, but walking that path requires commitment from every stakeholder in the AI ecosystem. The work is challenging, the stakes are high, and the need is urgent. Through collaborative effort guided by frameworks like LENS, we can build an AI future characterized not by spectacular failures but by reliable, trustworthy systems that enhance human capability while respecting human values.

The journey from recognizing hallucinations as a critical challenge to developing comprehensive evaluation frameworks has revealed the depth and complexity of ensuring AI reliability. LENS represents not the end of this journey but a crucial waypoint, demonstrating what is possible while illuminating the path ahead. As we continue to push the boundaries of AI capability, we must equally advance our ability to evaluate and ensure the trustworthiness of these systems. The future depends not on choosing between capability and reliability but on achieving both through rigorous, comprehensive evaluation that matches the sophistication of the systems we seek to deploy.

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